

Stochastic Analysis, Modeling, and Simulation (SAMS) Version 2007

USER's MANUAL

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December, 2007



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Department of Civil and Environmental Engineering
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Fort Collins, Colorado**

TECHNICAL REPORT No.11

Stochastic Analysis, Modeling, and Simulation (SAMS) Version 2007 - User's Manual

by

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PREFACE

Several computer packages have been developed since the 1970's for analyzing the stochastic characteristics of time series in general and hydrologic and water resources time series in particular. For instance, the LAST package was developed in 1977-1979 by the US Bureau of Reclamation (USBR) in Denver, Colorado. Originally the package was designed to run on a mainframe computer, but later it was modified for use on personal computers. While various additions and modifications have been made to LAST over the past twenty years, the package has not kept pace with either advances in time series modeling or advances in computer technology. These facts prompted USBR to promote the initial development of SAMS, a computer software package that deals with the Stochastic Analysis, Modeling, and Simulation of hydrologic time series, for example annual and seasonal streamflow series. It is written in C, Fortran, and C++, and runs under modern windows operating systems such as WINDOWS XP. This manual describes the current version of SAMS denoted as SAMS 2007.

ACKNOWLEDGEMENTS

SAMS has been developed as a cooperative effort between USBR and Colorado State University (CSU) under USBR Advanced Hydrologic Techniques Research Project through an Interagency Personal Agreement with Professor Jose D. Salas as Principal Investigator. Drs. W.L. Lane and D.K. Frevert provided additional expert guidance and supervision on behalf of USBR. Further enhancements were made in collaboration with the International Joint Commission for Lake Ontario, HydroQuebec, Canada, and the Great Lakes Environmental Research Laboratory (NOAA), Ann Arbor Michigan. Currently further improvements are being made in collaboration with the USBR Lower Colorado Region, Boulder City, Nevada. Several former CSU graduate students collaborated in various parts of this project including, M.W. AbdelMohsen, who developed many of the Fortran codes, M. Ghosh who initiated the programming in C language followed by Mr. Bradley Jones, Nidhal M. Saada, and Chen-Hua Chung. The latest version has been reprogrammed by Oli G. B. Sveinsson. Acknowledgements are due to the funding agency and to the several students who collaborated in this project.

STOCHASTIC ANALYSIS, MODELING, AND SIMULATION

(SAMS 2007)

1. INTRODUCTION

Stochastic simulation of water resources time series in general and hydrologic time series in particular has been widely used for several decades for various problems related to planning and management of water resources systems. Typical examples are determining the capacity of a reservoir, evaluating the reliability of a reservoir of a given capacity, evaluation of the adequacy of a water resources management strategy under various potential hydrologic scenarios, and evaluating the performance of an irrigation system under uncertain irrigation water deliveries (Salas et al, 1980; Loucks et al, 1981).

Stochastic simulation of hydrologic time series such as streamflow is typically based on mathematical models. For this purpose a number of stochastic models have been suggested in literature (Salas, 1993; Hipel and McLeod, 1994). Using one type of model or another for a particular case at hand depends on several factors such as, physical and statistical characteristics of the process under consideration, data availability, the complexity of the system, and the overall purpose of the simulation study. Given the historical record, one would like the model to reproduce the historical statistics. This is why a standard step in streamflow simulation studies is to determine the historical statistics. Once a model has been selected, the next step is to estimate the model parameters, then to test whether the model represents reasonably well the process under consideration, and finally to carry out the needed simulation study.

The advent of digital computers several decades ago led to the development of computer software for mathematical and statistical computations of varied degree of sophistication. For instance, well known packages are IMSL, STATGRAPHICS, ITSM, MINITAB, SAS/ETS, SPSS, and MATLAB. These packages can be very useful for standard time series analysis of hydrological processes. However, despite of the availability of such general purpose programs, specialized software for simulation of hydrological time series such as streamflow, have been attractive because of several reasons. One is the particular nature of hydrological processes in which periodic properties are important in the mean, variance, covariance, and skewness. Another one is that some hydrologic time series include complex characteristics such as long term dependence and memory.

Still another one is that many of the stochastic models useful in hydrology and water resources have been developed specifically oriented to fit the needs of water resources, for instance temporal and spatial disaggregation models. Examples of specific oriented software for hydrologic time series simulation are HEC-4 (U.S Army Corps of Engineers, 1971), LAST (Lane and Frevert, 1990), and SPIGOT (Grygier and Stedinger, 1990).

The LAST package was developed during 1977-1979 by the U. S. Bureau of Reclamation (USBR). Originally, the package was designed to run on a mainframe computer (Lane, 1979) but later it was modified for use on personal computers (Lane and Frevert, 1990). While various additions and modifications have been made to LAST over the past 20 years, the package has not kept pace with either advances in time series modeling or advances in computer technology. This is especially true of the computer graphics. These facts prompted USBR to promote the initial development of the SAMS package. The first version of SAMS (SAMS-96.1) was released in 1996. Since then, corrections and modifications were made based on feedback received from the users. In addition, new functions and capabilities have been implemented leading to SAMS 2000, which was released in October, 2000.

The most current version is SAMS 2007, which includes new modeling approaches and data analysis features. SAMS 2007 has the following capabilities:

1. Analyze the stochastic features of annual and seasonal data.
2. It includes several types of transformation options to transform the original data into normal.
3. It includes a number of single site, multisite, and disaggregation stochastic models that have been widely used in hydrologic literature.
4. It includes two major modeling schemes for data generation of complex river network systems.
5. The number of samples that can be generated is unlimited.
6. The number of years that can be generated is unlimited.

The main purpose of SAMS is to generate synthetic hydrologic data. It is not built for hydrologic forecasting although data generation for some of the models can be conditioned on most recent historical observations.

The purpose of this manual is to provide a detailed description of the current version of SAMS developed for the stochastic simulation of hydrologic time series such as annual and monthly streamflows.

2. DESCRIPTION OF SAMS

In section 2.1, a general description of SAMS is presented in which different operations undertaken by SAMS are briefly explained. Then, each operation is explained and illustrated in subsequent sections more thoroughly.

2.1 General Overview

SAMS is a computer software package that deals with the stochastic analysis, modeling, and simulation of hydrologic time series. It is written in C, Fortran and C++, and runs under modern windows operating systems such as WINDOWS XP. The package consists of many menu options which enables the user to choose between different options that are currently available. SAMS 2005 is a modified and expanded version of SAMS-96.1 and SAMS 2000. It consists of three primary application modules: 1) Data Analysis, 2) Fit a Model, and 3) Generate Series. Figure 2.1 shows SAMS's main window. The main menu bar indicates "Model" next to "Fit Model" where the model parameters can be shown. It also allows resetting the model. In addition, "Plot Properties" is shown next to "Generate Series", which enables one selecting some useful plotting features grid and zoom.

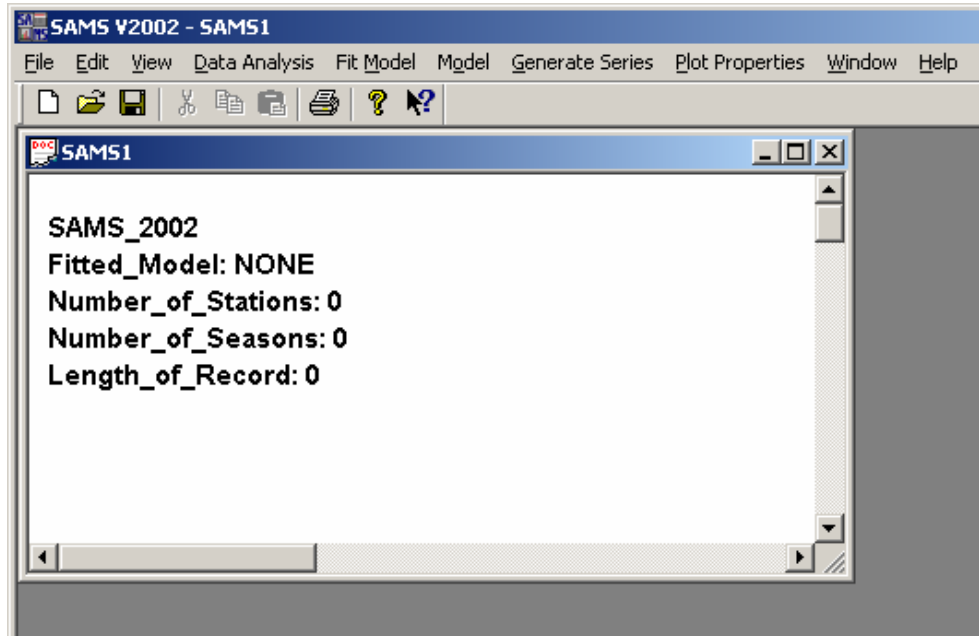


Figure 2.1 The software SAMS main window menu.

Before running the applications, the user must import a file that contains the (historical) input data to be analyzed. This can be done by clicking on "File Menu" then choosing the "Import Flow File" option as shown in Fig. 2.2.

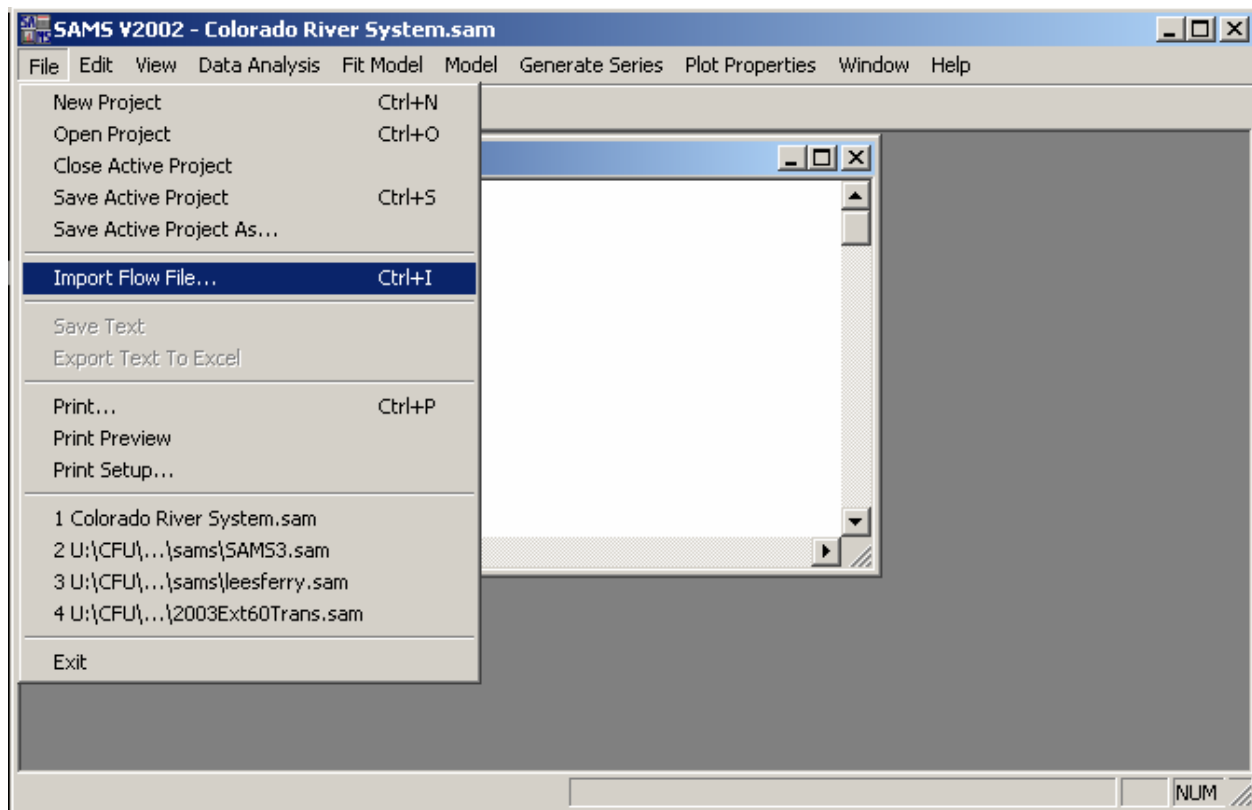


Figure 2.2 Import input data file.

The “Data Analysis” is one of the main applications of SAMS. The functions of this module consist of data plotting, checking the normality of the data, data transformation, and computing and displaying the statistical (stochastic) characteristics of the data. Plotting the data may help detecting trends, shifts, outliers, or errors in the data. Probability plots are included for verifying the normality of the data. The data can be transformed to normal by using different transformation techniques. Currently, logarithmic, power, gamma, and Box-Cox transformations are available. SAMS determines a number of statistical characteristics of the data. These include basic statistics such as mean, standard deviation, skewness, serial correlations (for annual data), spectrum, season-to-season correlations (for seasonal data), annual and seasonal cross-correlations for multisite data, and drought, surplus, and storage related statistics. These statistics are important in investigating the stochastic characteristics of the data.

The second main application of SAMS “Fit Model” includes parameter estimation and model testing for alternative univariate and multivariate stochastic models. The following models are included: (1) univariate $ARMA(p,q)$ model, where p and q can vary from 1 to 10, (2) univariate GAR(1) model, (3) univariate periodic PARMA(p,q) model, (4) univariate shifting-mean SM model,

(5) univariate seasonal disaggregation, (6) multivariate autoregressive $MAR(p)$ model, (7) contemporaneous multivariate $CARMA(p, q)$ model, where p and q can vary from 1 to 10, (8) multivariate periodic $MPAR(p)$ model, (9) multivariate $CSM-CARMA(p, q)$ model, (10) multivariate annual (spatial) disaggregation model, and (11) multivariate temporal disaggregation model.

Two estimation methods are available, namely the method of moments (MOM) and the least squares method (LS). MOM is available for most of the models while LS is available only for univariate ARMA, PARMA, and CARMA models. For CARMA models, both the method of moments (MOM) and the method of maximum likelihood (MLE) are available for estimation of the variance-covariance (G) matrix. Regarding multivariate annual (spatial) disaggregation models, parameter estimation is based on Valencia-Schaake or Mejia-Rousselle methods, while for annual to seasonal (temporal) disaggregation Lane's condensed method is applied.

For stochastic simulation at several sites in a stream network system a direct modeling approach based on multivariate autoregressive and CARMA processes are available for annual data and multivariate periodic autoregressive process is available for seasonal data. In addition, two schemes based on disaggregation principles are available. For this purpose, it is convenient to divide the stations into *key stations*, *substations*, *subsequent stations*, etc. Generally the key stations are the farthest downstream stations, substations are the next upstream stations, and subsequent stations are the next further upstream stations etc. In the first scheme, the annual flows at the key stations are added creating an annual flow data at an “*artificial or index station*”. Subsequently, a univariate $ARMA(p,q)$ model is fitted to the annual flows of the *index* station. Then, a spatial disaggregation model relating the annual flows of the index station to the annual flows of the key stations is fitted. Further, one or more statistical disaggregation models relating the annual flows of the key stations to those of the substations are fitted. This process can be repeated as long as there are any unmodeled stations left, where each modeled station can be defined as key station at the next disaggregation level and each unmodeled station can be defined as substation. In the second scheme a multivariate model is fitted to the annual data of the key stations, then the rest of the model relating the annual flows at the key station, substations, and subsequent stations are conducted in a similar manner as in the first scheme. Furthermore, if the objective of the modeling exercise is to generate seasonal data by using disaggregation approaches, then an additional temporal disaggregation model is fitted that

relates the annual flows of a group of stations with the corresponding seasonal flows. The foregoing schemes of modeling and generation at the annual time scale with spatial disaggregation as needed and then performing the temporal disaggregation can also be reversed, i.e. starting with temporal disaggregation of key station annual flows to seasonal flows followed by spatial disaggregation.

The third main application of SAMS is “Generate Series”, i.e. simulating synthetic data. Data generation is based on the models, approaches, and schemes as mentioned above. The model parameters for data generation are those that are estimated by SAMS. The user also has the option of importing annual series at key stations (e.g. series generated using a software other than SAMS). The statistical characteristics of the generated data are presented in graphical or tabular forms along with the historical statistics of the data that was used in fitting the generating model. The generated data including the "generated" statistics can be displayed graphically or in table form, and be printed and/or written on specified output files. As a matter of clarification, we will summarize here the overall data generation procedure for generating seasonal data based on scheme 2:

- (a) a multivariate model, such as $AR(p)$, is utilized to generate the annual flows at the key stations;
- (b) a spatial disaggregation model is used to disaggregate the generated annual flows at the key stations into annual flows at the substations, followed by additional spatial disaggregations until all upstream stations are taken into account;
- (c) a temporal disaggregation model is used to disaggregate the annual flows at one or more groups of stations into the corresponding seasonal flows at those stations.

2.2 Statistical Analysis of Data

Figure 2.3 shows the “Data Analysis” menu. By selecting this menu the user can carry out statistical analysis on the annual or seasonal data, either original or transformed data. The following are the four operations that the user may choose:

1. Plot Time Series.
2. Transform.
3. Show Statistics.
4. Plot Statistics.

We will examine and illustrate each of these options below.

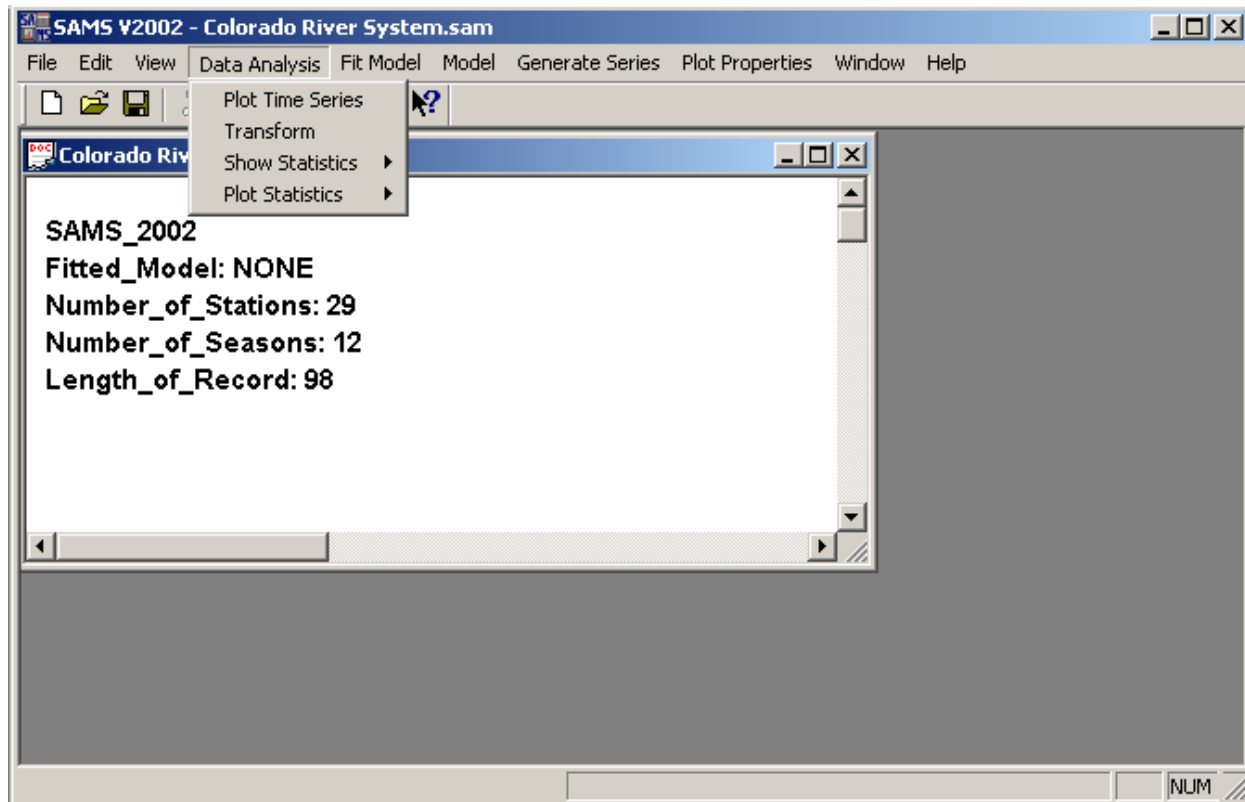


Figure 2.3 Data Analysis menu.

Plot Time Series

Plotting the data can help detecting trends, shifts, outliers, and errors (in the data.) Figure 2.4 shows the menu after choosing the “Plot Time Series” function. Annual or seasonal time series may be plotted in the original or transformed domain. Figure 2.5 illustrates a time series plot for annual data. The user may plot either the entire time series or just part of it. To do so, one must activate the “Plot Properties” menu (also shown in Fig. 2.3) and chose “Range” or “Rectangle” under the menu “ZOOM”. The time series plots and any other plots produced by SAMS can be easily transferred into other word/image processing or spreadsheet applications such as MS Word,

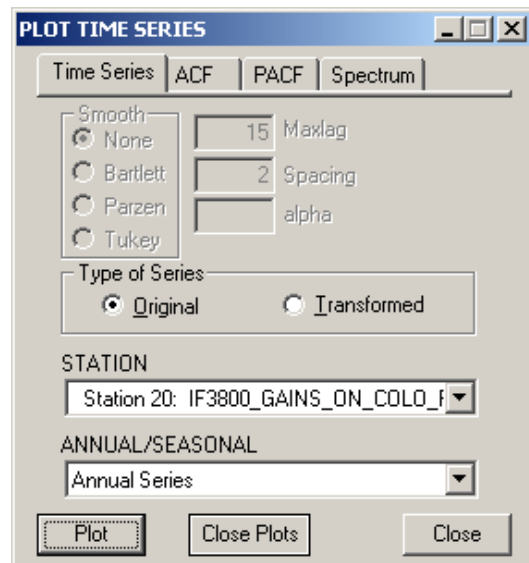


Figure 2.4 Plot Time Series menu.

Excel, and Adobe Photoshop. The transferring can be done by using the “Copy to Clipboard” function, which is also available under the “Plot Properties” menu and then paste the plot into other applications.

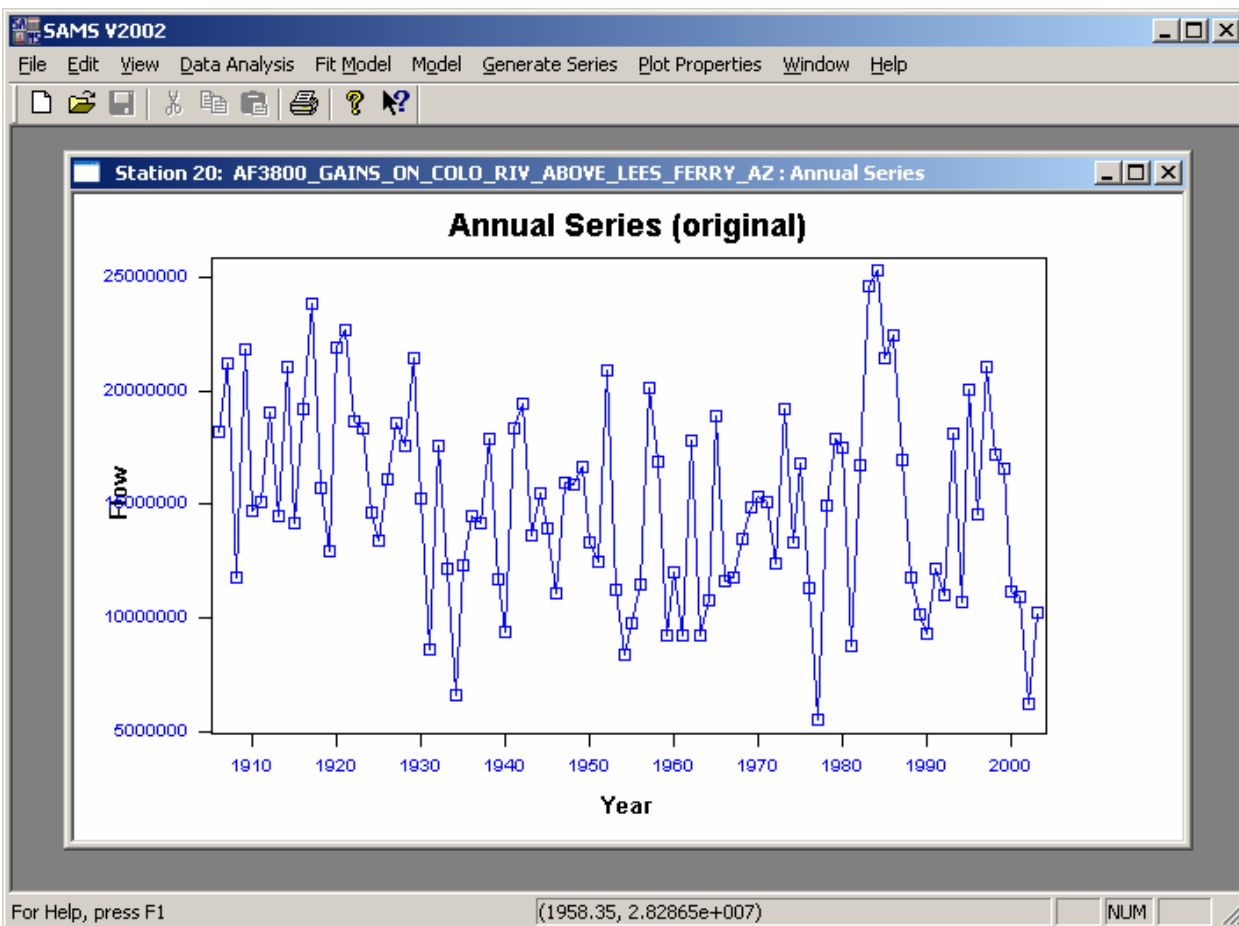


Figure 2.5 Time series of the annual flows of the Colorado River at site 20

Transform Time series

SAMS tests the normality of the data by plotting the data on normal probability paper and by using the skewness and the Filliben tests of normality. To examine the adequacy of the transformation, the comparison of the theoretical distribution based on the transformation and the counterpart historical sample distribution is shown. Meanwhile the critical values and the results of the test are displayed in table format. Figure 2.6 is the display obtained after clicking on the “Transform” menu. The user can test the annual or seasonal data of any site by selecting proper options of “Data Type” and “Station #” on the left hand side panel. To plot the empirical frequency

distribution the user may select either the Cunnane's or the Weibull's plotting position equations.

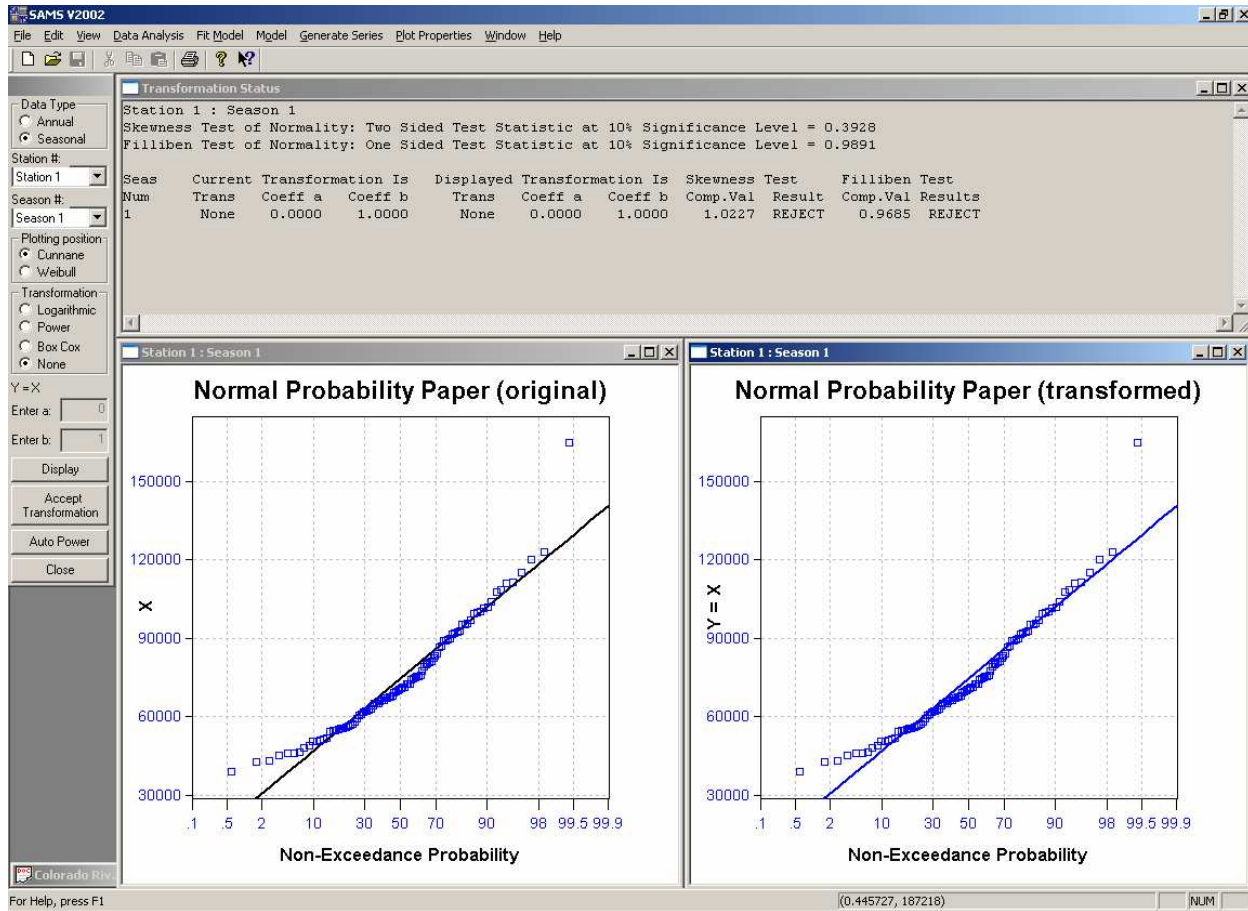


Figure 2.6 Plot of the data on normal probability paper and test of normality

If the data at hand is not normal, one may try using a transformation function. The transformation methods available in SAMS include: logarithmic, power, and Box-Cox transformations as shown in the left panel in Fig. 2.6. After selecting the type of transformation method one must click on the “Accept Transformation” button. The results of the transformation are displayed in graphical forms where the plot of the frequency distribution of the original and the transformed data may be shown on the normal probability paper. The graphical results include the theoretical distribution as well as numerical values of the tests of normality. Figure 2.7 displays the results after a logarithm transformation for site 1 and season (month) 1 of the data.

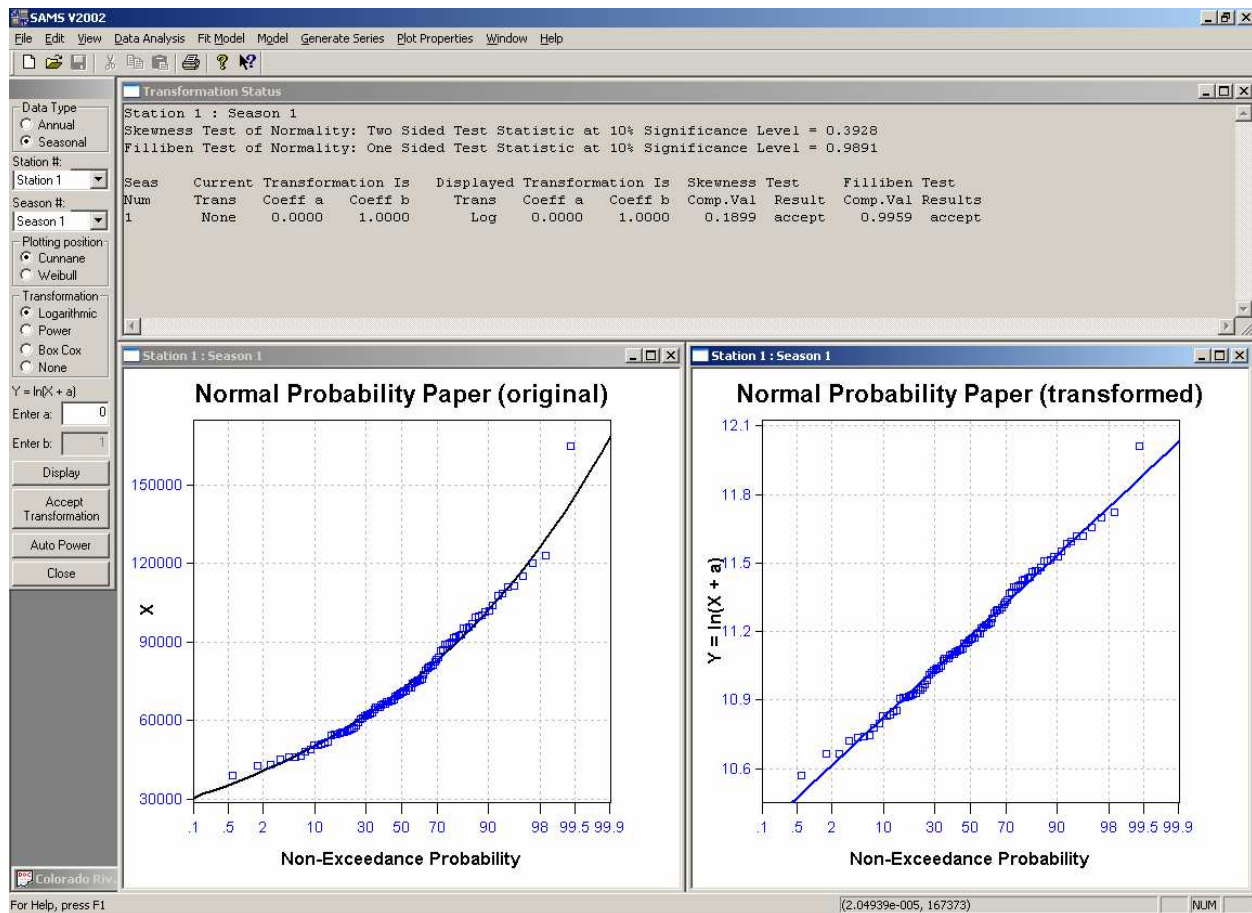


Figure 2.7 Plot of the transformed data on normal probability paper and test of normality

Show Statistics

A number of statistical characteristics can be calculated for the annual and seasonal data either original or transformed. The results can be displayed in tabular formats and can be saved in a file. These calculations can be done by choosing the “Show Statistics” under the “Data Analysis” menu. The statistics include: (1) Basic Statistics such as mean, standard deviation, skewness coefficient, coefficient of variation, maximum, and minimum values, autocorrelation coefficients, season-to season correlations, spectrum, and cross-correlations. The equations utilized for the calculations are described in section 3.1. Figure 2.8 shows an example of some of the calculated basic statistics. (2) Storage, Drought, and Surplus Related Statistics such as the longest deficit period, maximum deficit volume, longest surplus period, maximum surplus volume, storage capacity, rescaled range, and the Hurst coefficient. The equations used for the calculation are shown in section 3.2.

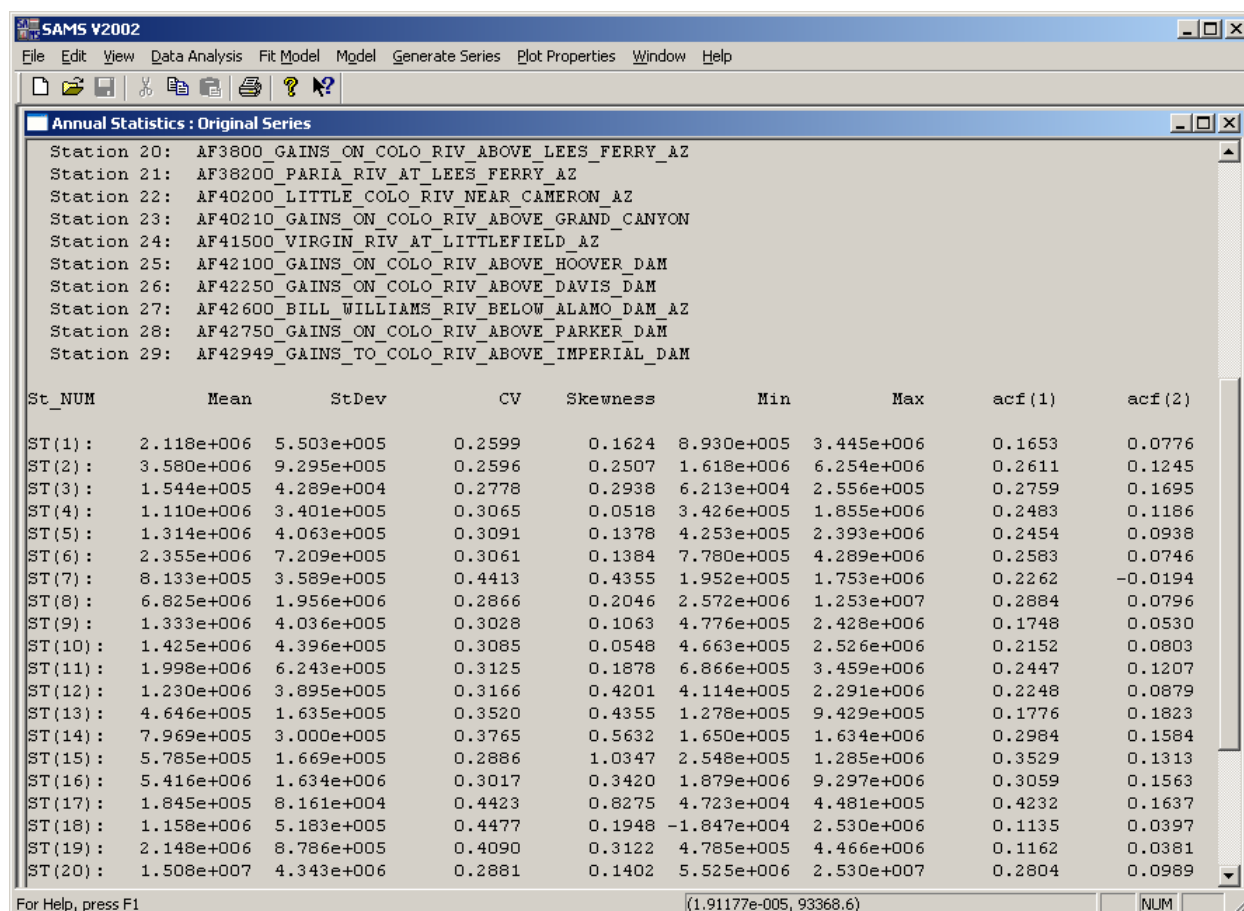


Figure 2.8 Calculated basic statistics for the annual flows of the Colorado River at 29 stations (the results for the first 20 stations are shown).

To calculate the drought statistics, the user needs to specify a demand level. Figure 2.9 shows the menu where the demand level has been specified as a fraction of the sample mean, and the results of the various storage, drought, and surplus related statistic also displayed.

Any tabular displays in SAMS all can be easily saved to a text file. Just highlight the window of the tabular displays and then to the “File” menu and using the “Save Text” function. Some users may prefer to use MS Excel to further process the results of the calculations done by SAMS. This can be done by using the “Export to Excel” function also under the “File” menu.

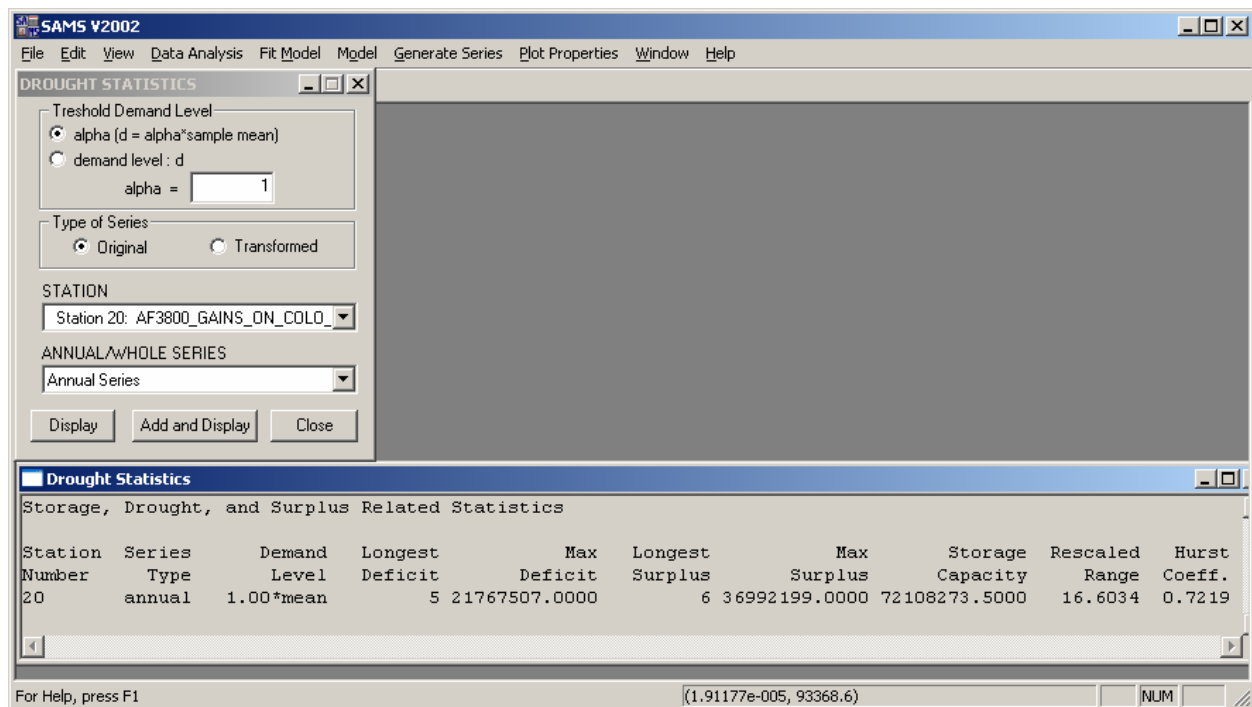


Figure 2.9 The menu for selecting the demand level (upper left corner) and the results of storage, drought, and surplus related statistics.

Plot Statistics

Some of the statistical characteristics may be displayed in graphical formats. These statistics include annual and seasonal correlation (autocorrelation) coefficients, season-to-season correlations, cross correlation coefficient between different sites, spectrum, and seasonal statistics including mean, standard deviation, skewness coefficient, coefficient of variation, maximum, and minimum values.

Figures 2.10 and 2.11 show the menu for plotting the serial correlation coefficient and the cross correlation coefficient, respectively along with some examples. The left hand side window in Fig. 2.10 shows 15 as the maximum number of lags for calculating the autocorrelation function. It also shows whether the calculation will be done for the original of the transformed series. And the bottom part of the window shows the slots for selecting the station number to be analyzed and the type of data, i.e. annual or seasonal. The correlogram shown corresponds to the annual flows for station 1 (Colorado River near Glenwood Springs). Figure 2.11 shows the menu for calculating the cross-correlation function between (two) sites 19 and 20. The plot of the spectrum (spectral density function) against the frequency is displayed in Fig. 2.12. The left hand side of the figure has slots for

selecting the smoothing function (window), the maximum number of lags (in terms of a fraction of the sample size N), and the spacing. The right hand side of the figure shows the spectrum for the annual flows of the Colorado River at site 20. In addition, the various seasonal statistics may be seen graphically. Figure 2.13 shows the monthly means for the monthly streamflows of the Colorado River at site 20.

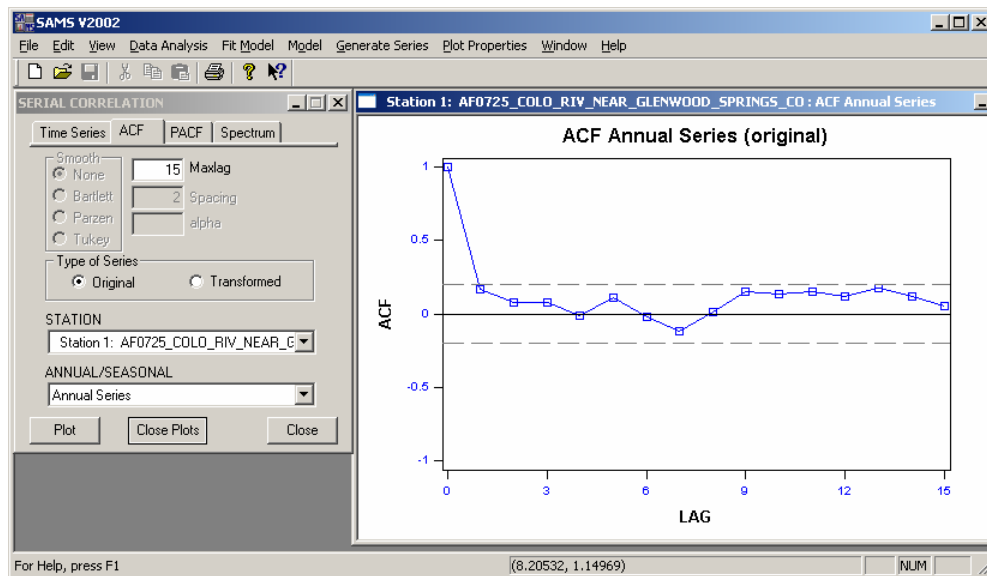


Figure 2.10: The menu for plotting the serial correlation coefficient (up-left panel), and the results of the plot.

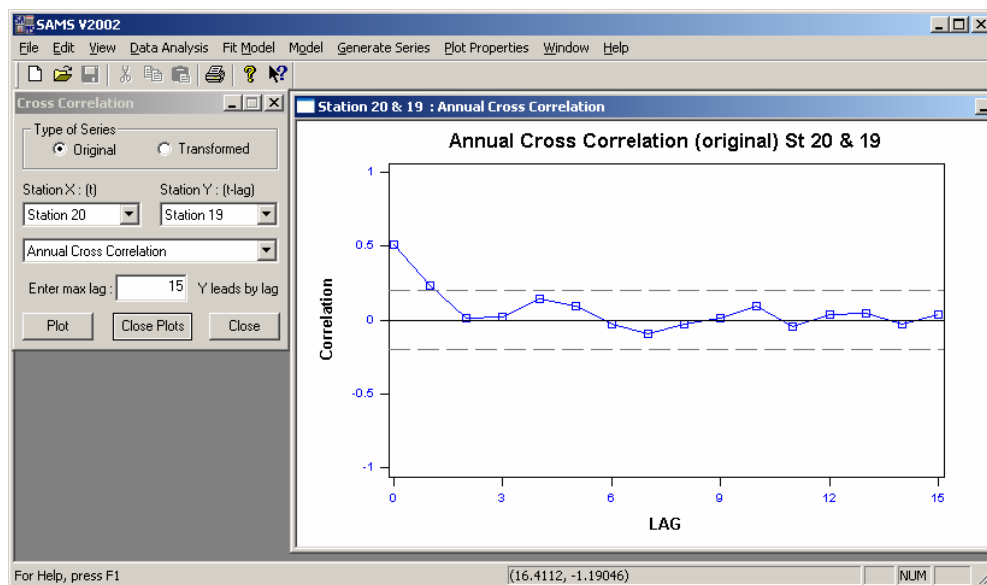


Figure 2.11: The menu for plotting the cross correlation coefficient (up-left panel), and the results of the plot

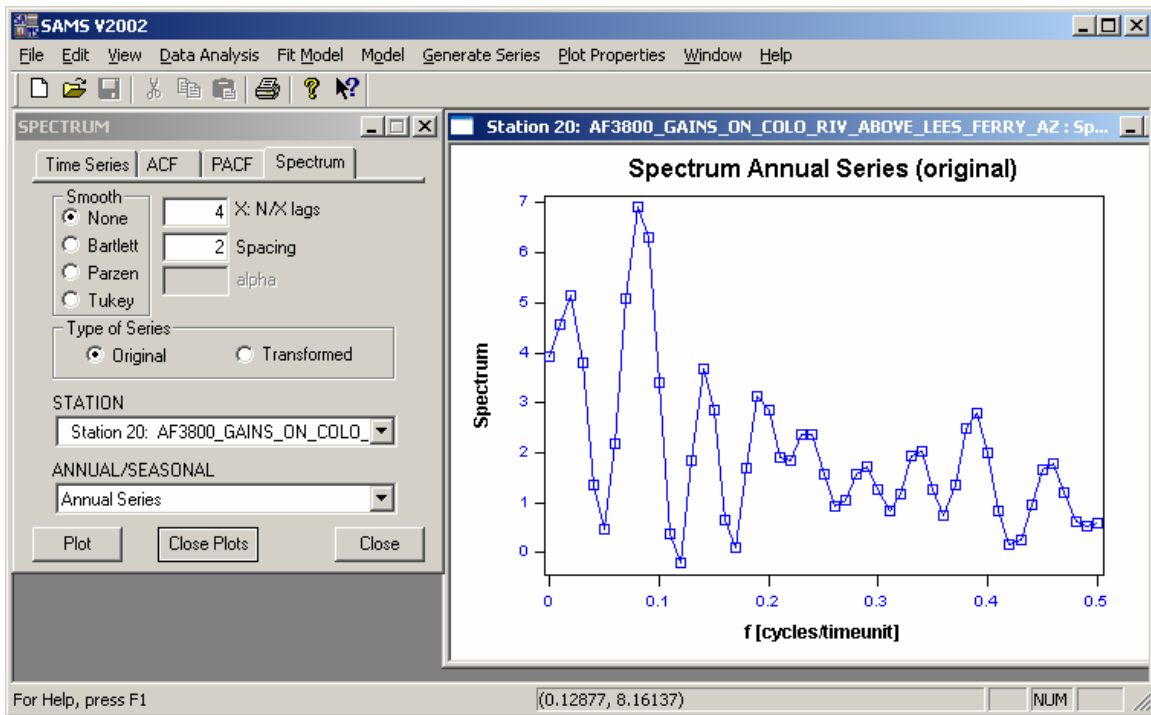


Figure 2.12 The menu for plotting the spectrum (up-left panel), and the spectrum for the annual flows of the Colorado River at site 20.

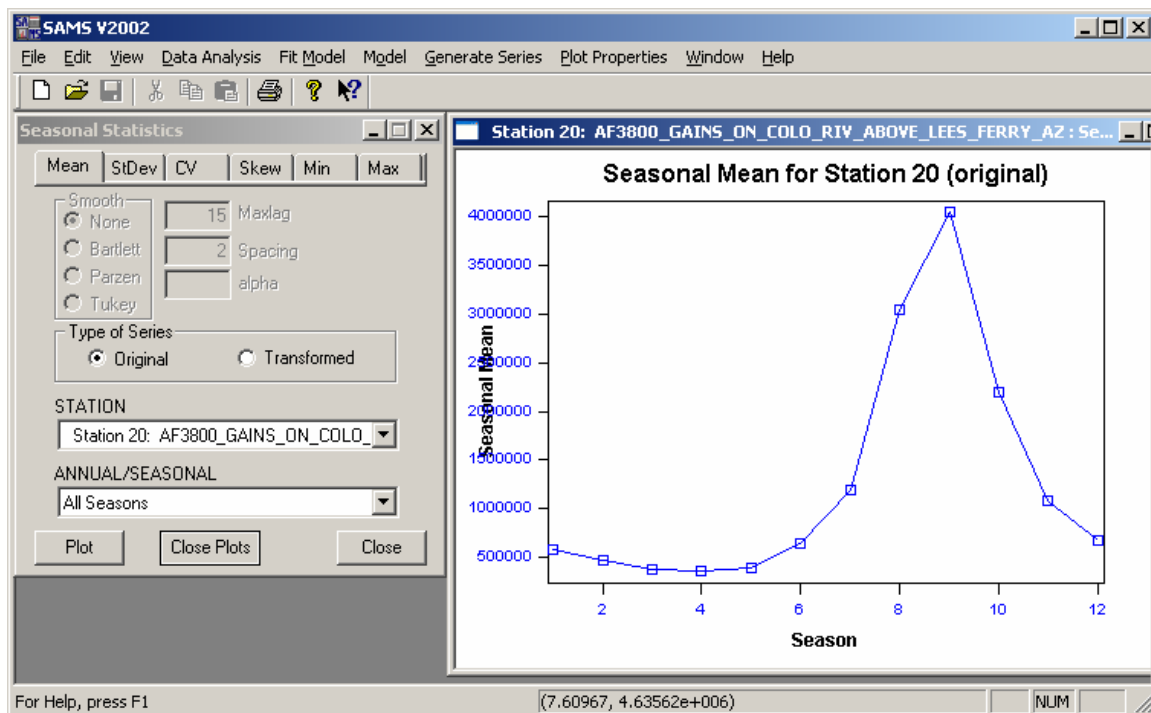


Figure 2.13 The menu for plotting the seasonal statistics (up-left panel) and the seasonal (monthly) mean of the monthly flows of the Colorado River at site 20.

Any plot produced in SAMS can be shown in tabular format (i.e. display the values that are used for making the plots). This can be done by using the “Show Plot Values” function under the “Plot Properties” menu. These values can be further saved to a text file or transferred into Excel. Figure 2.14 shows an example of the values used in the plot for the serial correlation coefficients.

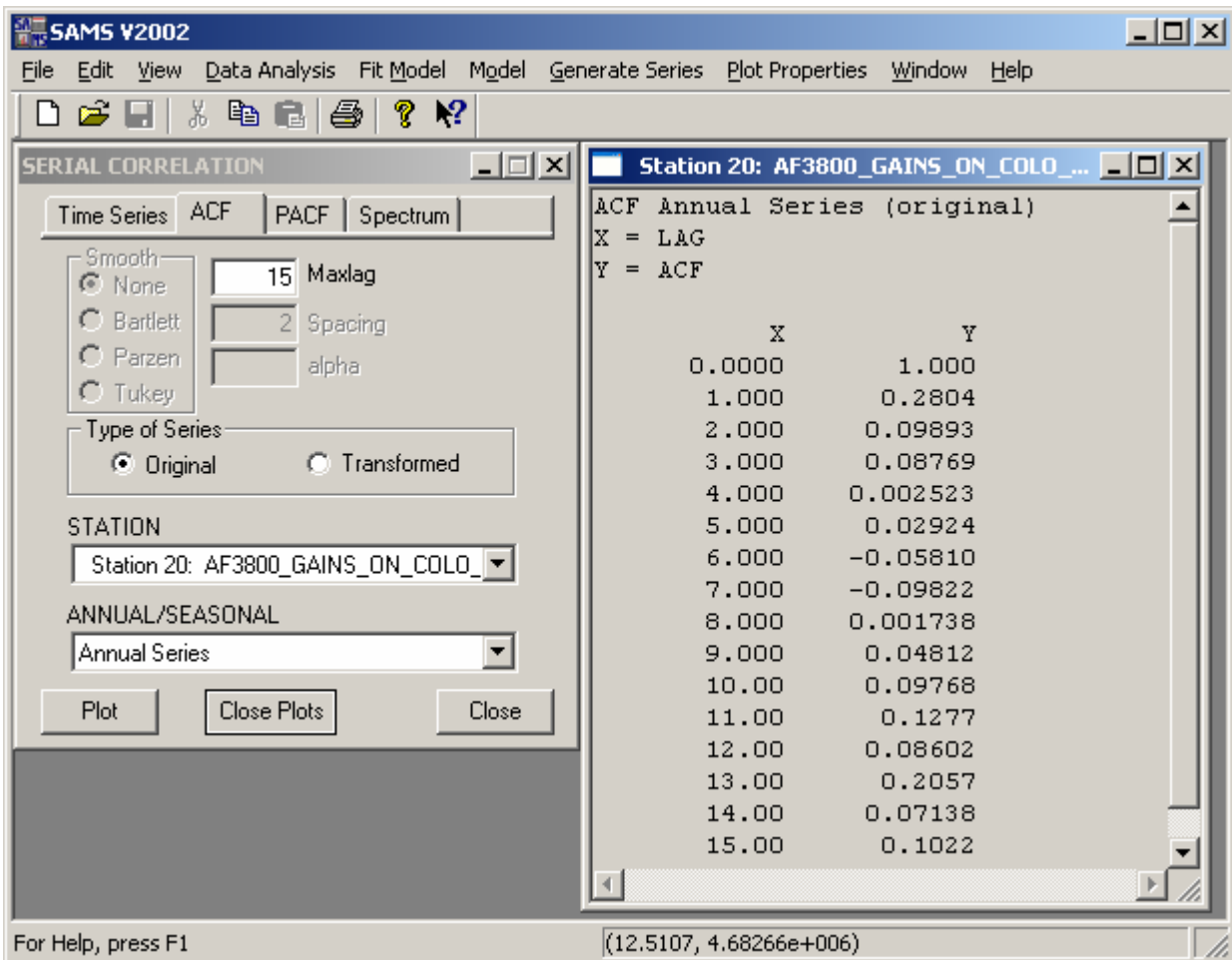


Figure 2.14 Example displaying the values that are used for the plot of the correlogram for the annual flows of the Colorado River at station 20.

2.3 Fitting a Stochastic Model

The LAST package included a number of programs to perform several objectives regarding stochastic modeling of time series. The basic procedure involved modeling and generating the annual time series using a multivariate AR(1) or AR(2) model, then using a disaggregation model to disaggregate the generated annual flows to their corresponding seasonal flows. In contrast, SAMS has two major modeling strategies which may be categorized as direct and indirect modeling. Direct

modeling means fitting an stationary model (e.g. univariate ARMA or multivariate AR, CARMA or CSM-CARMA) directly to the annual data or fitting a periodic (seasonal) model (e.g. univariate PARMA or multivariate PAR) directly to the seasonal data of the system at hand. Disaggregation modeling, on the other hand, is an indirect procedure because the modeling of the annual data for a site can rely on the modeling of the annual data of another site (key station), and the modeling of seasonal data involves also modeling the corresponding annual data as well before the seasonal data are obtained by temporal disaggregation. SAMS categorizes the models into those for the annual data and for the seasonal data. In each category, there are univariate, multivariate, and disaggregation models. The following specific models are currently available in SAMS under each category:

1. For annual data:

- Univariate ARMA(p,q) model.
- Univariate GAR(1) model.
- Univariate Shifting Mean (SM) model.
- Multivariate AR(p) model (MAR).
- Contemporaneous ARMA(p,q) model (CARMA).
- CSM-CARMAR(p,q) model.
- Multivariate annual (spatial) disaggregation.

2. For seasonal data:

- Univariate PARMA(p,q) model.
- Multivariate PAR(p) model (MPAR(p)).
- Univariate seasonal disaggregation model.
- Multivariate spatial-seasonal disaggregation model.
- Multivariate seasonal-spatial disaggregation model.

The operation for fitting the models rather than a disaggregation model is basically the same. After clicking on the “Fit Model” menu and choosing the desired model, a menu for fitting the chosen model will appear where the site number, the model order, etc. can be specified. The user needs to specify the station (site) number(s). If standardization of the data is desired, one must click on the "Standardize Data" button. Generally, the modeling is performed with data in which the mean

is subtracted. Thus, standardization implies that not only the mean is subtracted but in addition the data will be further transformed to have standard deviation equal to one. For example, for monthly data the mean for month 5 is subtracted and the result is divided by the standard deviation for that month. As a result, the mean and the standard deviation of the standardized data for month 5 become equal to zero and one, respectively. Then, the order of the model to be fitted is selected, for instance for ARMA models, one must enter p and q . In the case of MAR or MPAR models, one must key in the order p only. Subsequently, the method of estimation of the model parameters must be selected.

Currently SAMS provides two methods of estimation namely the method of moments (MOM) and the least squares (LS) method. MOM is available for the ARMA(p,q), GAR(1), SM, MAR(p), CSM part of the CSM-CARMA, PARMA($p,1$), and MPAR(p) models while LS is available for ARMA(p,q), CARMA(p,q), and PARMA(p,q) models. The LS method is often iterative and may require some initial parameters estimates (starting points). These starting points are either based on fitting a high order simpler model using LS or by using the MOM parameters estimates as starting points. For cases where the MOM estimates are not available such as for the PARMA(p,q) model where $q>1$, the MOM parameter estimates of the closest model will be used instead. For fitting CARMA(p,q) models, the residual variance-covariance G matrix can be estimated using either the method of moments (MOM) or the maximum likelihood estimation (MLE) method (Stedinger et al., 1985). Figure 2.15 shows an example of fitting a CARMA(1,0) model.

In the case of fitting the CSM-CARMA(p,q) model a special dialog box will appear, and the user need to key in proper information for the model setup (see Fig. 2.16). The mixed model can be used to fit a CSM model only or a CARMA model only and is recommended over using the single CARMA model option.

Fitting disaggregation models needs additional operations. Before explaining these operations, it is necessary to describe briefly the concept in setting up disaggregation models in SAMS. In disaggregation modeling, the user should conduct the process to setup the model configuration step by step. The configuration depends upon the orders and positions of the stations in the system relative to each other. The system structure means defining for each main river system the sequence of stations (sites) that conform the river network. SAMS uses the concept of key stations and substations. A key station is a downstream station along a main stream. It could be the

farthest downstream station or any other station depending on the particular problem at hand. For instance, referring to the Colorado River system shown in Fig. 2.17, station 29 is a key station if one is interested in modeling the entire river system. On the other hand, if station 29 is not used in the analysis, station 28 will become the key station. Also there could be several key stations. Let us continue the explanations assuming that stations 8 and 16 are key stations for the Upper Colorado River Basin. Substations are the next upstream stations draining to a key station. For instance, stations 2, 6, and 7 are substations draining to key station 8. Likewise, stations 11, 12, 13, 14, and 15 are substations for key station 16. Subsequent stations are the next upstream stations draining into a substation. For instance, stations 1, 5, and 10 are subsequent stations relative to substations 2, 6, and 11, respectively.

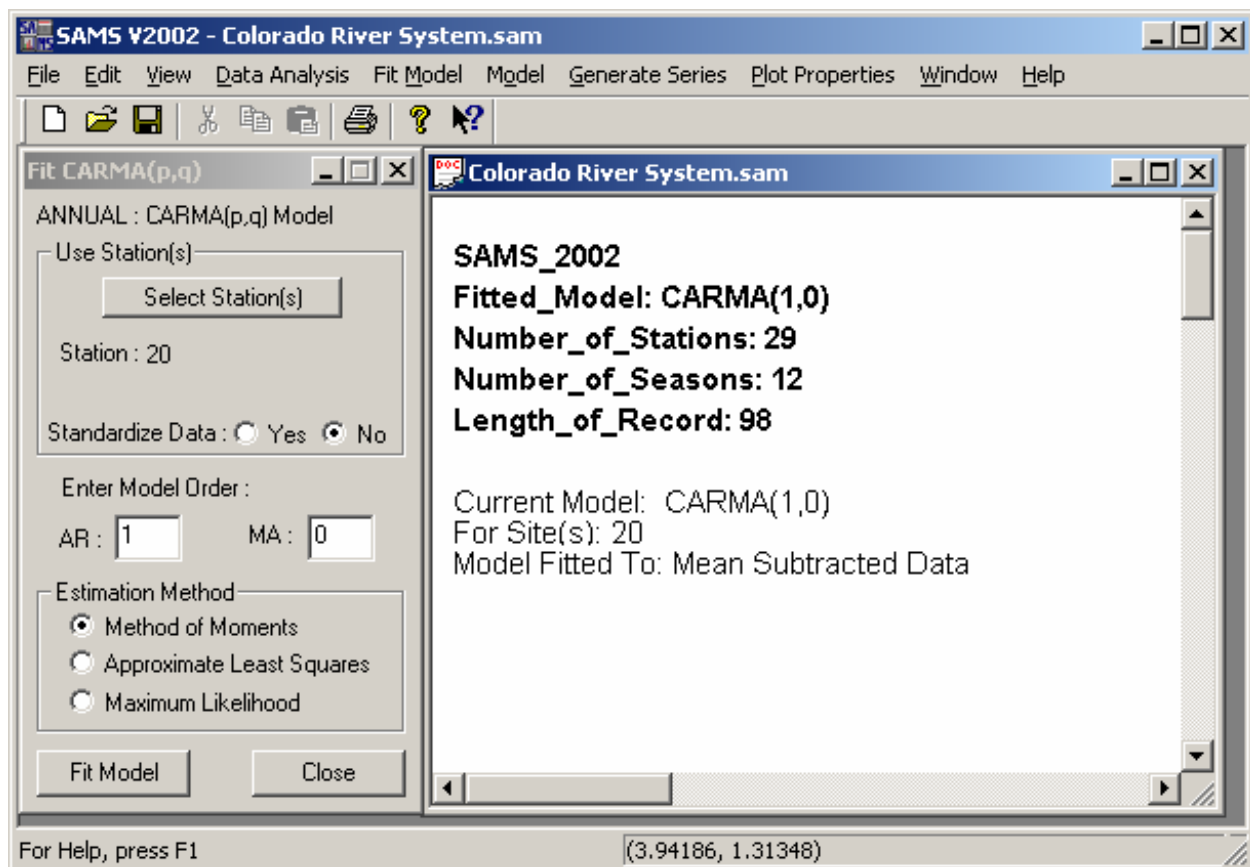


Figure 2.15 An example of fitting a CARMA(1,0) model.

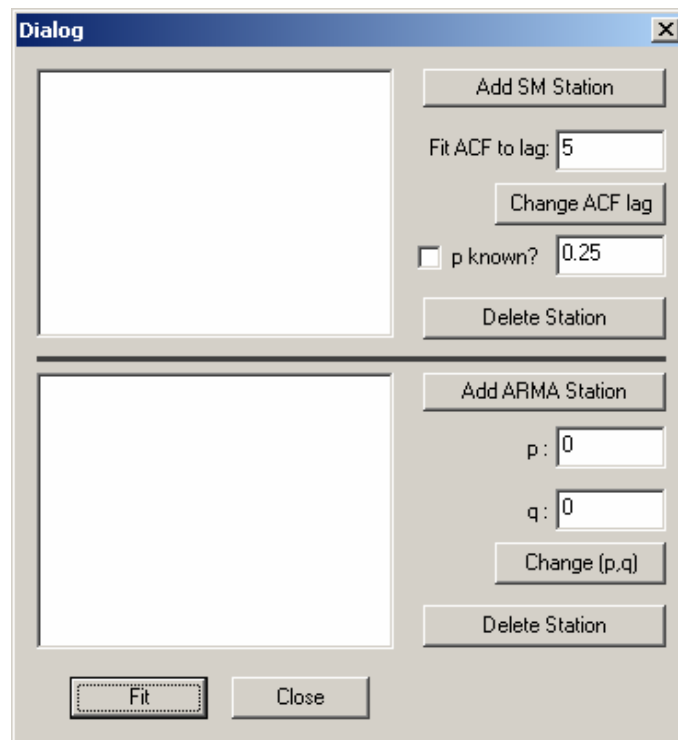


Figure 2.16 The menu for fitting CSM-CARMA(p,q) models.

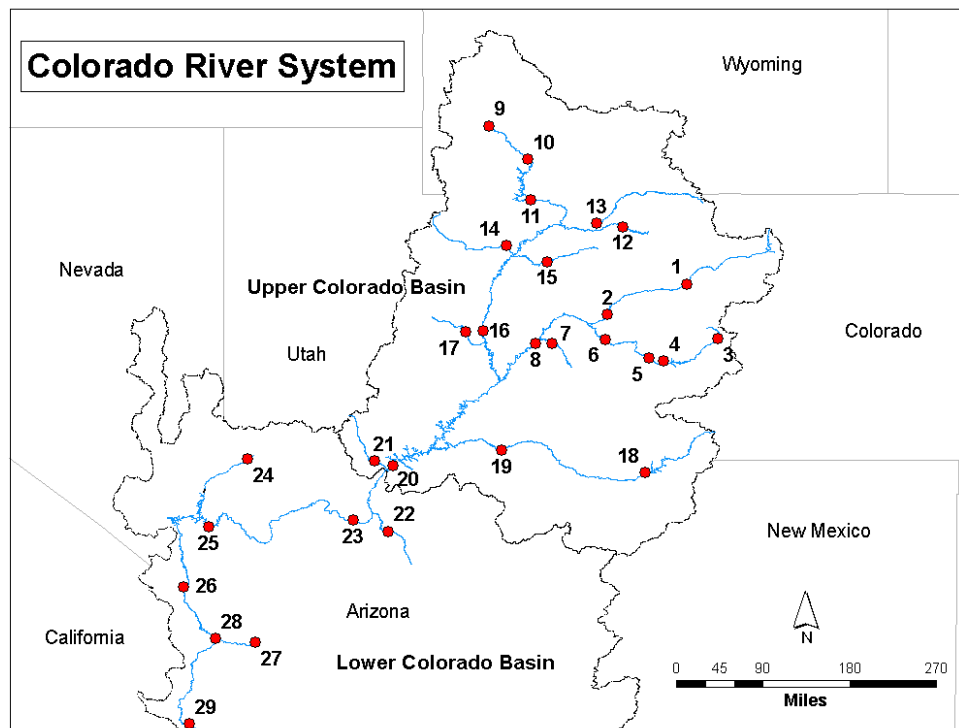


Figure 2.17 Schematic representation of the Colorado River stream network

In addition, for defining a "disaggregation procedure" SAMS uses the concept of groups. A group consists of one or more key stations and their corresponding substations. Groups must be defined in each disaggregation step. Each group contains a certain number of stations to be modeled in a multivariate fashion, i.e. jointly, in order to preserve their cross-correlations. For instance, if a certain group has two key stations and three substations, then the disaggregation process will preserve the cross-correlations between all stations (key and substations.) On the other hand, if two separate groups are selected, then the cross-correlations between the stations that belong to the same group will be preserved, but the cross-correlations between stations belonging to different groups will not be preserved.

The definition of a group is important in the disaggregation process. For instance, referring to Fig. 2.17, key station 8 and substations 2, 6, and 7 may form one group in which the flows of all these stations are modeled jointly in a multivariate framework, while key station 16 and its substations 11, 12, 13, 14, and 15 may form another group. In this case, the cross-correlations between the stations within each group will be preserved but the cross-correlations among stations of the two different groups will not be preserved. For example, the cross-correlations between stations 8 and 16 will not be preserved but the cross-correlations between stations 8 and 2 will be preserved. On the other hand, if all the stations are defined in a single group, then the cross-correlations between all the stations will be preserved. After modeling and generating the annual flows at the desired stations, the annual flows can be disaggregated into seasonal flows. This is handled again by using the concept of groups as explained above. The user, for example, may choose stations 11, 12, 13, 14, 15, and 16 as one group. Then, the annual flows for these stations may be disaggregated into seasonal flows by a multivariate disaggregation model so as to preserve the seasonal cross-correlations between all the stations.

Figure 2.18 shows the menu available for "fitting the model". The user must choose whether the model (and generation thereof) is for annual or for seasonal data. Figure 2.18 shows the selection for seasonal data. The options to choose depend whether the modeling (and generation) problem is for 1 site (1 series) or for several sites (more than 1 series). Accordingly the model may be either univariate or multivariate, respectively. Choosing a univariate or multivariate model implies fitting the model using a direct modeling approach, e.g. for 3 sites using a trivariate periodic (seasonal) model based on the seasonal data available for the three sites. On the other hand, one may generate

seasonal flows indirectly using aggregation and disaggregation methods. When using disaggregation methods two broad options are available (Fig. 2.18), i.e. spatial-seasonal and seasonal-spatial. The first option defines a modeling approach whereby annual flow are generated first at key stations, subsequently, spatial disaggregation is applied to generate annual flows at upstream stations, then seasonal flow are obtained using temporal disaggregation. Alternatively, the second option defines a modeling approach where annual flows are generated at key stations, which are then disaggregated into seasonal flows based on temporal disaggregation models. And the final step is to disaggregate such seasonal flows spatially to obtain the seasonal flows at all stations in the system at hand.

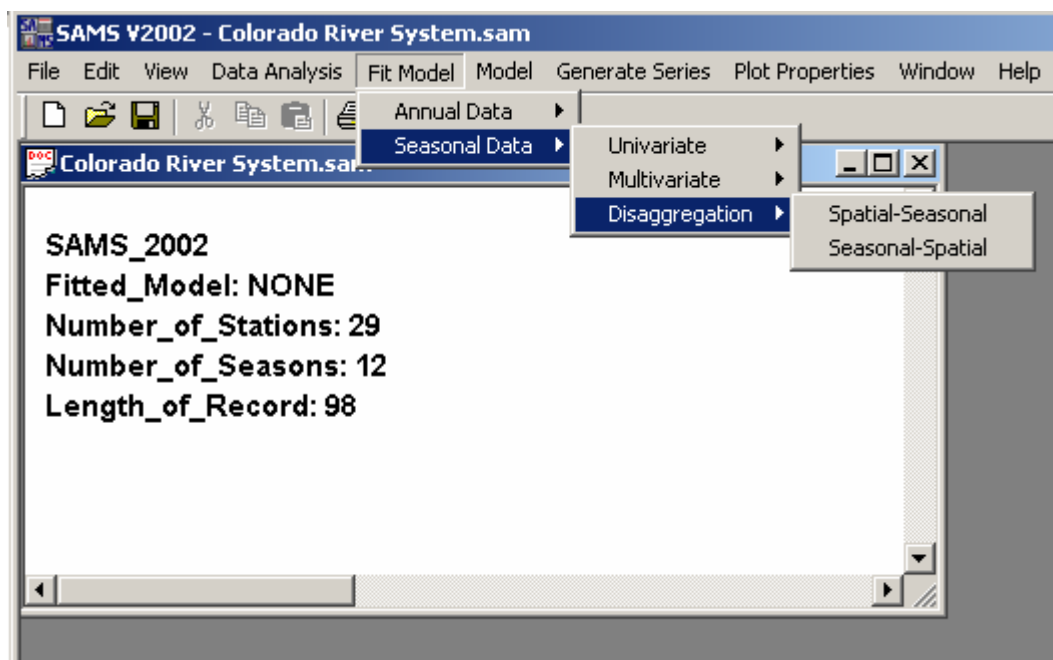


Figure 2.18 The menu for model fitting. Note that “seasonal data” and “disaggregation” options are selected (highlighted) and the options “Spatial-temporal” and “Temporal-Spatial” are shown.

SAMS has two schemes for modeling the key stations. In the first scheme, denoted as Scheme 1, the annual flows of the key stations that belong to a given group are aggregated to form an “index station”, then a univariate ARMA(p,q) model is used to model the aggregated flows (of the index station.) The aggregated annual flows are then disaggregated (spatially) back to each key station by using the Valencia and Schaake or the Mejia and Rouselle disaggregation methods. Then the annual flows at the key stations are disaggregated spatially to obtain the annual flows at the substations and then to the subsequent stations, etc. The second scheme, denoted as Scheme 2, uses

a multivariate model to represent (generate) the annual flows of the key stations belonging to a given group and then disaggregate those flows spatially to obtain the annual flows for the substations, subsequent stations, etc. For either Scheme 1 or 2, temporal disaggregation may be carried out if seasonal flows are desired. The mathematical description of the disaggregation methods is presented in chapter 4, and examples of disaggregation modeling applied to real streamflow data are presented in chapter 5.

In applying disaggregation methods the user needs to choose the specific disaggregation models for both spatial and temporal disaggregation. For example, when modeling seasonal data the user may select either the “spatial-temporal” or the “temporal-spatial” option. In any selection one must determine the type of disaggregation models. Figure 2.19 shows the windows option after choosing the “spatial-temporal” option. The modeling scheme as either 1 or 2 (as noted above) must model) be chosen, as well as the type of spatial disaggregation (either the Valencia-Schaake or Mejia-Rousselle model) and the type of temporal disaggregation (for this purpose only Lane’s model is available). The option “Temporal-Spatial” is slightly different where the user has a choice between two temporal disaggregation models, namely Lane’s model and Grygier and Stedinger model.

As illustration some of the steps and options followed in using a disaggregation approach are shown in Figs. 2.19 to 2.23. They are summarized as:

- In Fig. 2.19 Scheme 1 is selected along with the V-S model for spatial disaggregation and Lane’s model for temporal disaggregation.
- In Fig. 2.20 stations 8 and 16 (refer to Fig. 2.17) are selected as key stations and an index station will be formed (the aggregation of the annual flows for sites 8 and 16). Then the ARMA(1,0) model was chosen to generate the annual flows of the index station.
- The spatial disaggregation of the annual flows for key to substations must be carried out by groups. For example, this could be accomplished by considering key station 8 and 16 and their corresponding substations 2, 6, and 7 and 11, 12, 13, 14, and 15, respectively into a single group or by forming two or more groups. For instance, 2 groups were formed one per key station and Figs. 2.21 and 2.22 show the procedure for selecting the group corresponding to key station 8.
- The temporal disaggregation (from annual into seasonal flows) is also performed by groups

(of stations) as shown in Fig. 2.23. The specifications for the disaggregation modeling are completed by pressing the “Finish” button shown in Fig. 2.23.

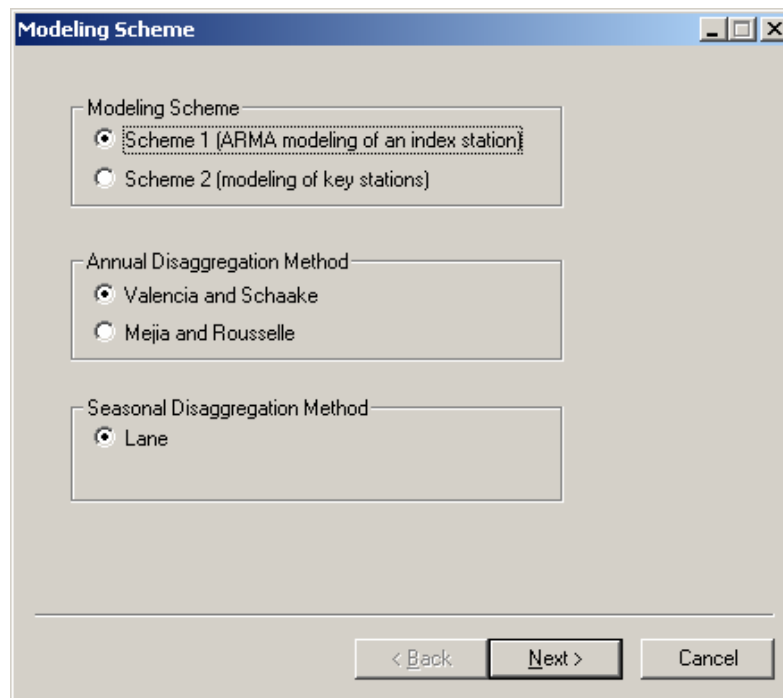


Figure 2.19 The menu for the modeling scheme for seasonal data after selecting the spatial-temporal option as shown in Fig. 2.18.

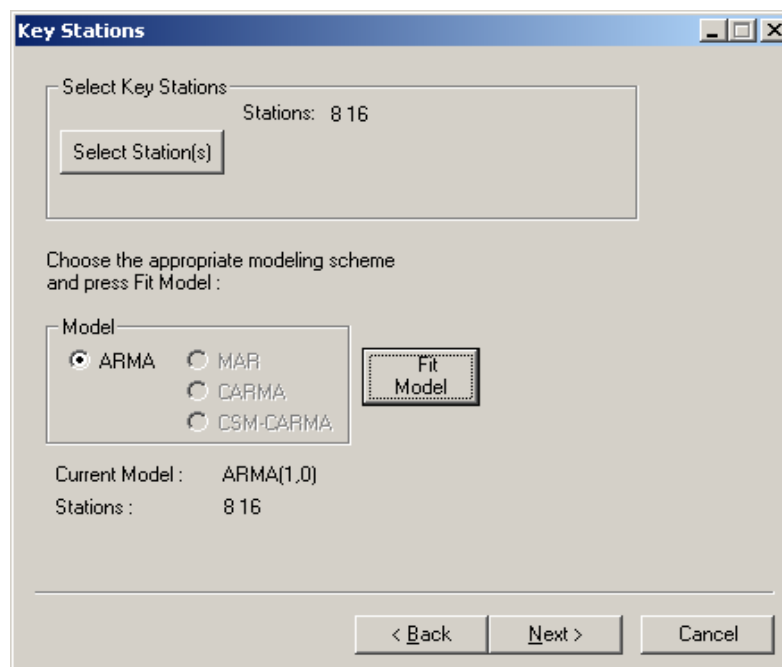


Figure 2.20 The menu for selecting the key stations that will be used for defining the index station. Also the definition of the model for the index station is shown.

Spatial Disaggregation

Key Stations :

Sub Stations :

Key	Sub

Figure 2.21 The menu for selecting the key stations and substations that will form a group.

Spatial Disaggregation

Key Stations :

Sub Stations :

Key	Sub
8	2 6 7

Figure 2.22: Definition of the spatial disaggregation groups

Temporal Disaggregation

Key Stations :

Temporal Disaggregation Groups

11 12 13 14 15 16
2 6 7 8

Figure 2.23: Definition of the temporal disaggregation groups

After fitting a stochastic model, one may view a summary of the model parameters by using the “Show Parameters” function under the “Model” menu. Figure 2.24 shows part of the model parameters regarding the simulation of seasonal flows using disaggregation methods as described above.

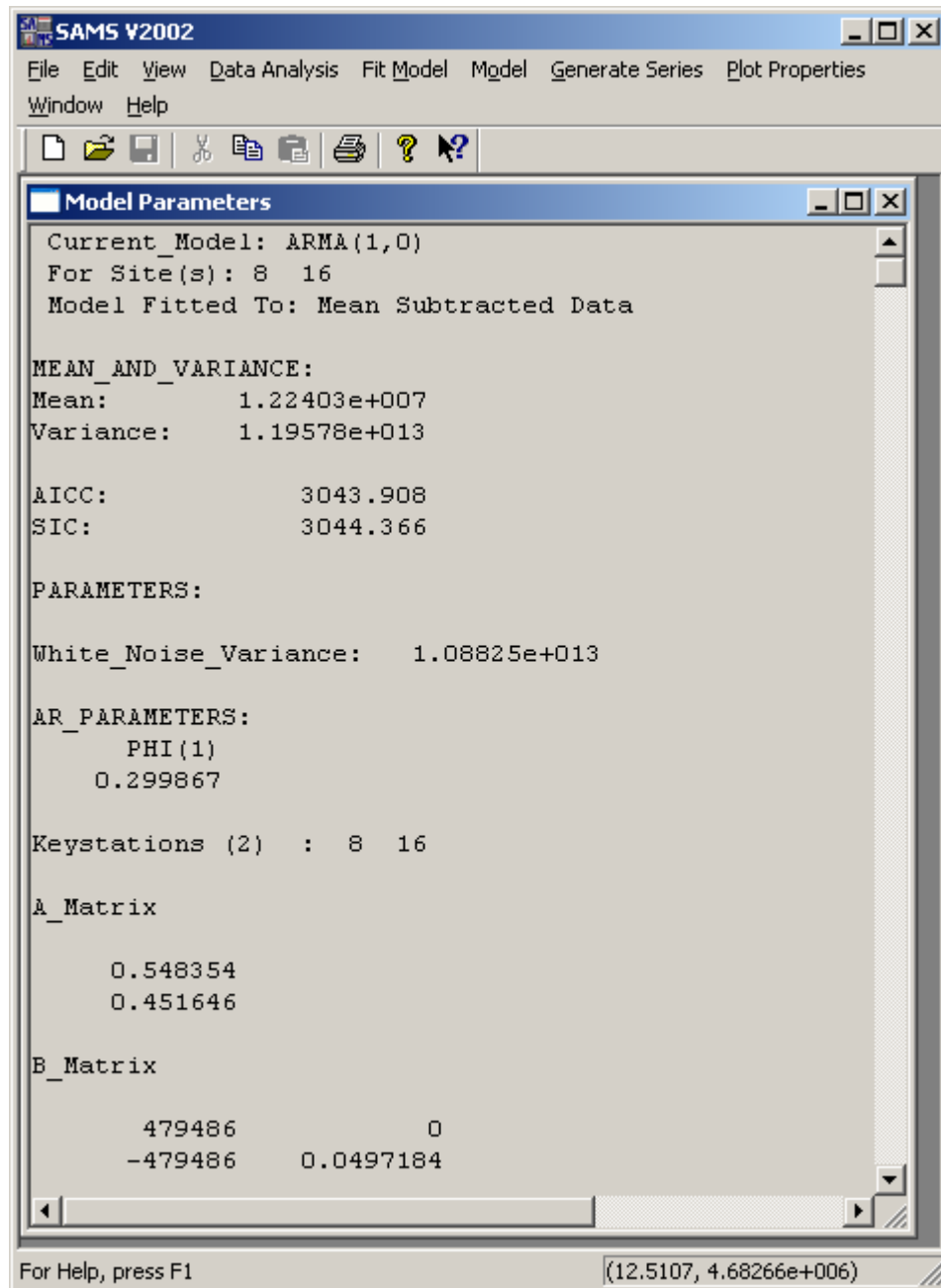


Figure 2.24 Summary of the model parameters for the index stations and for disaggregating the annual flows of the index station and disaggregating the annual flows at stations 8 and 16. Other features of the model and parameters thereof are not shown.

2.4 Generating Synthetic Series

Data generation is an important subject in stochastic hydrology and has received a lot of attention in hydrologic literature. Data generation is used by hydrologists for many purposes. These include, for example, reservoir sizing, planning and management of an existing reservoir, and reliability of a water resources system such as a water supply or irrigation system (Salas et al,1980). Stochastic data generation can aid in making key management decisions especially in critical situations such as extended droughts periods (Frevert et al, 1989). The main philosophy behind synthetic data generation is that synthetic samples are generated which preserve certain statistical properties that exist in the natural hydrologic process (Lane and Frevert, 1990). As a result, each generated sample and the historic sample are equally likely to occur in the future. The historic sample is not more likely to occur than any of the generated samples (Lane and Frevert, 1990).

Generation of synthetic time series is based on the models, approaches and schemes. Once the model has been defined and the parameters have been estimated, one can generate synthetic samples based on this model. SAMS allows the user to generate synthetic data and eventually compare important statistical characteristics of the historical and the generated data. Such comparison is important for checking whether the model used in generation is adequate or not. If important historical and generated statistics are comparable, then one can argue that the model is adequate. The generated data can be stored in files. This allows the user to further analyze the generated data as needed. Furthermore, when data generation is based on spatial or temporal disaggregation, one may like to make adjustments to the generated data. This may be necessary in many cases to enforce that the sum of the disaggregated quantities will add up to the original total quantity.

Dialog

Generate Synthetic Time Series
Using Current Model

Length of Series: 50

Number of Series: 10

Initial Seed [1-1000]: 1

☐ Force Generated Data >= 0

Adjustments

☒ Spatial Adjustments view/change

☐ Temporal Adjustments view/change

Options

☐ Import flows for key sites

☐ Store all Generated Series

☒ Store only Last Generated Series

Options for calculating statistics

Save

☒ Save Generated Data

☒ Save Statistics of all Generated Data

Generate Cancel

Figure 2.25: Menu for generate data.

For example, spatial adjustments may be necessary if the annual flows at a key station is exactly the sum of the annual flows at the corresponding substations. Likewise, in the case of temporal disaggregation, one may like to assure that the sum of monthly values will add up to the annual value. Various options of adjustments are included in SAMS. Further descriptions on spatial and temporal adjustments are described in later sections of this manual.

Figure 2.25 shows the data generation menu. In this menu the user must specify necessary information for the generation process. For example, the length of the generated data, how many samples will be generated, and whether the generated data or the statistics of the generated data will be saved to files should be specified by the user. Figure 2.26 show the window for the adjustment. The user can chose a method for the spatial adjustment.

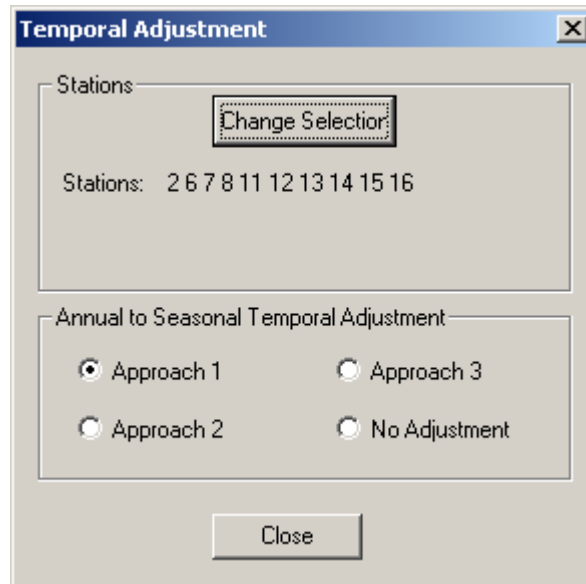


Figure 2.26: The window for temporal adjustment.

After the generation of data, the user can compare the generated data to the historical record by using the “Compare” function under the “Generate” menu. The comparison can be made between the basic statistics, drought statistics, autocorrelations, and the time series plots. Figure 2.27 shows the menu for the comparison, and the comparison of the basic statistics. Figure 2.28 shows the comparison of the time series.

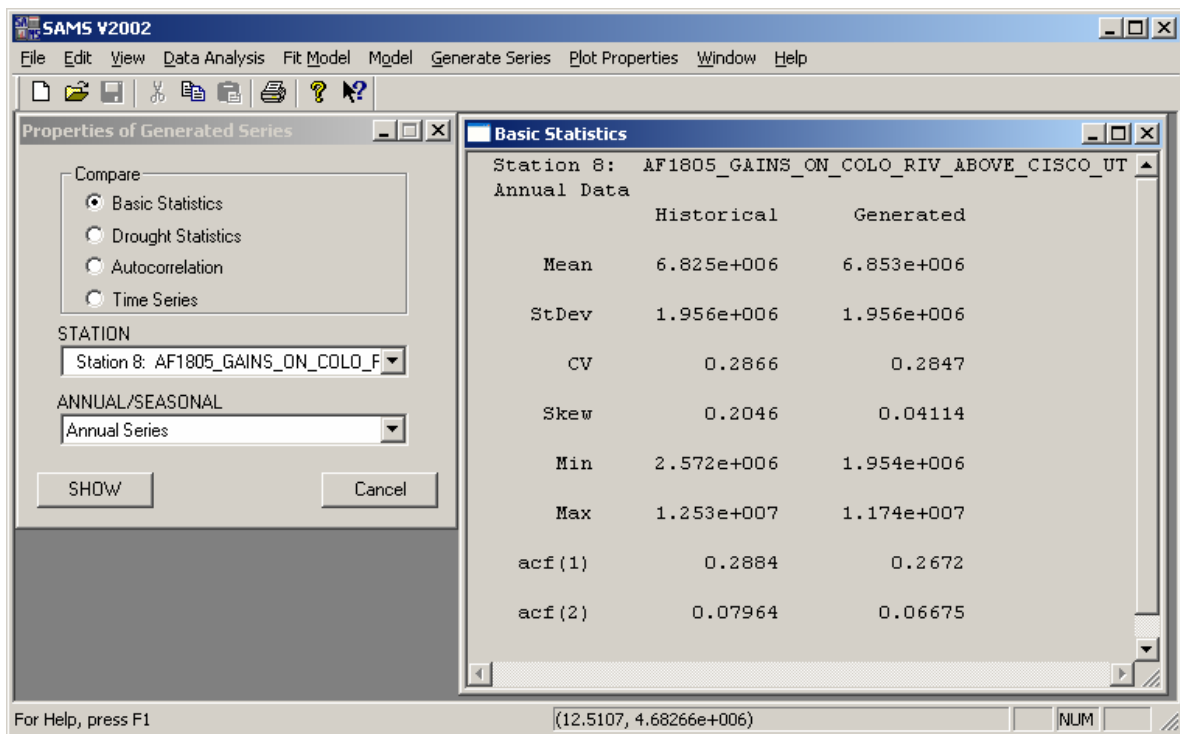


Figure 2.27: Comparison of the basic statistics of the generated data and the historical record.

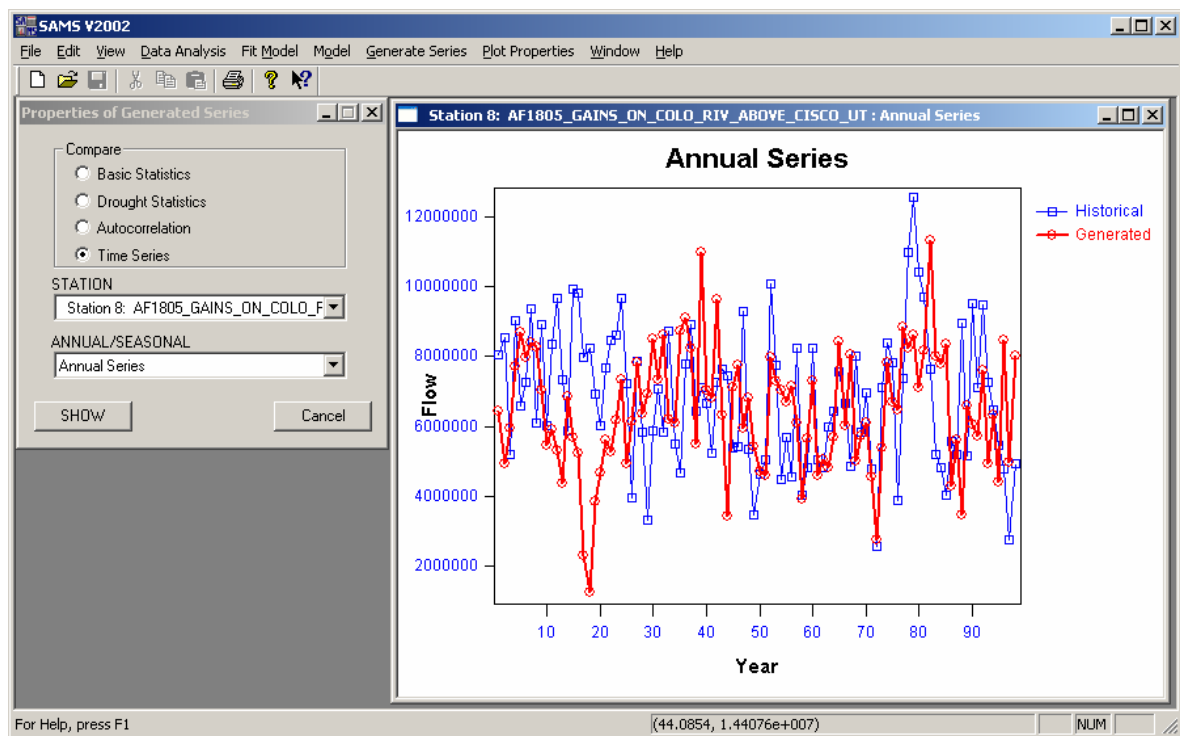


Figure 2.28: Comparison of the time series.

3 DEFINITION OF STATISTICAL CHARACTERISTICS

A time series process can be characterized by a number of statistical properties such as the mean, standard deviation, coefficient of variation, skewness coefficient, season-to-season correlations, autocorrelations, cross-correlations, and storage and drought related statistics. These statistics are defined for both annual and seasonal data as shown below.

3.1 Basic Statistics

3.1.1 Annual Data

The mean and the standard deviation of a time series y_t are estimated by

$$\bar{y} = \frac{1}{N} \sum_{t=1}^N y_t \quad (3.1)$$

and

$$s = \sqrt{\frac{1}{N} \sum_{t=1}^N (y_t - \bar{y})^2} \quad (3.2)$$

respectively, where N is the sample size. The coefficient of variation is defined as $cv = s / \bar{y}$.

Likewise, the skewness coefficient is estimated by

$$g = \frac{\frac{1}{N} \sum_{t=1}^N (y_t - \bar{y})^3}{s^3} \quad (3.3)$$

The sample autocorrelation coefficients r_k of a time series may be estimated by

$$r_k = \frac{m_k}{m_0} \quad (3.4)$$

where

$$m_k = \frac{1}{N} \sum_{t=1}^{N-k} (y_{t+k} - \bar{y})(y_t - \bar{y}) \quad (3.5)$$

and k = time lag. Likewise, for multisite series, the lag- k sample cross-correlations between site i and site j , denoted by r_k^{ij} , may be estimated by

$$r_k^{ij} = \frac{m_k^{ij}}{\sqrt{m_0^{ii} m_0^{jj}}} \quad (3.6)$$

where

$$m_k^{ij} = \frac{1}{N} \sum_{t=1}^{N-k} (y_{t+k}^{(i)} - \bar{y}^{(i)})(y_t^{(j)} - \bar{y}^{(j)}) \quad (3.7)$$

in which m_0^{ii} is the sample variance for site i .

3.1.2 Seasonal data

Seasonal hydrologic time series, such as monthly flows, are better characterized by seasonal statistics. Let $y_{v,\tau}$ be a seasonal time series, where $v = 1, \dots, N$ represents years with N being the number of years, and $\tau = 1, \dots, \omega$ seasons with ω being the number of seasons. The mean and standard deviation for season τ can be estimated by

$$\bar{y}_\tau = \frac{1}{N} \sum_{v=1}^N y_{v,\tau} \quad (3.8)$$

and

$$s_\tau = \sqrt{\frac{1}{N} \sum_{v=1}^N (y_{v,\tau} - \bar{y}_\tau)^2} \quad (3.9)$$

respectively. The seasonal coefficient of variation is $cv_\tau = s_\tau / \bar{y}_\tau$. Similarly, the seasonal skewness coefficient is estimated by

$$g_\tau = \frac{\frac{1}{N} \sum_{v=1}^N (y_{v,\tau} - \bar{y}_\tau)^3}{s_\tau^3} \quad (3.10)$$

The sample lag- k season-to-season correlation coefficient may be estimated by

$$r_{k,\tau} = \frac{m_{k,\tau}}{\sqrt{m_{0,\tau} m_{0,\tau-k}}} \quad (3.11)$$

where

$$m_{k,\tau} = \frac{1}{N} \sum_{v=1}^N (y_{v,\tau} - \bar{y}_\tau)(y_{v,\tau-k} - \bar{y}_{\tau-k}) \quad (3.12)$$

in which $m_{0,\tau}$ represents the sample variance for season τ . Likewise, for multisite series, the lag- k sample cross-correlations between site i and site j , for season τ , $r_{k,\tau}^{ij}$ may be estimated by

$$r_{k,\tau}^{ij} = \frac{m_{k,\tau}^{ij}}{\sqrt{m_{0,\tau}^{ii} m_{0,\tau-k}^{jj}}} \quad (3.13)$$

and

$$m_{k,\tau}^{ij} = \frac{1}{N} \sum_{\nu=1}^N (y_{\nu,\tau}^{(i)} - \bar{y}_{\tau}^{(i)})(y_{\nu,\tau-k}^{(j)} - \bar{y}_{\tau-k}^{(j)}) \quad (3.14)$$

in which $m_{0,\tau}^{ii}$ represents the sample variance for season τ and site i . Note that in Eqs. (3.11) through (3.14) when $\tau - k < 1$, the terms, $\nu = 1, y_{\nu,\tau-k}, \bar{y}_{\tau-k}, m_{0,\tau-k}, y_{\nu,\tau-k}^{(j)}, \bar{y}_{\tau-k}^{(j)}$, and $m_{0,\tau-k}^{jj}$ are replaced by $\nu = 2, y_{\nu-1,\omega+\tau-k}, \bar{y}_{\omega+\tau-k}, m_{0,\omega+\tau-k}, y_{\nu,\omega+\tau-k}^{(j)}, \bar{y}_{\omega+\tau-k}^{(j)}$, and $m_{0,\omega+\tau-k}^{jj}$, respectively.

3.2 Storage, Drought, and Surplus Related Statistics

3.2.1 Storage Related Statistics

The storage-related statistics are particularly important in modeling time series for simulation studies of reservoir systems. Such characteristics are generally functions of the variance and autocovariance structure of a time series. Consider the time series $y_i, i = 1, \dots, N$ and a subsample $y_1, \dots, y_n$ with $n \leq N$. Form the sequence of partial sums S_i as

$$S_i = S_{i-1} + (y_i - \bar{y}_n) \quad , i = 1, \dots, n \quad (3.15)$$

where $S_0 = 0$ and $\bar{y}_n$ is the sample mean of $y_1, \dots, y_n$ which is determined by Eq. (3.1). Then, the adjusted range R_n^* and the rescaled adjusted range R_n^{**} can be calculated by

$$R_n^* = \max(S_0, S_1, \dots, S_n) - \min(S_0, S_1, \dots, S_n) \quad (3.16)$$

and

$$R_n^{**} = \frac{R_n^*}{s_n} \quad (3.17)$$

respectively, in which s_n is the standard deviation of $y_1, \dots, y_n$ which is determined by Eq. (3.2).

Likewise, the Hurst coefficient for a series is estimated by

$$K = \frac{\ln(R_n^{**})}{\ln(n/2)} \quad , n > 2 \quad (3.18)$$

The calculation of the storage capacity is based on the sequent peak algorithm (Loucks, et al., 1981) which is equivalent to the Rippl mass curve method. The algorithm, applied to the time series $y_i, i = 1, \dots, N$ may be described as follows. Based on y_i and the demand level d , a new sequence S_i' can be determined as

$$S'_i = \begin{cases} S'_{i-1} + d - y_i & \text{if positive} \\ 0 & \text{otherwise} \end{cases} \quad (3.19)$$

where $S'_0 = 0$. Then the storage capacity is obtained as

$$S_c = \max(S'_1, \dots, S'_N) \quad (3.20)$$

Note that algorithms described in Eqs.(3.15) to (3.20) apply also to seasonal series. In this case, the underlying seasonal series $y_{v,\tau}$ is simply denoted as y_t .

3.2.2 Drought Related Statistics

The drought-related statistics are also important in modeling hydrologic time series (Salas, 1993). For the series y_i , $i = 1, \dots, N$, the demand level d may be defined as $\alpha \cdot \bar{y}$, $0 < \alpha < 1$ (for example, for $\alpha = 1$, $d = \bar{y}$). A deficit occurs when $y_i < d$ consecutively during one or more years until $y_i > d$ again. Such a deficit can be defined by its duration L , by its magnitude M , and by its intensity $I = M/L$. Assume that m deficits occur in a given hydrologic sample, then the maximum deficit duration (longest drought or maximum run-length) is given by

$$L_n^* = \max(L_1, \dots, L_m) \quad (3.21)$$

and the maximum deficit magnitude (maximum run-sum) is defined by

$$M_n^* = \max(M_1, \dots, M_m) \quad (3.22)$$

In SAMS, the longest drought duration and the maximum deficit magnitude are estimated for both annual and seasonal series.

3.2.3 Surplus Related Statistics

For our purpose here, surplus related statistics are simply the opposite of drought related statistics. Considering the same threshold level d , a surplus occurs when $y_i > d$ consecutively until $y_i < d$ again. Then, assuming that m surpluses occur during a given time period N , the maximum surplus period L^* and maximum surplus magnitude M^* may be determined also from Eqs. (3.21) and (3.22).

4 MATHEMATICAL MODELS

The following univariate and multivariate models are available in SAMS for modeling of annual and seasonal data.

1. For Annual Modeling:

- Univariate ARMA(p,q) model.
- Univariate GAR(1) model.
- SM (shifting mean) model.
- Multivariate AR(p) model (MAR).
- Contemporaneous ARMA(p,q) model (CARMA(p,q)).
- Mixture of contemporaneous shifting mean and ARMA(p,q) models (CSM – CARMA(p,q)).

2. For Seasonal Modeling:

- Univariate PARMA(p,q) model.
- Multivariate PAR(p) model (MPAR).

3. Disaggregation Models

- Spatial Valencia and Schaake.
- Spatial Mejia and Rousselle.
- Temporal Lane.
- Temporal Grygier and Stedinger.

All models, except the GAR(1), assume that the underlying data is normally distributed. The GAR(1) model assumes that the process being modeled follows a gamma distribution. Thus for all other models than the GAR(1) it is necessary to transform the data into normal.

4.1 Data Transformations and Scaling

In cases where the normality tests in SAMS indicate that the observed series are not normally distributed, the data has to be transformed into normal before applying the models. To normalize the data, the following transformations $Y = f(X)$ are available in SAMS:

Logarithmic

$$Y = \ln(X + a) \quad (4.1)$$

Gamma

$$Y = \text{Gamma}(X) \quad (4.2)$$

Power

$$Y = (X + a)^b \quad (4.3)$$

Box-Cox

$$Y = \frac{(X + a)^b - 1}{b}, b \neq 0 \quad (4.4)$$

where Y is the normalized series, X is the original observed series, and a and b are transformation coefficients. The variables Y and X represent either annual or seasonal data, where for seasonal data a and b vary with the season. Note that the logarithmic transformation is simply the limiting form of the Box-Cox transform as the coefficient b approaches zero. Also, the power transformation is a shifted and scaled form of the Box-Cox transform.

Scaling and Standardization

Scaling of normally distributed data is an option in SAMS. This option is intended for use for multivariate disaggregation models when normalized data for different stations or different seasons have values that differ from each other by couple of orders of magnitude which can cause problems in parameter estimation of multivariate models. This can happen when some of the historical time series

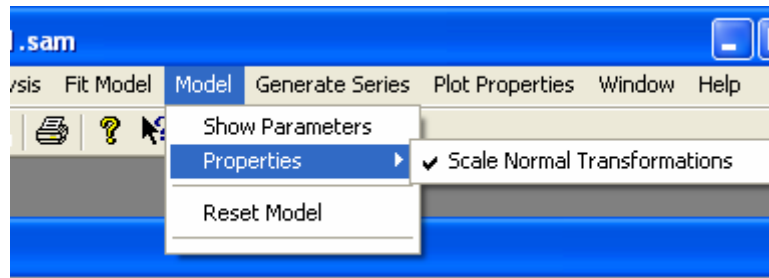


Figure 4.1: Scaling of normally distributed data.

are normally distributed and do not need to be transformed to normal while others do. To use this option select “*Scale Normal Transformations*” from the SAMS menu as is illustrated in Fig. 4.1. If this option is selected then all time series that have not been transformed by any of the transformations in Eqs. (4.1)-(4.4) are scaled by dividing by the standard deviation.

In addition, for most of the univariate and multivariate models (except disaggregation models and the CSM-CARMA) the normalized data can then be standardized by subtracting the mean and dividing by the standard deviation. This option is usually offered in the model estimation dialogs in SAMS. For example, for seasonal series, the standardization may be expressed as:

$$Y_{\nu,\tau} = \frac{X_{\nu,\tau} - \bar{X}_{\tau}}{S_{\tau}(X)} \quad (4.5)$$

where $Y_{\nu,\tau}$ is the scaled normally distributed variable with standard deviation one and mean zero for year ν of the seasonal series for season τ . $S_{\tau}(X)$ and $\bar{X}_{\tau}$ are the mean and the standard deviation of the transformed series for month τ .

The transformation bar

The transformation bar in SAMS is shown in Fig. 4.2. Data can be transformed one station or one season at a time, or one station and all seasons for that station, or all stations and all seasons at the same time. There are two plotting position formulas that are available for plotting of the empirical frequency curve: (1) the Cunnane plotting position, and (2) the Weibull plotting position. The Cunnane plotting position is approximately quantile-unbiased while the Weibull plotting position has unbiased exceedance probabilities for all distributions (Stedinger et al., 1993). In general the Cunnane plotting position should be preferred.

The parameters of the transformation can be entered manually if working with a single station or a single season. In that case, the final transformation

Figure 4.2: The transformation bar.

must be accepted by pressing on the “Accept Transf” button. The functionality of the buttons on the transformation bar are as follows:

Display Displays the currently defined transformation.

Accept Transf Accepts the currently displayed transformation.

Auto Log/Power Searches for the best Log or Power transformation for multiple stations and/or seasons.

Best Transf Searches for the best overall transformation for multiple stations and/or seasons

Refer to Appendix A for further information on how SAMS selects between different transformations. There are various tests for normality available in the literature. In SAMS two normality tests are available, namely the skewness test of normality (Salas et al., 1980; Snedecor and Cochran, 1980) and Filliben probability plot correlation test (Filliben, 1975). These two test are described in Appendix A.

Generation

During generation, synthetic time series are generated in the transformed domains, and then brought into the original domain using an inverse transformation $X = f^{-1}(Y)$.

4.2 Univariate Models

Various univariate models are available in SAMS. The annual models are the traditional ARMA(p,q) for modeling of autoregressive moving average processes, the GAR(1) for modeling of gamma distributed process, the SM for modeling of processes having a shifting pattern in the mean, and the PARMA(p,q) for modeling of seasonal processes.

4.2.1 Univariate ARMA(p,q)

The ARMA(p,q) model of autoregressive order p and moving average order q is expressed as:

$$Y_t = \sum_{i=1}^p \phi_i Y_{t-i} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j} \quad (4.6)$$

where Y_t represents the streamflow process for year t , it is normally distributed with mean zero and variance $\sigma^2(Y)$, ε_t is the uncorrelated normally distributed noise term with mean zero and variance

$\sigma^2(\varepsilon)$, $\{\phi_1, \dots, \phi_p\}$ are the autoregressive parameters and $\{\theta_1, \dots, \theta_q\}$ are the moving average parameters. The characteristics of the autocorrelation function (ACF) and the partial autocorrelation function (PACF) of the ARMA(p,q) model for different p and q are given in Table 4.1 below:

Table 4.1 Properties of the ACF and PACF of ARMA(p,q) processes.

	AR(1)	AR(p)	MA(q)	ARMA(p,q)
ACF	Decays geometrically	Tails off	Zero at lag > q	Tails off
PACF	Zero at lag > 1	Zero at lag > p	Tails off	Tails off

Two methods are available for estimation of the model parameters, namely the method of moments (MOM) and the least squares method (LS). These two estimation methods are described in Appendix A.

4.2.2 Univariate GAR(1)

The gamma-autoregressive model GAR(1) is similar to the well known AR(1) model except that the underlying process being modeled is assumed to follow the gamma distribution instead of the normal distribution. Thus if the intent is to use the GAR(1) model, then the underlying data should not be transformed to normal by SAMS. The GAR(1) model can be expressed as (Lawrence and Lewis, 1981)

$$X_t = \phi X_{t-1} + \varepsilon_t \quad (4.7)$$

where X_t is a gamma variable defined at time t , ϕ is the autoregression coefficient, and ε_t is the independent noise term. X_t is a three-parameter gamma distributed variable with marginal density function given by:

$$f_X(x) = \frac{\alpha^\beta (x - \lambda)^{\beta-1} \exp[-\alpha(x - \lambda)]}{\Gamma(\beta)} \quad (4.8)$$

where λ , α and β are the location, scale, and shape parameters, respectively. Lawrence (1982) found that the independent noise term, ε_t , can be obtained by the following scheme:

$$\varepsilon = \lambda(1 - \phi) + \eta, \text{ where } \begin{cases} \eta = 0 & \text{if } M = 0 \\ \eta = \sum_{j=1}^M Y_j \phi^{U_j} & \text{if } M > 0 \end{cases} \quad (4.9)$$

where M is an integer random variable distributed as Poisson with mean $[-\beta \ln(\phi)]$, $U_j, j=1,2,\dots$ are independent identically distributed (iid) random variables with uniform (0,1) distribution, and, $Y_j, j=1,2,\dots$ are iid random variables distributed as exponential with mean $(1/\alpha)$. The stationary GAR(1)

process of Eq. (4.7) has four parameters, namely $\{\phi, \lambda, \alpha, \beta\}$. The model parameters are estimated based on a procedure suggested by Fernandez and Salas (1990), as illustrated in Appendix A.

4.2.3 Univariate SM

The shifting mean (SM) model is characterized by sudden shifts or jumps in the mean. More precisely, the underlying process is assumed to be characterized by multiple stationary states, which only differ from each other by having different means that vary around the long term mean of the process. The process is autocorrelated, where the autocorrelation arises only from the sudden shifting pattern in the mean. A general definition of the SM model is given by (Sveinsson et al., 2003 and 2005)

$$X_t = Y_t + Z_t \quad (4.10)$$

where $\{X_t\}$ is a sequence of random variables representing the hydrologic process of interest; $\{Y_t\}$ is a sequence of *iid* random variables normally distributed with mean μ_Y and variance σ_Y^2 ; and $\{Z_t\}$ is a sequence with mean zero and variance σ_Z^2 . The sequences $\{Y_t\}$ and $\{Z_t\}$ are assumed to be mutually independent of each other. The X_t process is characterized by multiple “stationary” states each of random length N_i , $i = 1, 2, \dots$ as shown in Fig. 4.3. The Z_t process represents the shifting pattern from one state to another, and the different states are referred to as noise levels. The noise level process $\{Z_t\}$ can be written as

$$Z_t = \sum_{i=1}^t M_i I_{(S_{i-1}, S_i]}(t) \quad (4.11)$$

Where $\{M_i\}_{i=1}^{\infty} \sim iid \ N(0, \sigma_M^2 = \sigma_Z^2)$, $S_i = N_1 + N_2 + \dots + N_i$ with $S_0 = 0$, and $I_{(a,b)}(t)$ is the indicator function equal to one if $t \in (a, b)$ and zero otherwise. The $\{N_t\}_{i=1}^{\infty}$ is a discrete, stationary, delayed-renewal sequence on the positive integers, with $\{N_t\}_{i=1}^{\infty} \sim iid \ \text{Positive Geometric}(p)$ (Sveinsson et al., 2003 and 2005). Thus the average length of each state of the process is the inverse of the parameter of the positive Geometric distribution or $1/p$. The estimation of model parameters is described in Appendix A.

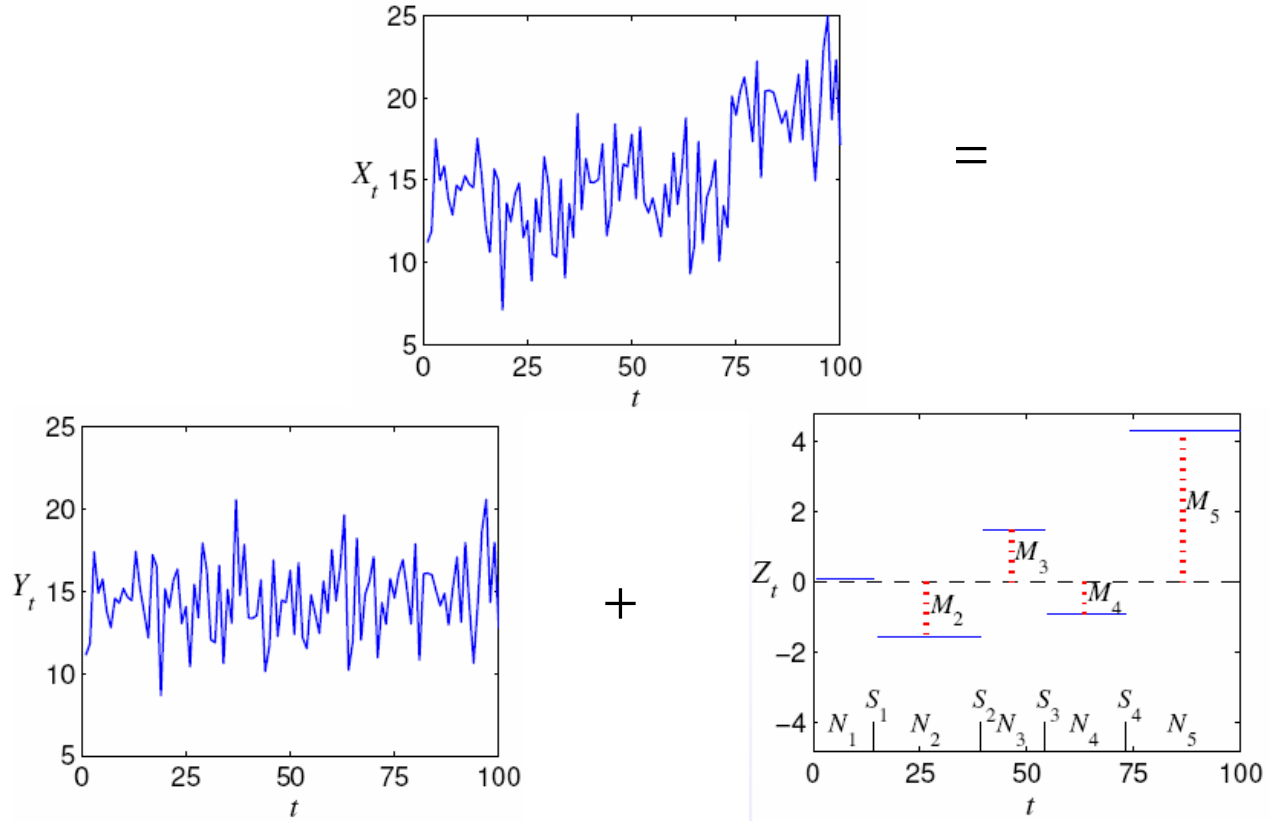


Figure 4.3: The processes in the SM model.

4.2.4 Univariate Seasonal PARMA(p,q)

Stationary ARMA models have been widely applied in stochastic hydrology for modeling of annual time series where the mean, variance, and the correlation structure do not depend on time. For seasonal hydrologic time series, such as monthly series, seasonal statistics such as the mean and standard deviation may be reproduced by a stationary ARMA model by means of standardizing the underlying seasonal series. However, this procedure assumes that season-to-season correlations are the same for a given lag. Hydrologic time series, such as monthly streamflows, are usually characterized by different dependence structure (month-to-month correlations) depending on the season (e.g. spring or fall). Periodic ARMA (PARMA) models have been suggested in the literature for modeling such periodic dependence structure. A PARMA(p,q) model may be expressed as (Salas, 1993):

$$Y_{\nu,\tau} = \sum_{i=1}^p \phi_{i,\tau} Y_{\nu,\tau-i} + \varepsilon_{\nu,\tau} - \sum_{j=1}^q \theta_{j,\tau} \varepsilon_{\nu,\tau-j} \quad (4.12)$$

where $Y_{\nu,\tau}$ represents the streamflow process for year ν and season τ . For each season, τ , this process is normally distributed with mean zero and variance $\sigma_{\tau}^2(Y)$. The $\varepsilon_{\nu,\tau}$ is the uncorrelated

noise term which for each season is normally distributed with mean zero and variance $\sigma_\tau^2(\varepsilon)$. The $\{\phi_1, \phi_2, \dots, \phi_p, \tau\}$ are the periodic autoregressive parameters and the $\{\theta_1, \theta_2, \dots, \theta_q, \tau\}$ are the periodic moving average parameters. If the number of seasons or the period is ω , then a PARMA(p,q) model consists of ω number of individual ARMA(p,q) models, where the dependence is across seasons instead of years. Parameters are estimated using MOM or LS as illustrated in Appendix A. The MOM method can only be used in SAMS for $q = 0$ or 1 .

4.3 Multivariate Models

Analysis and modeling of multiple time series is often needed in Hydrology. In SAMS full multivariate model are available for modeling complex dependence structure in space and time at multiple lags. Also in SAMS, contemporaneous models are available for preserving complex dependence structure within each site but simpler structure in space across sites. Typical property of contemporaneous models is diagonal parameter matrixes which simplify the parameters estimation by allowing the model to be decoupled into univariate models. The multivariate models available in SAMS are the multivariate autoregressive model MAR(p), the contemporaneous ARMA(p,q) model dubbed as CARMA(p,q), the mixed contemporaneous shifting mean and CARMA(p,q) model dubbed as CSM-CARMA(p,q), and the seasonal multivariate periodic autoregressive model MPAR(p).

4.3.1 Multivariate MAR(p)

The multivariate MAR(p) model for n sites can be expressed as:

$$\mathbf{Y}_t = \sum_{i=1}^p \Phi_i \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t \quad (4.13)$$

where $\mathbf{Y}_t$ is a $n \times 1$ column vector of normally distributed zero mean elements $Y_t^{(k)}$, $k = 1, 2, \dots, n$, representing the different sites. $\Phi_1, \Phi_2, \dots, \Phi_p$ are the $n \times n$ autoregressive parameter matrixes, and $\{\boldsymbol{\varepsilon}_t\} \sim iid \text{ MVN}(\mathbf{0}, \mathbf{G})$ is the $n \times 1$ vector of normally distributed noise terms with mean zero and variance-covariance matrix $\mathbf{G}$. The noise vector is independent in time and correlated in space at lag zero. In SAMS the following notation is used to simplify the generation process:

$$\boldsymbol{\varepsilon}_t = \mathbf{B} \mathbf{z}_t \quad (4.14)$$

where $\{\mathbf{z}_t\} \sim iid \text{ MVN}(\mathbf{0}, \mathbf{I})$, that is a $n \times 1$ vector of independent standard normally distributed

variables uncorrelated in both time and space. The $n \times n$ matrix $\mathbf{B}$ is a lower triangular matrix such that $\mathbf{G} = \mathbf{B}\mathbf{B}^T$, where $\mathbf{B}$ is the Cholesky decomposition of $\mathbf{G}$. The lag 0 spatial correlation across all sites is preserved through the matrix $\mathbf{B}$. In the MAR(p) model the correlation in time and space across all sites is preserved up to lag p. For further information on parameter estimation and generation refer to Appendix A.

4.3.2 Multivariate CARMA(p,q)

When modeling multivariate hydrologic processes based on the full multivariate ARMA model, often problems arise in parameter estimation. The CARMA (Contemporaneous Autoregressive Moving Average) model was suggested as a simpler alternative to the full multivariate ARMA model (Salas, et al., 1980). In the CARMA(p,q) model, both autoregressive and moving average parameter matrixes are assumed to be diagonal such that a multivariate model can be decoupled into univariate ARMA models. Thus, instead of estimating the model parameters jointly, they can be estimated independently for each single site by regular univariate ARMA model estimation procedures. This allows for identification of the best univariate ARMA model for each single station. Thus different dependence structure in time can be modeled for each site, instead of having to assume a similar dependence structure in time for all sites if a full multivariate ARMA model was used.

The CARMA(p,q) model for n sites can be expressed as:

$$\mathbf{Y}_t = \sum_{i=1}^p \Phi_i \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t - \sum_{j=1}^q \Theta_j \boldsymbol{\varepsilon}_{t-j} \quad (4.15)$$

where $\mathbf{Y}_t$ is a $n \times 1$ column vector of normally distributed zero mean elements $Y_t^{(k)}$, $k = 1, 2, \dots, n$, representing the different sites. $\Phi_1, \Phi_2, \dots, \Phi_p$ are the diagonal $n \times n$ autoregressive parameter matrixes and $\Theta_1, \Theta_2, \dots, \Theta_q$ are diagonal $n \times n$ moving average matrixes. $\{\boldsymbol{\varepsilon}_t\} \sim iid \text{ MVN}(\mathbf{0}, \mathbf{G})$ is the $n \times 1$ vector of normally distributed noise terms with mean zero and variance-covariance matrix $\mathbf{G}$. For information on parameter estimation and generation refer to Appendix A.

The CARMA model is capable of preserving the lag zero cross correlation in space between different sites, in addition to the time dependence structure for each site as defined by the parameters p and q.

4.3.3 Multivariate CSM – CARMA(p,q)

Analyses of multiple time series of different hydrologic variables may require mixing of models. For example shifts in time series of one hydrologic variable may not be present in a time series of another hydrologic variable. Or, if different geographic locations are used for analysis of a single hydrologic variable, then characteristics of the corresponding times series may be dependent on their geographic location. In such cases mixing of multiple SM models and other time series models, such as ARMA(p,q), may be desirable. Such mixed model is available in SAMS representing a mixture of one contemporaneous shifting mean model (CSM) with one CARMA(p,q) model, where the lag zero cross correlation function (CCF) in space is preserved between the CARMA(p,q) model and the CSM model. In the CSM part of the model is assumed that all sites exhibit shifts at the same time as is further discussed in Appendix A.

Lets assume that there are total of n sites, of which n_1 sites follow a CSM model and the remaining n_2 sites follow a CARMA(p,q) model. The model of the n sites can be presented by a vector version of Eq (4.10) for the SM model, where the first n_1 elements of $\mathbf{X}_t$ represent the CSM model and the remaining n_2 elements of $\mathbf{X}_t$ represent the CARMA(p,q) model (Sveinsson and Salas, 2006):

$$\begin{bmatrix} X_t^{(1)} \\ \vdots \\ X_t^{(n_1)} \\ X_t^{(n_1+1)} \\ \vdots \\ X_t^{(n)} \end{bmatrix} = \begin{bmatrix} Y_t^{(1)} \\ \vdots \\ Y_t^{(n_1)} \\ Y_t^{(n_1+1)} \\ \vdots \\ Y_t^{(n)} \end{bmatrix} + \begin{bmatrix} Z_t^{(1)} \\ \vdots \\ Z_t^{(n_1)} \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (4.16)$$

where the whole $n \times 1$ vector $\mathbf{Y}_t$ can be looked at as being modeled by a CARMA(p, q) model as in Eq (4.15). Each of the first n_1 elements of $\mathbf{Y}_t$ is an ARMA(0,0) process, and each of the remaining n_2 elements of $\mathbf{Y}_t$ follows some ARMA(p,q) process. That is, $Y_t^{(k)}$ is an ARMA(p_k, q_k) process, $k = 1, 2, \dots, n$, where the p_k s can be different and the q_k s can be different. The p and the q of the CARMA(p,q) model are $p = \max(p_1, p_2, \dots, p_n)$ and $q = \max(q_1, q_2, \dots, q_n)$. The parameter matrixes of the CARMA(p,q) are diagonal, thus estimation of parameters of the CSM-CARMA model is done by uncoupling the model into univariate SM and ARMA(p,q) models. The estimation of parameters and generation of synthetic time series is described in Appendix A. The estimation module in SAMS for the CSM-CARMA model can also be used for estimation of a pure CSM model and a pure CARMA model only.

The CSM-CARMA model is capable of preserving the lag zero cross correlation in space between different sites, in addition to the time dependence structure for each site as defined by the parameters p and q . In addition, the CSM portion of the model is capable of preserving a certain dependence structure both in time and space through the noise level process Z_t .

4.3.4 Multivariate Seasonal MPAR (p)

The MPAR(p) model for n sites can be expressed as:

$$\mathbf{Y}_{\nu,\tau} = \sum_{i=1}^p \Phi_{i,\tau} \mathbf{Y}_{\nu,\tau-i} + \boldsymbol{\varepsilon}_{\nu,\tau} \quad (4.17)$$

Where $\mathbf{Y}_{\nu,\tau}$ is a $n \times 1$ column vector of normally distributed zero mean elements representing the process for year ν and season τ . The $\Phi_{1,\tau}, \Phi_{2,\tau}, \dots, \Phi_{p,\tau}$ are the $n \times n$ autoregressive periodic parameter matrixes, and $\{\boldsymbol{\varepsilon}_{\nu,\tau}\} \sim iid \text{MVN}(\mathbf{0}, \mathbf{G}_\tau)$ is the $n \times 1$ vector of normally distributed noise terms with mean zero and periodic $n \times n$ variance-covariance matrix $\mathbf{G}_\tau$. The noise vector is independent in time and correlated in space at lag zero. For estimation of parameters and generation of synthetic time series refer to Appendix A.

4.4 Disaggregation Models

Valencia and Schaake (1973) and later extension by Mejia and Rousselle (1976) introduced the basic disaggregation model for temporal disaggregation of annual flows into seasonal flows. However, the same model can also be used for spatial disaggregation. For example, the sum of flows of several stations can be disaggregated into flows at each of these stations or the total flows at key stations can be disaggregated into flows at substations which usually, but not necessarily, sum to form the flows of the key stations. The Valencia and Schaake and the Mejia and Rousselle models require many parameters to be estimated in the case of temporal disaggregation. For example, Valencia and Schaake model requires 156 parameters for the case of disaggregating annual flows into 12 seasons for one station. Mejia and Rousselle model require 168 parameters. For 3 sites, the above models require 1,404 and 1,512 for both models, respectively. Lane (1979) introduced the condensed model for temporal disaggregation which reduces the number of parameters required drastically. For example, for the cases mentioned above, Lane's model requires 36 parameters for the one site case and 324 parameters for the 3 site case. Later Grygier and Stedinger (1990) introduced a contemporaneous temporal disaggregation model which requires 48 parameters for the above one

site case and 216 parameters for the above 3 site case.

In SAMS, Lane's model and Grygier and Stedinger model are used for temporal (seasonal) disaggregation, and the Valencia and Schaake model and Mejia and Rousselle model are used for spatial disaggregation of annual and seasonal data.

In using disaggregation models for data generation, adjustments may be needed to ensure additivity constraints. For instance, in spatial disaggregation, to ensure that the generated flows at substations (or at subsequent stations) add to the total or a fraction (depending on the particular case at hand) of the corresponding generated flow at a key station (or subkey station) or, in temporal disaggregation, to ensure that the generated seasonal values add exactly to the generated annual value, three methods of adjustment based on Lane and Frevert (1990) are provided in SAMS. These methods will be described in the following sections.

4.4.1 Spatial Disaggregation of Annual Data

For spatial disaggregation of annual data from N key stations to M sub stations there are two models available, namely the Valencia and Schaake (VS) model (Valencia and Schaake, 1973)

$$\mathbf{Y}_\nu = \mathbf{A} \mathbf{X}_\nu + \mathbf{B} \boldsymbol{\varepsilon}_\nu \quad (4.18)$$

and the Mejia and Rousselle (MR) model (Mejia and Rousselle, 1976)

$$\mathbf{Y}_\nu = \mathbf{A} \mathbf{X}_\nu + \mathbf{B} \boldsymbol{\varepsilon}_\nu + \mathbf{C} \mathbf{Y}_{\nu-1} \quad (4.19)$$

where $\mathbf{X}_\nu$ is the $N \times 1$ column vector of observations in year ν at the N key sites, $\mathbf{Y}_\nu$ is the corresponding $M \times 1$ column vector at the sub sites, $\boldsymbol{\varepsilon}_\nu$ is the $M \times 1$ column noise vector uncorrelated in space and time with each element distributed as standard normal, and $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are full $M \times N$, $M \times M$, and $M \times M$ parameter matrixes, respectively. The differences between the VS and MR models is that the VS model is designed to preserve the lag 0 correlation coefficient in space between all sub stations through the matrix $\mathbf{B}$, and the lag 0 correlation in space between all sub and key stations through the matrix $\mathbf{A}$. The MR model additionally preserves the lag 1 correlation coefficient in space between all sub stations through the matrix $\mathbf{C}$, i.e. the correlations between current year values with past year values. For estimation of parameters refer to Appendix A.

4.4.2 Spatial Disaggregation of Seasonal Data

For spatial disaggregation of seasonal data from N key stations to M sub stations only the MR model is made available in SAMS although the simpler VS model could also be used. The reason

for this is that almost all hydrological data do shown seasonal dependence structure. Although not available in SAMS the VS model for spatial disaggregation of seasonal data becomes

$$\mathbf{Y}_{\nu,\tau} = \mathbf{A}_\tau \mathbf{X}_{\nu,\tau} + \mathbf{B}_\tau \boldsymbol{\varepsilon}_{\nu,\tau} \quad (4.20)$$

and the MR model becomes

$$\mathbf{Y}_{\nu,\tau} = \mathbf{A}_\tau \mathbf{X}_{\nu,\tau} + \mathbf{B}_\tau \boldsymbol{\varepsilon}_{\nu,\tau} + \mathbf{C}_\tau \mathbf{Y}_{\nu,\tau-1} \quad (4.21)$$

where the data vector and parameter matrixes are seasonal with τ representing the current season. I.e. $\mathbf{X}_{\nu,\tau}$ is the $N \times 1$ column vector of observations in year ν season τ at the N key sites, $\mathbf{Y}_{\nu,\tau}$ is the corresponding $M \times 1$ column vector at the sub sites, $\mathbf{Y}_{\nu,\tau-1}$ is the previous season $M \times 1$ column vector at the sub sites, $\boldsymbol{\varepsilon}_{\nu,\tau}$ is the *iid* standard normal $M \times 1$ column noise vector for year ν season τ , and $\mathbf{A}_\tau, \mathbf{B}_\tau$, and $\mathbf{C}_\tau$ are the seasonal parameter matrixes of the same dimensions as in the models for spatial disaggregation of annual data. The VS model preserves for each season the lag 0 correlation coefficient in space between all sub stations through the matrix $\mathbf{B}$, and lag 0 correlations in space between all sub and key stations through the matrix $\mathbf{A}$. The MR model additionally preserves the lag 1 correlation coefficient in space between all sub stations through the matrix $\mathbf{C}$, i.e. the correlations between current season values with the previous season values. For estimation of parameters refer to Appendix A.

4.4.3 Temporal Disaggregation

For temporal disaggregation of annual data from N stations to seasonal data at the same N stations the available models are the temporal Lane model (Lane and Frevert, 1990) and the temporal Grygier and Stedinger model (Grygier and Stedinger, 1990). The temporal Lane model can be summarized by

$$\mathbf{Y}_{\nu,\tau} = \mathbf{A}_\tau \mathbf{Y}_\nu + \mathbf{B}_\tau \boldsymbol{\varepsilon}_{\nu,\tau} + \mathbf{C}_\tau \mathbf{Y}_{\nu,\tau-1} \quad (4.22)$$

where $\mathbf{A}_\tau, \mathbf{B}_\tau$, and $\mathbf{C}_\tau$ are full $N \times N$ parameter matrixes, $\mathbf{Y}_\nu$ is the $N \times 1$ column vector of observations in year ν at the N sites, $\mathbf{Y}_{\nu,\tau}$ is the corresponding $N \times 1$ column vector of observations in the same year ν season τ , and $\mathbf{Y}_{\nu,\tau-1}$ is the previous season $N \times 1$ column vector. $\boldsymbol{\varepsilon}_{\nu,\tau}$ is the *iid* standard normal $N \times 1$ column noise vector for year ν season τ

The Grygier and Stedinger model (Grygier and Stedinger, 1990) is a contemporaneous model

$$\mathbf{Y}_{\nu,\tau} = \mathbf{A}_\tau \mathbf{Y}_\nu + \mathbf{B}_\tau \boldsymbol{\varepsilon}_{\nu,\tau} + \mathbf{C}_\tau \mathbf{Y}_{\nu,\tau-1} + \mathbf{D}_\tau \boldsymbol{\Lambda}_{\nu,\tau} \quad (4.23)$$

where $\mathbf{A}_\tau$, $\mathbf{C}_\tau$, and $\mathbf{D}_\tau$ are diagonal $N \times N$ parameter matrixes (i.e. contemporaneous), $\mathbf{B}_\tau$ is a full $N \times N$ parameter matrix, and $\mathbf{Y}_v$, $\mathbf{Y}_{v,\tau}$, $\mathbf{Y}_{v,\tau-1}$ and $\boldsymbol{\varepsilon}_{v,\tau}$ are the same as in the Lane model. $\boldsymbol{\Lambda}_{v,\tau} = \mathbf{W}_\tau \mathbf{Y}_{v,\tau-1}$ are weighted seasonal flows, where the weights $\mathbf{W}_\tau$ (a diagonal $N \times N$ matrix) depend on the type of transformations used to transform the historical seasonal data to normal and the seasonal historical data themselves.. This term $\boldsymbol{\Lambda}_{v,\tau}$ ensures that additivity of the model is approximately preserved, i.e. the seasonal flows summing to the annual flows. For the first season $\mathbf{C}_1$ and $\mathbf{D}_1$ are null matrixes, and for the second season $\mathbf{C}_2$ is a null matrix. For further technical description of the model the reader is referred to Grygier and Stedinger (1990).

Both models preserve the correlations of the annual data with same year season data through the matrix $\mathbf{A}_\tau$ for each season, and the lag 1 season to season correlations through the matrix $\mathbf{C}_\tau$ for each season. Since the parameter matrixes in the Lane model are full these correlations are preserved across all sites, while in the Grygier and Stedinger model they are preserved only within each site (diagonal parameter matrixes). In addition the Grygier and Stedinger model does not preserve the lag 1 correlation between the first season of a given year and the last season of the previous year. For estimation of parameters refer to Appendix A.

4.5 Unequal Record Lengths

When working with different length records difficulties can arise in the use of multivariate procedures that require the records to be of same lengths. Record extension can be a tedious task and if not done properly it can do more damage than good. Several models in SAMS have been formulated to deal with unequal record lengths at different sites. In these models all available data are used for parameter estimation in such a way that synthetic generated series will preserve the overall mean and the variance of each record and either the cross-covariance or the cross-correlation of the common period of records. The models in SAMS capable of dealing with unequal record lengths are the:

- Multivariate CSM – CARMA(p,q).
- The Valencia and Schaake model and the Mejia and Rousselle model for spatial disaggregation of annual and seasonal data.
- The Lane model and the Grygier and Stedinger model for temporal disaggregation.

The CSM-CARMA(p,q) model can also be used to fit a CSM model only or a CARMA(p,q) model

only to data from multiple sites having different record lengths.

When the mean and the variance of each different length record is preserved then a choice has to be made whether to preserve the cross-covariance or the cross-correlation of the common period of records (Sveinsson, 2004). In SAMS the cross-correlation coefficients of the common period of records are preserved for the VS and the MR spatial disaggregation models and the Lane temporal disaggregation model, while the cross-covariance coefficients of the common period of records are preserved for the CSM-CARMA(p,q) model and the Grygier and Stedingar temporal disaggregation model. For further information on how SAMS deals with unequal record lengths refer to Sveinsson (2004) and Appendix A.

4.6 Adjustment of Generated Data

When using transformed data in disaggregation models, the constraint that the seasonal (or spatial) flows should sum to the given value of the annual flow is lost. Thus, the generated annual flows calculated as the sum of the generated seasonal flows, will deviate from the value of the generated annuals produced by the annual models. These small differences can be ignored, or can be corrected, scaling somehow each year's seasonal flows so their sum equals the specified value of the annual flow. Three approaches are available in SAMS for the adjustment of spatial and temporal disaggregated data based on Lane and Frevert (1990). The options for these adjustments are set in the “Generation” dialog in SAMS.

Spatial adjustment

Three approaches are available to spatially adjust annual or seasonal disaggregated data based on the modeling choice in SAMS. More precisely for the modeling option “Annual Data” → “Disaggregation” and “Seasonal Data” → “Disaggregation” → “Spatial-Seasonal”, the spatial adjustment is intended to be done on annual data.

Annual Data

approach 1:

$$\hat{q}_v^{*(i)} = \hat{q}_v^{(i)} + (r \hat{q}_v - \sum_{j=1}^n \hat{q}_v^{(j)}) \frac{|\hat{q}_v^{(i)} - \hat{\mu}^{(i)}|}{\sum_{j=1}^n |\hat{q}_v^{(j)} - \hat{\mu}^{(j)}|} \quad (4.24)$$

approach 2:

$$\hat{q}_v^{*(i)} = \hat{q}_v^{(i)} \frac{r \hat{q}_v}{\sum_{j=1}^n \hat{q}_v^{(j)}} \quad (4.25)$$

approach 3:

$$\hat{q}_v^{*(i)} = \hat{q}_v^{(i)} + (r \hat{q}_v - \sum_{j=1}^n \hat{q}_v^{(j)}) \frac{(\hat{\sigma}^{(i)})^2}{\sum_{j=1}^n (\hat{\sigma}^{(j)})^2} \quad (4.26)$$

where:

$$r = \frac{1}{N} \sum_{v=1}^N r_v \quad (4.27a)$$

$$r_v = \frac{1}{q_v} \sum_{j=1}^n q_v^{(j)} \quad (4.27b)$$

and N is the number of observations, n is the number of substations, q_v is the v -th observed value at a key station (or substation), $q_v^{(j)}$ is the v -th observed value at substation (or subsequent station) j , $\hat{q}_v$ is the generated value at the key station, $\hat{q}_v^{(i)}$ is the generated value at substation i , $\hat{q}_v^{*(i)}$ is the adjusted generated value at substation i , $\hat{\mu}^{(i)}$ is the estimated mean of $\hat{q}_v^{(i)}$ for site i , and $\hat{\sigma}^{(i)}$ is the estimated standard deviation of $\hat{q}_v^{(i)}$ for site i .

Similarly for spatial adjustment of seasonal data when the modeling option “Seasonal Data” → “Disaggregation” → “Seasonal-Spatial” is used.

Seasonal Data

approach 1:

$$\hat{q}_{v,\tau}^{*(i)} = \hat{q}_{v,\tau}^{(i)} + (r_\tau \hat{q}_{v,\tau} - \sum_{j=1}^n \hat{q}_{v,\tau}^{(j)}) \frac{|\hat{q}_{v,\tau}^{(i)} - \hat{\mu}_\tau^{(i)}|}{\sum_{j=1}^n |\hat{q}_{v,\tau}^{(j)} - \hat{\mu}_\tau^{(j)}|} \quad (4.28)$$

approach 2:

$$\hat{q}_{v,\tau}^{*(i)} = \hat{q}_{v,\tau}^{(i)} \frac{r_\tau \hat{q}_{v,\tau}}{\sum_{j=1}^n \hat{q}_{v,\tau}^{(j)}} \quad (4.29)$$

approach 3:

$$\hat{q}_{v,\tau}^{*(i)} = \hat{q}_{v,\tau}^{(i)} + (r_\tau \hat{q}_{v,\tau} - \sum_{j=1}^n \hat{q}_{v,\tau}^{(j)}) \frac{(\hat{\sigma}_\tau^{(i)})^2}{\sum_{j=1}^n (\hat{\sigma}_\tau^{(j)})^2} \quad (4.30)$$

where:

$$r_\tau = \frac{1}{N} \sum_{v=1}^N r_{v,\tau} \quad (4.31a)$$

$$r_{v,\tau} = \frac{\sum_{j=1}^n q_{v,\tau}^{(j)}}{q_{v,\tau}} \quad (4.31b)$$

and N is the length of the available sample in years, n is the number of substations, $q_{v,\tau}$ is the observed value at key station in year v , season τ , $q_{v,\tau}^{(i)}$ is the observed value at substation i in year v , month τ , $\hat{q}_{v,\tau}$ is the generated value at key station, $\hat{q}_{v,\tau}^{(i)}$ is the generated at substation i , $\hat{q}_{v,\tau}^{*(i)}$ is the adjusted generated value at substation i , $\hat{\mu}_\tau^{(i)}$ is the estimated mean of $q_{v,\tau}^{(i)}$ for season τ and $\hat{\sigma}_\tau^{(i)}$ is the estimated standard deviation of $q_{v,\tau}^{(i)}$ for season τ .

Adjustment for temporal disaggregation

Three approaches are also available for the adjustment of temporal disaggregated data. This adjustment is done for one station at a time.

approach 1:

$$\hat{q}_{v,\tau}^{*(i)} = \hat{q}_{v,\tau}^{(i)} + (\hat{Q}_v - \sum_{t=1}^{\omega} \hat{q}_{v,t}) \frac{|\hat{q}_{v,\tau} - \hat{\mu}_\tau|}{\sum_{t=1}^n |\hat{q}_{v,t} - \hat{\mu}_t|} \quad (4.32)$$

approach 2:

$$\hat{q}_{v,\tau}^* = \hat{q}_{v,\tau} \frac{\hat{Q}_v}{\sum_{t=1}^{\omega} \hat{q}_{v,t}} \quad (4.33)$$

approach 3:

$$\hat{q}_{v,\tau}^* = \hat{q}_{v,\tau} + (\hat{Q}_v - \sum_{t=1}^{\omega} \hat{q}_{v,t}) \frac{\hat{\sigma}_\tau^2}{\sum_{t=1}^{\omega} \hat{\sigma}_t^2} \quad (4.34)$$

where ω is the number of seasons, $\hat{Q}_v$ is the generated annual value, $\hat{q}_{v,\tau}$ is the generated seasonal value, $\hat{q}_{v,\tau}^*$ is the adjusted generated seasonal value, $\hat{\mu}_\tau$ is the estimated mean of $\hat{q}_{v,\tau}$ for season τ , and $\hat{\sigma}_\tau$ is the estimated standard deviation of $\hat{q}_{v,\tau}$ for season τ .

4.7 Model Testing

The fitted model must be tested to determine whether the model complies with the model assumptions and whether the model is capable of reproducing the historical statistical properties of the data at hand. Essentially the key assumptions of the models refer to the underlying characteristics of the residuals such as normality and independence.

Aikaike Information Criteria for ARMA and PARMA Models

The ACF and PACF are often used to get an idea of the order of the ARMA(p,q) or the PARMA(p,q) model to fit. An alternative is to use information criteria for selecting the best-fit model. The two information criteria available in SAMS are the corrected Akaike information criterion (AICC) and the Schwarz information criterion (SIC) also often referred to as the Bayesian information criterion. To see the values of the criteria the user has to select “Show Parameters” from the “Model” menu in SAMS.

The AICC is given by (Hurvich and Tsai, 1989, Brockwell and Davis, 1996):

$$\text{AICC} = n \ln \hat{\sigma}^2(\varepsilon) + n + \frac{2(k+1)n}{n-k-2} \quad (4.35)$$

where n is the size of the sample used for fitting, k is the number of parameters excluding constant terms ($k = p + q$ for the ARMA(p,q) model), and $\hat{\sigma}^2(\varepsilon)$ is the maximum likelihood estimate of the residual variance (biased). The AICC statistic is efficient but not consistent and is good for small samples but tends to overfit for large samples and large k .

The SIC is given by (Hurvich and Tsai, 1993, Shumway and Stoffer, 2000):

$$\text{SIC} = n \ln \hat{\sigma}^2(\varepsilon) + n + k \ln n \quad (4.36)$$

where n , k and $\hat{\sigma}^2(\varepsilon)$ are defined in the same way as for the AICC statistic. In general the SIC is good for large samples, but tends to underfit for small samples. Efficiency is usually more important than consistency since the true model order is not known for real world data.

Testing the properties of the process

Testing the properties of the process generally means comparing the statistical properties

(statistics) of the process being modeled, for instance, the process $Y_{v,\tau}$, with those of the historical sample. In general, one would like the model to be capable of reproducing the necessary statistics that affect the variability of the data. Furthermore, the model should be capable of reproducing certain statistics that are related to the intended use of the model.

If $Y_{v,\tau}$ has been previously transformed from $X_{v,\tau}$, the original non-normal process, then one must test, in addition to the statistical properties of Y , some of the properties of X . Generally, the properties of Y include the seasonal mean, seasonal variance, seasonal skewness, and season-to-season correlations and cross-correlations (in the case of multisite processes), and the properties of X include the seasonal mean, variance, skewness, correlations, and cross-correlations (for multisite systems). Furthermore, additional properties of $X_{v,\tau}$ such as those related to low flows, high flows, droughts, and storage may be included depending on the particular problem at hand.

In addition, it is often the case that not only the properties of the seasonal processes $Y_{v,\tau}$ and $X_{v,\tau}$, must be tested but also the properties of the corresponding annual processes AY and AX . For example, this case arises when designing the storage capacity of reservoir systems or when testing the performance of reservoir systems of given capacities, in which one or more reservoirs are for over year regulation. In such cases the annual properties considered are usually the mean, variance, skewness, autocorrelations, cross-correlations (for multisite systems), and more complex properties such as those related to droughts and storage.

The comparison of the statistical properties of the process being modeled versus the historical properties may be done in two ways. Depending on the type of model, certain properties of the Y process such as the mean(s), variance(s), and covariance(s), can be derived from the model in close form. If the method of moments is used for parameter estimation, the mean(s), variance(s), and some of the covariance should be reproduced exactly, however, except for the mean, that may not be the case for other estimation methods. Finding properties of the Y process in closed form beyond the first two moments, for instance, drought related properties, are complex and generally are not available for most models. Likewise, except for simple models, finding properties in close form for the corresponding annual process AY , is not simple either. In such cases, the required statistical properties are derived by data generation.

Data generation studies for comparing statistical properties of the underlying process Y (and other derived processes such as AY , X and AX) are generally undertaken based on samples of equal

length as the length of the historical record and based on a certain number of samples which can give enough precision for estimating the statistical properties of concern. While there are some statistical rules that can be derived to determine the number of samples required, a practical rule is to generate say 100 samples which can give an idea of the distribution of the statistic of interest say θ . In any case, the statistics $\theta(i)$, $i = 1, \dots, 100$ are estimated from the 100 samples and the mean $\bar{\theta}$ and variance $s^2(\theta)$ are determined. Then, the mean deviation, $MD(\theta)$

$$MD(\theta) = \bar{\theta} - \theta(H) \quad (4.37)$$

and the relative root mean square deviations, $RRMSD(\theta)$

$$RRMSD(\theta) = \frac{1}{\theta(H)} \sqrt{\sum_{i=1}^{100} [\theta_i - \theta(H)]^2} \quad (4.38)$$

are obtained in which $\theta(H)$ is the statistic derived from the historical sample (historical statistic). The statistics $MD(\theta)$ and $RRMSD(\theta)$ are useful for comparing between the historical and model statistics derived by data generation. In addition, one can observe where $\theta(H)$ falls relative to $\bar{\theta} - s(\theta)$ and $\bar{\theta} + s(\theta)$. Also graphical comparisons such as the Box-Cox diagrams can be useful.

5 EXAMPLES

5.1 Statistical Analysis of Data

In this section, SAMS operations will be used to model actual hydrologic data. The data used is the monthly data of the Colorado River basin. The data will be read from the file Colorado_River.dat which can be obtained from the diskette accompanying this manual. The file contains data for 29 stations in the Colorado River basin. Each station's data consists of 12 seasons and is 98 years long (1905 -2003). As an illustration a sample of the data file is shown in Appendix B. SAMS was used to analyze the statistics of the seasonal and annual data. Some of the statistics calculated by SAMS are shown below.

Annual Statistics

Site Number 20: IF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ
Historical

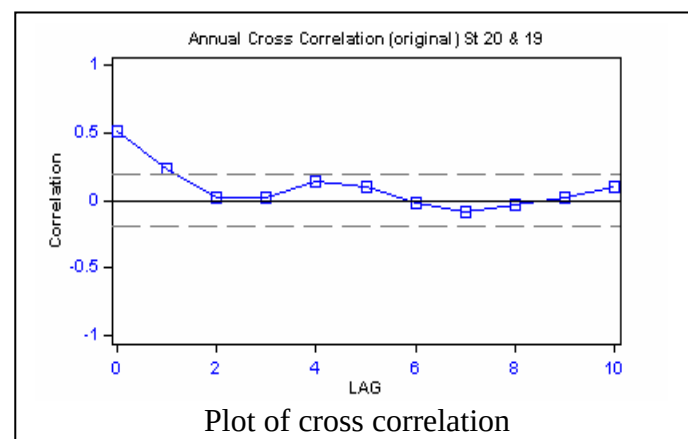
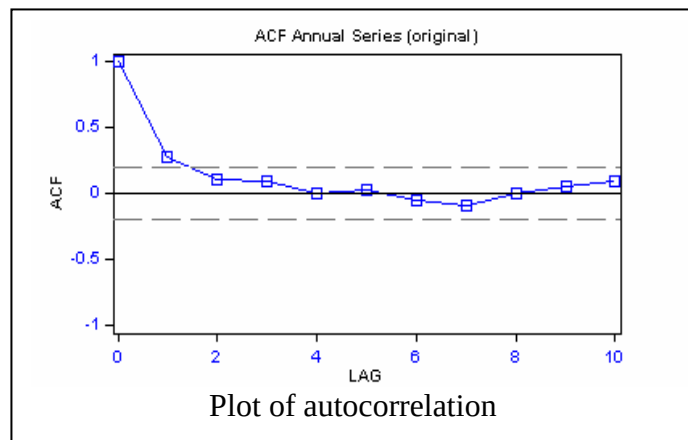
Mean	15,080,000
StDev	4,343,000
CV	0.2881
Skewness	0.1402
Min	5,525,000
Max	25,300,000
acf(1)	0.2804
acf(2)	0.0989

Correlation Structure

LAG	Autocorr.
0	1
1	0.280
2	0.099
3	0.088
4	0.003
5	0.029
6	-0.058
7	-0.098
8	0.002
9	0.048
10	0.098

Cross Correlations Sites 29 and 19

LAG	Autocorr.
0	0.511
1	0.230
2	0.016
3	0.018
4	0.142
5	0.094



6	-0.026
7	-0.090
8	-0.032
9	0.016
10	0.097

Storage and Drought Statistics

Demand Level	1.00×mean
Longest Deficit	5
Max Deficit	21,767,507
Longest Surplus	6
Max Surplus	36,992,199
Storage Capacity	72,108,274
Rescaled Range	16.603
Hurst Coeff.	0.722

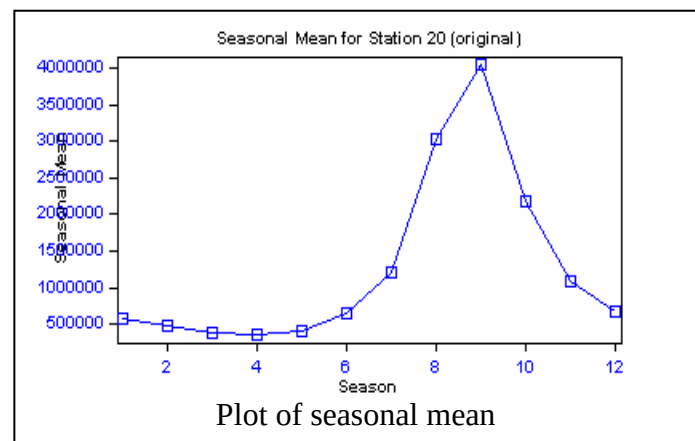
Seasonal Statistics

Site Number 20: IF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ

Season #	Month	Mean	StDev	CV	Skewness	Min	Max	acf(1)	acf(2)
1	Oct	580,900	270,600	0.466	1.641	193,800	1,814,000	0.16	0.22
2	Nov	480,800	140,800	0.293	1.215	181,400	999,100	0.31	0.28
3	Dec	382,500	95,370	0.249	1.223	226,900	730,200	0.54	0.36
4	Jan	356,600	78,230	0.219	0.590	200,300	588,800	0.52	0.36
5	Feb	393,800	97,080	0.247	1.419	252,700	774,700	0.25	0.01
6	Mar	645,200	210,300	0.326	1.081	279,600	1,404,000	0.28	0.15
7	Apr	1,200,000	509,800	0.425	0.961	362,900	2,929,000	0.07	0.04
8	May	3,037,000	1,141,000	0.376	0.271	621,000	6,051,000	0.19	-0.05
9	Jun	4,054,000	1,564,000	0.386	0.427	948,900	8,467,000	0.13	0.05
10	Jul	2,190,000	1,007,000	0.460	1.133	655,400	5,275,000	0.01	0.09
11	Aug	1,083,000	421,800	0.389	0.946	438,400	2,390,000	0.15	0.17
12	Sep	671,400	308,100	0.459	1.953	284,800	2,117,000	-0.01	0.40

Lag-0 Season to Season Cross Correlations Site 20 and site 19

Season #	Month	Cross Corr. Coeff.
1	Oct	0.528
2	Nov	0.553
3	Dec	0.394
4	Jan	0.046
5	Feb	0.145
6	Mar	-0.078
7	Apr	-0.347
8	May	-0.120
9	Jun	0.325
10	Jul	0.613
11	Aug	0.549



Storage and Drought Statistics

Demand Level	1.00×mean
Longest Deficit	22
Max Deficit	16,181,417
Longest Surplus	6

Max Surplus	13,728,208
Storage Capacity	77,644,242
Rescaled Range	58.069
Hurst Coeff.	0.637

5.2 Stochastic Modeling and Generation of Streamflow Data

SAMS was used to model the annual and monthly flows of site 20 of Colorado River basin (refer to file Colorado_River.dat). Both annual and monthly data used in the following examples are transformed using logarithmic transformation and the transformation coefficients are shown in Appendix D.

5.2.1 Univariate ARMA(p,q) Model

SAMS was used to model the annual flows of site 20 with an ARMA(1,1) model. The MOM was used to estimate the model parameters. SAMS was also used to generate 100 samples each 98 years long using the estimated parameters. The following is a summary of the results of the model fitting and generation by using the ARMA(1,1) model.

Results of fitting an ARMA(1,1) model to the transformed and standardized annual flows of site 20:

Model: ARMA

Model Parameters

Current_Model: ARMA(1,1)

For Site(s): 20

Model Fitted To: Mean Subtracted Data

MEAN_AND_VARIANCE:

Mean: 15,076,300

Variance: 1.886×10^{13}

AICC: 3091.860

SIC: 3094.775

PARAMETERS:

White_Noise_Variance: 1.737×10^{13}

AR_PARAMETERS:

PHI(1): 0.352827

MA_PARAMETERS:

THT(1): 0.078648

Results of statistical analysis of the data generated from the ARMA(1,1) model:

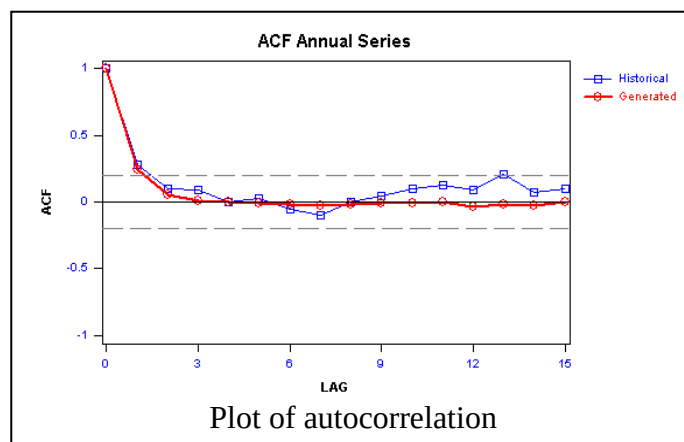
Site Number 20: IF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ

Statistics	Historical	Generated
Mean	15,080,000	15090000
StDev	4,343,000	4264000

CV	0.2881	0.2821
Skewness	0.1402	-0.04098
Min	5,525,000	4255000
Max	25,300,000	25550000
acf(1)	0.2804	0.2463
acf(2)	0.0989	0.05785

Correlation Structure

Lag	Historical	Generated
0	1	1
1	0.2804	0.2463
2	0.09893	0.05785
3	0.08769	0.005489
4	0.002523	0.0032
5	0.02924	-0.0124
6	-0.0581	-0.0216
7	-0.09822	-0.02472
8	0.001738	-0.01838
9	0.04812	-0.00682
10	0.09768	-0.01279



Storage and Drought Statistics

Statistics	Historical	Generated
Demand Level	1.00×mean	1.00×mean
Longest Deficit	22	7.6
Max Deficit	16,181,417	31780000
Longest Surplus	6	7.39
Max Surplus	13,728,208	32170000
Storage Capacity	77,644,242	61560000
Rescaled Range	58.069	13.78
Hurst Coeff.	0.637	0.6672

SAMS was also used to model the transformed and standardized annual flows of site 29 with an ARMA(2,2) model using the Approximate LS method. The results of modeling for this site are shown below:

Model: ARMA

Model Parameters

Current_Model: ARMA(2,2)

For Site(s): 29

Model Fitted To: Mean Subtracted Data

MEAN_AND_VARIANCE:

Mean: 1.64E+07

Variance: 2.05E+13

AICC: 3104.354

SIC: 3112.042

PARAMETERS:

White_Noise_Variance: 1.89E+13

AR_PARAMETERS:

PHI(1) PHI(2)

	-0.220024	0.487627
MA_PARAMETERS:		
	THT(1)	THT(2)
	-0.476987	0.338792

100 samples each 98 years long were generated using these estimated parameters. The statistical analysis results of the generated data are shown below:

Model: Univariate ARMA, (Statistical Analysis of Generated Data)

Site Number: 29

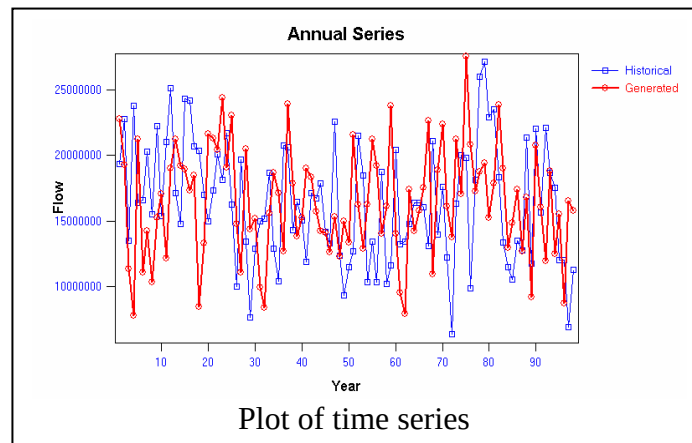
Statistics	Historical	Generated
Mean	1.64E+07	1.64E+07
StDev	4.53E+06	4.51E+06
CV	0.2767	0.2743
Skewness	0.1349	-0.05187
Min	6.34E+06	5.01E+06
Max	2.72E+07	2.73E+07
acf(1)	0.2694	0.2522
acf(2)	0.1173	0.09072

Correlation Structure

Lag	Historical	Generated
0	1	1
1	0.269	0.252
2	0.117	0.091
3	0.106	0.085
4	0.034	0.014
5	0.063	0.025
6	-0.034	-0.022
7	-0.088	-0.014
8	0.003	-0.028
9	0.051	-0.004
10	0.103	-0.021

Storage and Drought Statistics

Statistics	Historical	Generated
Demand Level	7	8.22
Longest Deficit	2.33E+07	3.76E+07
Max Deficit	6	7.8
Longest Surplus	3.78E+07	3.58E+07
Max Surplus	7.85E+07	6.78E+07
Storage Capacity	17.31	14.95
Rescaled Range	0.7327	0.6875
Hurst Coeff.	7	8.22



5.2.2 Univariate GAR(1) Model

An GAR(1) model was fitted to the annual data of site 20. Based on this model, the skewness coefficient of the historical data can be preserved without data transformation. The estimated parameters of the model are shown below:

Model: GAR

Model Parameters

Current_Model: GAR(1)

For Site(s): 20

Model Fitted To: Standardized Data

MEAN_AND_VARIANCE:

Mean: 1.50763e+007

Variance: 1.88614e+013

PARAMETERS:

lambda	alpha	beta	phi
-13.422091	13.167813	176.739581	0.302968

100 samples each 98 years long were generated using these estimated parameters. The statistical analysis results of the generated data are shown below:

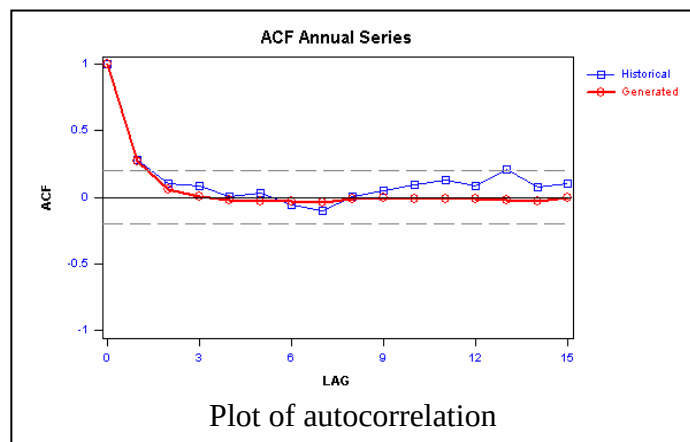
Model: Univariate GAR(1), (Statistical Analysis of Generated Data)

Site Number 20: IF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ

Statistics	Historical	Generated
Mean	15080000	15050000
StDev	4343000	4310000
CV	0.2881	0.2858
Skewness	0.1402	0.1388
Min	5525000	4869000
Max	25300000	26560000
acf(1)	0.2804	0.2752
acf(2)	0.09893	0.05583

Correlation Structure

Lag	Historical	Generated
0	1	1
1	0.2804	0.2752
2	0.09893	0.05583
3	0.08769	0.001677
4	0.002523	-0.02237
5	0.02924	-0.02995
6	-0.0581	-0.02983
7	-0.09822	-0.03643
8	0.001738	-0.01775
9	0.04812	-0.00772
10	0.09768	-0.01058



Storage and Drought Statistics

Statistics	Historical	Generated
Demand Level	1.00×mean	1.00×mean
Longest Deficit	5	7.38
Max Deficit	21770000	31470000
Longest Surplus	6	7.44
Max Surplus	36990000	33270000
Storage Capacity	72110000	63400000
Rescaled Range	16.6	14.44
Hurst Coeff.	0.7219	0.6806

5.2.3 Univariate PARMA(p,q) Model

A PARMA (1,1) model was fitted to the transformed and standardized monthly data of site 20 of the Yakima basin using MOM. Part of the modeling results obtained by SAMS are shown below:

Model:PARMA

Model Parameters

Current_Model: PARMA(1,1)

For Site(s): 1

Model Fitted To: Mean Subtracted Data

MEAN_AND_VARIANCE:

Season	Mean	Variance	AICC	AIC
1	580893	7.32E+10	2519.33	2522.25
2	480821	1.98E+10	2338.84	2341.75
3	382530	9.10E+09	2239.37	2242.29
4	356611	6.12E+09	2245.4	2248.31
5	393776	9.42E+09	2309.17	2312.09
6	645201	4.42E+10	2472.58	2475.5
7	1.20E+06	2.60E+11	2634.89	2637.81
8	3.04E+06	1.30E+12	2780.08	2783
9	4.05E+06	2.45E+12	2848.44	2851.36
10	2.19E+06	1.01E+12	2695.92	2698.84
11	1.08E+06	1.78E+11	2545.1	2548.01
12	671371	9.49E+10	2530.26	2533.18

PARAMETERS:

White_Noise_Variance:

Season

1	5.04E+10
2	7.99E+09
3	2.90E+09
4	3.08E+09
5	5.91E+09
6	3.13E+10
7	1.64E+11

8	7.21E+11
9	1.45E+12
10	3.06E+11
11	6.56E+10
12	5.64E+10

PAR_PARAMETERS:

Season PHI(1)

1	0.636097
2	0.510793
3	0.560785
4	0.602475
5	1.013047
6	1.733109
7	2.59168
8	2.226865
9	0.657275
10	0.465891
11	0.366904
12	0.45941

PMA_PARAMETERS:

Season THT(1)

1	0.27852
2	0.16926
3	0.00413
4	0.08044
5	0.65302
6	1.09952
7	2.05308
8	1.4291
9	-0.3606
10	-0.1168
11	0.1314
12	-0.0166

The estimated parameters were used to generate 100 samples of seasonal (12 seasons) data each sample 98 years long. The statistical analysis results of the generated data are shown below:

Model: Univariate PARMA, (Statistical Analysis of Generated Data)

Site Number: 20

Stats	Season 1		Season 2		Season 3		Season 4		Season 5		Season 6	
	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen
Mean	5.81E+05	5.81E+05	4.81E+05	4.81E+05	3.83E+05	3.83E+05	3.57E+05	3.57E+05	3.94E+05	3.93E+05	6.45E+05	6.43E+05
StDev	2.71E+05	2.66E+05	1.41E+05	1.40E+05	9.54E+04	9.45E+04	7.82E+04	7.71E+04	9.71E+04	9.64E+04	2.10E+05	2.06E+05

CV	0.4659	0.4572	0.2928	0.2915	0.2493	0.2466	0.2194	0.2156	0.2465	0.2449	0.326	0.3203
Skew	1.641	0.09918	1.215	-0.00575	1.223	-0.01048	0.59	0.009797	1.419	-0.03459	1.081	-0.01211
Min	1.94E+05	6487	1.81E+05	1.34E+05	2.27E+05	1.46E+05	2.00E+05	1.63E+05	2.53E+05	1.47E+05	2.80E+05	1.21E+05
Max	1.81E+06	1.25E+06	9.99E+05	8.30E+05	7.30E+05	6.21E+05	5.89E+05	5.52E+05	7.75E+05	6.37E+05	1.40E+06	1.17E+06
acf(1)	0.162	0.03165	0.3074	0.03939	0.5401	0.04022	0.5161	0.01767	0.2453	0.02729	0.2781	0.008951
acf(2)	0.2198	-0.00261	0.2829	-0.00458	0.3606	-0.01844	0.3645	-0.00886	0.01406	-0.01188	0.1519	-0.0054

Stats	Season 7		Season 8		Season 9		Season 10		Season 11		Season 12	
	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen
Mean	1.20E+06	1.20E+06	3.04E+06	3.03E+06	4.05E+06	4.06E+06	2.19E+06	2.20E+06	1.08E+06	1.08E+06	6.71E+05	6.71E+05
StDev	5.10E+05	5.01E+05	1.14E+06	1.13E+06	1.56E+06	1.54E+06	1.01E+06	9.80E+05	4.22E+05	4.16E+05	3.08E+05	3.02E+05
CV	0.4249	0.417	0.3756	0.3719	0.3858	0.3798	0.4598	0.446	0.3894	0.3845	0.4589	0.45
Skew	0.9605	0.01532	0.2713	0.02451	0.4266	-0.01703	1.133	0.085	0.9464	0.0802	1.953	0.06588
Min	3.63E+05	3.85E+04	6.21E+05	3.56E+05	9.49E+05	3.14E+05	6.55E+05	5.47E+04	4.38E+05	8.79E+04	2.85E+05	1.12E+04
Max	2.93E+06	2.44E+06	6.05E+06	5.89E+06	8.47E+06	7.90E+06	5.28E+06	4.70E+06	2.39E+06	2.19E+06	2.12E+06	1.44E+06
acf(1)	0.0659	0.009512	0.1908	0.04776	0.1275	0.04073	0.01063	0.01979	0.149	0.03047	-0.01465	0.01447
acf(2)	0.03616	-0.02452	-0.05232	-0.00213	0.04517	-0.01752	0.09168	-0.01678	0.1703	-0.02413	0.3992	-0.01069

Storage and Drought Statistics (for season 1)

Statistics	Historical	Generated
Demand Level	1.00×mean	1.00×mean
Longest Deficit	9	7.82
Max Deficit	4.04E+06	3.09E+06
Longest Surplus	1.79E+06	1.42E+06
Max Surplus	14.94	11.44
Storage Capacity	6	4.99
Rescaled Range	0.6949	0.6185
Hurst Coeff.	2.31E+06	1.48E+06

5.2.4 Multivariate MAR(p) Model

SAMS was also used to model the transformed and standardized annual data of sites 2, 6, 7 and 8 of the Colorado Rive basin using the MAR (1) model. The modeling results are shown below:

Model : MAR

Model Parameters

Current_Model: MAR(1)

For Site(s): 2 6 7 8

Model Fitted To: Standardized Data

MEAN_AND_VARIANCE:

Mean	Variance
3.58E+0	8.64E+1
6	1
2.36E+0	5.20E+1
6	1
	1.29E+1
813287	1

6.82E+0	3.83E+1
6	2

PARAMETERS:

White_Noise_Variance:

		0.59111	0.85335
0.911179	0.818236	4	4
		0.77416	0.87901
0.818236	0.904426	8	3
		0.92342	
0.591114	0.774168	9	0.75131
			0.88464
0.853354	0.879013	0.75131	3

Cholesky_of_White_Noise_Variance:

0.954557	0	0	0
0.857189	0.411889	0	0
		0.43691	
0.619255	0.590812	3	0
		0.08250	0.06136
0.893979	0.273627	3	4

AR_PARAMETERS:

PHI(1)	-	-	-	1.25979
-0.1776	-0.83115	-0.0085		8
				1.63507
-0.46771	-0.82542	-0.11557		8
		0.06664		1.50869
-0.39943	-0.98603	9		1
				2.15407
-0.63134	-1.151	-0.15781		6

These estimated parameters were used to generate 100 samples annual data each of 98 years long for the three sites. The statistical analysis result of the generated data is shown below:

Model: Multivariate AR (MAR), (Statistical Analysis of Generated Data)

Site Number: 2

Statistics	Historical	Generated
Mean	3.58E+06	3.58E+06
StDev	9.30E+05	9.17E+05
CV	0.2596	0.2558
Skewness	0.2507	0.008755
Min	1.62E+06	1.27E+06
Max	6.25E+06	5.90E+06
acf(1)	0.2611	0.2469
acf(2)	0.1245	0.04975

Correlation Structure

Lag	Historical	Generated
0	1	1

1	0.261	0.247
2	0.125	0.050
3	0.083	-0.009
4	-0.024	-0.015
5	0.055	-0.003
6	-0.053	-0.006
7	-0.145	-0.014
8	-0.013	-0.018
9	0.143	-0.031
10	0.163	-0.009

Storage and Drought Statistics

Statistics	Historical	Generated
Demand Level	6	7.2
Longest Deficit	4.83E+06	6.73E+06
Max Deficit	5	7.01
Longest Surplus	7.41E+06	6.41E+06
Max Surplus	1.70E+07	1.32E+07
Storage Capacity	18.23	14.15
Rescaled Range	0.746	0.6731
Hurst Coeff.	6	7.2

Site Number: 8

Statistics	Historical	Generated
Mean	6.83E+06	6.81E+06
StDev	1.96E+06	1.93E+06
CV	0.2866	0.2826
Skewness	0.2046	0.01049
Min	2.57E+06	2.03E+06
Max	1.25E+07	1.16E+07
acf(1)	0.2884	0.2588
acf(2)	0.07964	0.06486

Correlation Structure

Lag	Historical	Generated
0	1	1
1	0.288	0.259
2	0.080	0.065
3	0.051	0.001
4	-0.012	-0.006
5	0.032	-0.004
6	-0.087	-0.005
7	-0.175	-0.011
8	-0.024	-0.018
9	0.082	-0.028
10	0.103	-0.005

Storage and Drought Statistics

Statistics	Historical	Generated
Demand Level	5	7.51
Longest Deficit	9.71E+06	1.43E+07
Max Deficit	6	7.37
Longest Surplus	1.77E+07	1.44E+07

Max Surplus	3.16E+07	2.89E+07
Storage Capacity	16.13	14.62
Rescaled Range	0.7145	0.6819
Hurst Coeff.	5	7.51

5.2.5 Multivariate CARMA(p,q) Model

A CARMA(2,2) model was also fitted to sites 2, 6, 7 and 8 of the Yakima basin. The modeling results are shown below:

Model: CARMA

Model Parameters

Current_Model: CARMA(1,1)
For Site(s): 2 6 7 8
Model Fitted To: Mean Subtracted Data

MEAN_AND_VARIANCE:

Mean Variance
3.58E+06 8.64E+11
2.36E+06 5.20E+11
813287 1.29E+11
6.82E+06 3.83E+12

PARAMETERS:

White_Noise_Variance:
8.02E+11 5.68E+11 2.11E+11 1.60E+12
5.68E+11 4.85E+11 2.08E+11 1.28E+12
2.11E+11 2.08E+11 1.21E+11 5.52E+11
1.60E+12 1.28E+12 5.52E+11 3.51E+12

Cholesky_of_White_Noise_Variance:
895514 0 0 0
633977 288106 0 0
235294 205428 154532 0
1.79E+06 518898 161559 127078

AR_PARAMETERS:

PHI(1)
0.476986 0 0 0
0 0.288962 0 0
0 0 -0.085889 0
0 0 0 0.276098

MA_PARAMETERS:

THETA(1)
0.232579 0 0 0
0 0.03285 0 0
0 0 -0.330913 0
0 0 0 -0.01346

These estimated parameters were used to generate 100 samples annual data each of 98 years long for the three sites. The statistical analysis result of the generated data is shown below:

Model: Contemporaneous ARMA (CARMA), (Statistical Analysis of Generated Data)

Site Number: 2

Statistics	Historical	Generated
Mean	3.58E+06	3.59E+06

StDev	9.30E+05	9.24E+05
CV	0.2596	0.2568
Skewness	0.2507	-0.00927
Min	1.62E+06	1.26E+06
Max	6.25E+06	5.91E+06
acf(1)	0.2611	0.2477
acf(2)	0.1245	0.1032

Correlation Structure

Lag	Historical	Generated
0	1	1
1	0.261	0.248
2	0.125	0.103
3	0.083	0.038
4	-0.024	0.006
5	0.055	-0.001
6	-0.053	-0.017
7	-0.145	-0.012
8	-0.013	-0.034
9	0.143	-0.030
10	0.163	-0.011

Storage and Drought Statistics

Statistics	Historical	Generated
Demand Level	1.00×mean	1.00×mean
Longest Deficit	6	7.6
Max Deficit	4.83E+06	7.28E+06
Longest Surplus	5	7.56
Max Surplus	7.41E+06	7.23E+06
Storage Capacity	1.70E+07	1.28E+07
Rescaled Range	18.23	14.84
Hurst Coeff.	0.746	0.6864

Site Number: 8

Statistics	Historical	Generated
Mean	6.83E+06	6.84E+06
StDev	1.96E+06	1.94E+06
CV	0.2866	0.2832
Skewness	0.2046	0.01695
Min	2.57E+06	1.99E+06
Max	1.25E+07	1.18E+07
acf(1)	0.2884	0.272
acf(2)	0.07964	0.06459

Correlation Structure

Lag	Historical	Generated
0	1	1
1	0.288	0.272
2	0.080	0.065
3	0.051	0.007
4	-0.012	-0.010
5	0.032	-0.011
6	-0.087	-0.017
7	-0.175	-0.004

8	-0.024	-0.027
9	0.082	-0.025
10	0.103	-0.005

Storage and Drought Statistics

Statistics	Historical	Generated
Demand Level	1.00×mean	1.00×mean
Longest Deficit	5	7.57
Max Deficit	9.71E+06	1.47E+07
Longest Surplus	6	7.65
Max Surplus	1.77E+07	1.50E+07
Storage Capacity	3.16E+07	2.65E+07
Rescaled Range	16.13	14.63
Hurst Coeff.	0.7145	0.6842

5.2.6 Disaggregation Models

A spatial-temporal disaggregation modeling and generation example using SAMS based on multivariate data of the Colorado River basin is demonstrated here. In this example both annual and monthly data being modeled are transformed using logarithmic transformation. The stations' locations in the basin are shown in Fig. 5.1. In this example, the disaggregation modeling will be conducted for part of the Upper Colorado Basin. It can be seen from the map that the stations 8 and 16 control two major sources for the Upper Colorado Basin. Therefore both stations can be considered as key stations in this example. Further upstream, the stations 2, 6, 7, 11, 12, 13, 14, and 15 are the control stations for the tributaries. Therefore these stations are considered as the substations. Scheme 1 will be used to model the key stations so that the annual flows of the key stations will be added together to form one series of annual data as an index station. The index station data will be fitted with an ARMA(1,1) model and then a disaggregation model (either Valencia and Schaake or Mejia and Rousselle) will be used to disaggregate the annual flows of the index station into the annual flows at the key stations. The key station to substation disaggregation will be done using two groups. The first group contains key station 8 and substations 2, 6 and 7. The second group contains key station 16 and substations 11, 12, 13, 14, and 15. For temporal disaggregation, two group are used. The grouping is the same as the spatial grouping. The modeling results for the annual and monthly data are summarized below.

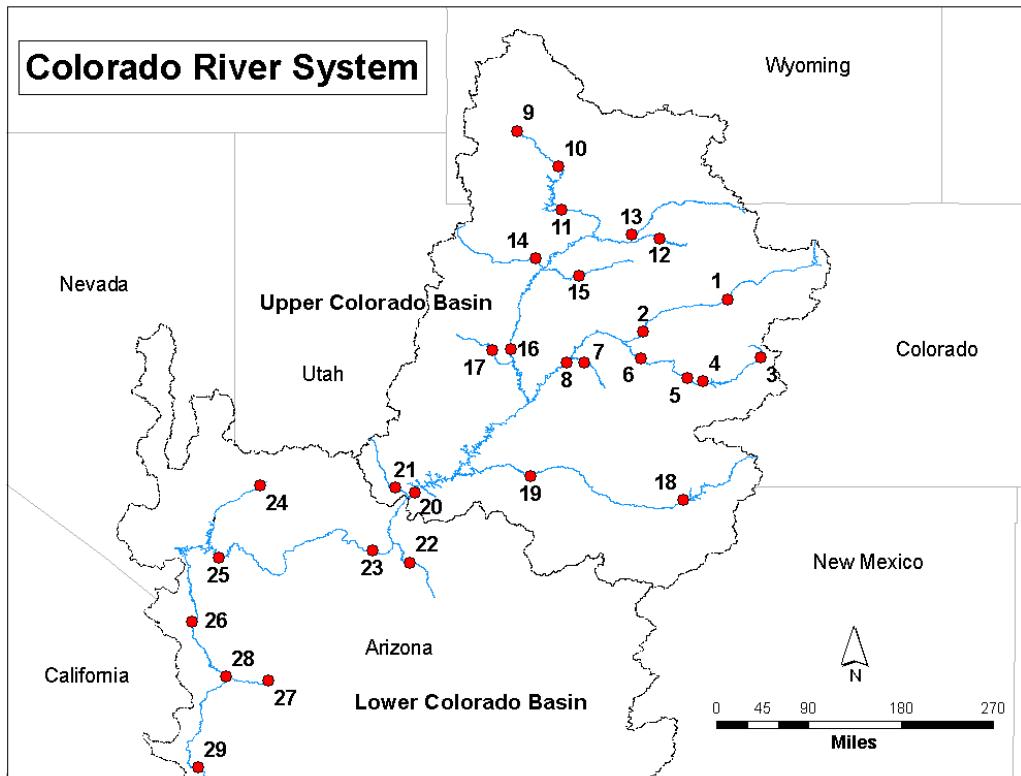


Figure 5.1: The location of the station in the Colorado River Basin

Seasonal (Spatial-Temporal) disaggregation

Model Parameters

Model Parameters

Current_Model: ARMA(1,0)

For Site(s): 8 16

Model Fitted To: Mean Subtracted Data

MEAN_AND_VARIANCE:

Mean: 1.22403e+007

Variance: 1.19578e+013

AICC: 3043.908

SIC: 3044.366

PARAMETERS:

White_Noise_Variance: 1.08825e+013

AR_PARAMETERS:

PHI(1)
 0.299867
 Keystations (2) : 8 16
 A_Matrix
 0.548354
 0.451646
 B_Matrix
 479486 0
 -479486 0.0497184
 G_Matrix
 2.29907e+011-2.29907e+011
 -2.29907e+011 2.29907e+011

 SPATIAL_DISAGGREGATION : # Groups = 2
 Group : 1

 Keystations (1) : 8
 Substations (3) : 2 6 7
 A_Matrix
 0.452577
 0.362358
 0.154347
 B_Matrix
 283537 0 0
 -64934.8 114533 0
 -156577 -26270.9 111572
 G_Matrix
 8.03931e+010-1.84114e+010-4.43953e+010
 -1.84114e+010 1.73344e+010 7.15838e+009
 -4.43953e+010 7.15838e+009 3.76549e+010
 Group : 2
 Keystations (1) : 16
 Substations (5) : 11 12 13 14 15
 A_Matrix
 0.351526
 0.215447
 0.093500

0.175401

0.087515

B_Matrix

244752	0	0	0	0
-93360.4	138228	0	0	0
-13778.5	-4861.83	56552.3	0	0
-9636.05	-62947.2	-13947.7	60399.3	0
-56008.6	20728.8	-24160.3	-7362.48	56760.4

G_Matrix

5.99037e+010-2.28502e+010-3.37232e+009-2.35845e+009-1.37082e+010
-2.28502e+010 2.78233e+010 6.14323e+008-7.80147e+009 8.0943e+009
-3.37232e+009 6.14323e+008 3.41165e+009-3.49965e+008-6.95385e+008
-2.35845e+009-7.80147e+009-3.49965e+008 7.89783e+009-8.72826e+008
-1.37082e+010 8.0943e+009-6.95385e+008-8.72826e+008 7.42632e+009

TEMPORAL_DISAGGREGATION : # Groups = 2

Group : 1

Keystations (4) : 2 6 7 8

Season : 1

A_Matrix

0.000000	-0.000000	0.000000	0.000000
0.000000	0.000001	0.000000	-0.000000
0.000001	0.000000	0.000002	-0.000001
0.000000	0.000000	0.000000	-0.000000

B_Matrix

0.165239	0	0	0
0.174246	0.188884	0	0
0.188922	0.0929113	0.388845	0
0.194451	0.0735582	0.0505985	0.0483824

C_Matrix

0.502	0.00601918	-0.0618478	0.2047
-0.00445861	0.202389	0.0441569	0.350722
-0.546917	0.0986539	0.413514	0.801098
0.0396133	-0.0925786	-0.00539379	0.701104

G_Matrix

0.027304	0.0287923	0.0312174	0.032131
0.0287923	0.0660387	0.0504684	0.0477763

0.0312174	0.0504684	0.195525	0.0632455
0.032131	0.0477763	0.0632455	0.0481231

Season : 2

A_Matrix

0.000000	0.000000	0.000000	-0.000000
-0.000000	0.000000	0.000000	-0.000000
0.000001	0.000001	0.000002	-0.000001
-0.000000	0.000000	0.000000	-0.000000

B_Matrix

0.115463	0	0	0
0.0683399	0.09938	0	0
0.191787	0.167487	0.515484	0
0.101526	0.0468169	0.0200979	0.0379594

C_Matrix

0.584598	0.295025	-0.0358156	-0.297984
0.195712	0.529944	-0.0559797	-0.104605
-1.11441	0.579704	-0.0267015	1.3718
0.101128	0.244169	-0.0635435	0.232122

G_Matrix

0.0133318	0.00789075	0.0221444	0.0117225
0.00789075	0.0145467	0.0297516	0.0115909
0.0221444	0.0297516	0.330558	0.0376727
0.0117225	0.0115909	0.0376727	0.0143442

Season : 3

A_Matrix

-0.000000	-0.000000	-0.000000	0.000000
0.000000	0.000000	0.000000	-0.000000
-0.000000	-0.000000	0.000001	0.000000
-0.000000	-0.000000	-0.000000	0.000000

B_Matrix

0.110573	0	0	0
0.0407358	0.117442	0	0
0.121705	0.14416	0.234975	0
0.0829946	0.0444141	0.0273941	0.0411484

C_Matrix

0.784109	0.221403	0.0265706	-0.251361
----------	----------	-----------	-----------

0.0745275	0.618018	-0.00898853	-0.0245939
-0.255243	0.622565	0.166933	0.428793
0.118908	0.125865	0.00968396	0.46957

G_Matrix

0.0122264	0.00450428	0.0134573	0.00917698
0.00450428	0.015452	0.0218882	0.00859692
0.0134573	0.0218882	0.0908075	0.0229405
0.00917698	0.00859692	0.0229405	0.0113044

Season : 4

A_Matrix

0.000000	0.000000	-0.000000	-0.000000
-0.000000	-0.000000	-0.000000	0.000000
-0.000000	-0.000000	0.000000	0.000000
-0.000000	-0.000000	-0.000000	0.000000

B_Matrix

0.104407	0	0	0
0.0280713	0.094133	0	0
0.0829529	0.0553162	0.160515	0
0.0817014	0.0393486	0.0192638	0.0520257

C_Matrix

0.611582	0.295256	0.0942009	-0.462559
0.368336	0.956194	0.0302847	-0.467767
0.247618	0.509665	0.620374	-0.59526
0.100135	0.343352	0.0268103	0.186882

G_Matrix

0.0109008	0.00293083	0.00866084	0.00853017
0.00293083	0.00964903	0.00753568	0.00599747
0.00866084	0.00753568	0.0357061	0.0120461
0.00853017	0.00599747	0.0120461	0.0113012

Season : 5

A_Matrix

-0.000000	-0.000000	-0.000000	0.000000
0.000000	0.000000	-0.000000	-0.000000
-0.000000	-0.000000	0.000000	0.000000
-0.000000	-0.000000	-0.000000	0.000000

B_Matrix

0.103777	0	0	0
----------	---	---	---

0.0580655	0.1107	0	0
0.120727	0.111002	0.181246	0
0.0845675	0.0520657	0.0392643	0.0386531

C_Matrix

0.508873	0.22648	-0.0207566	0.004536
0.00822624	0.736931	-0.0270794	-0.0161657
-0.348239	0.307361	0.599449	-0.277384
-0.0927785	0.189521	0.064263	0.385511

G_Matrix

0.0107697	0.00602586	0.0125287	0.00877616
0.00602586	0.0156261	0.0192979	0.0106741
0.0125287	0.0192979	0.0597464	0.0231054
0.00877616	0.0106741	0.0231054	0.0128982

Season : 6

A_Matrix

-0.000000	-0.000000	0.000000	0.000000
-0.000000	0.000000	0.000000	-0.000000
-0.000000	-0.000000	0.000000	0.000000
-0.000000	-0.000000	-0.000000	0.000000

B_Matrix

0.131858	0	0	0
0.157242	0.1192	0	0
0.307016	0.167227	0.220716	0
0.169115	0.0640698	0.027466	0.0335856

C_Matrix

0.832518	0.516669	-0.18311	-0.326776
0.112288	1.35199	-0.216668	-0.359371
0.35402	1.41921	0.271095	-1.04644
0.306075	0.870295	-0.18714	-0.15756

G_Matrix

0.0173864	0.0207336	0.0404824	0.0222991
0.0207336	0.0389337	0.0682094	0.0342292
0.0404824	0.0682094	0.170939	0.0686975
0.0222991	0.0342292	0.0686975	0.0345873

Season : 7

A_Matrix

0.000000	0.000000	0.000000	-0.000000
----------	----------	----------	-----------

0.000000	0.000001	0.000000	-0.000000
-0.000000	0.000000	0.000001	0.000000
-0.000000	0.000000	0.000000	-0.000000

B_Matrix

0.299458	0	0	0
0.355101	0.199638	0	0
0.278549	0.157015	0.266149	0
0.317119	0.112407	0.0616115	0.0401153

C_Matrix

0.261602	-0.738159	0.10459	1.0228
-0.314722	-0.758881	0.0986936	1.72177
-0.70911	-1.28352	0.240206	1.97092
-0.240315	-0.905514	0.0894443	1.66252

G_Matrix

0.0896748	0.106338	0.0834137	0.0949636
0.106338	0.165952	0.130259	0.13505
0.0834137	0.130259	0.173079	0.122381
0.0949636	0.13505	0.122381	0.118605

Season : 8

A_Matrix

0.000000	-0.000000	0.000000	-0.000000
0.000000	0.000001	-0.000000	-0.000000
-0.000000	0.000000	0.000001	0.000000
0.000000	0.000000	-0.000000	0.000000

B_Matrix

0.235985	0	0	0
0.186891	0.139408	0	0
0.095993	0.110841	0.144337	0
0.195934	0.0653626	0.0174017	0.0278394

C_Matrix

-0.214358	-0.0354761	-0.0495295	0.544537
-0.494318	0.0251764	0.183358	0.339971
-0.145423	0.214719	0.641238	-0.781904
-0.317377	0.0185208	0.118834	0.303672

G_Matrix

0.0556888	0.0441035	0.0226529	0.0462374
-----------	-----------	-----------	-----------

0.0441035	0.054363	0.0333924	0.0457304
0.0226529	0.0333924	0.0423335	0.0285648
0.0462374	0.0457304	0.0285648	0.0437401

Season : 9

A_Matrix

0.000001	-0.000001	0.000000	0.000000
0.000000	-0.000000	-0.000000	0.000000
0.000000	-0.000000	0.000000	0.000000
0.000000	-0.000000	0.000000	0.000000

B_Matrix

0.143215	0	0	0
0.131382	0.0852989	0	0
0.0904204	0.0745078	0.123244	0
0.134862	0.0353906	0.0102286	0.0146826

C_Matrix

-0.313999	1.0291	-0.00447895	-0.85982
-0.377288	1.13627	0.216743	-1.08805
0.140012	0.68938	0.640143	-1.54423
-0.408819	0.98173	0.0904113	-0.779671

G_Matrix

0.0205106	0.0188159	0.0129496	0.0193143
0.0188159	0.0245371	0.018235	0.0207372
0.0129496	0.018235	0.0289164	0.0160918
0.0193143	0.0207372	0.0160918	0.0197604

Season : 10

A_Matrix

0.000000	-0.000000	-0.000000	-0.000000
0.000000	-0.000000	-0.000000	0.000000
-0.000000	0.000000	-0.000000	0.000000
0.000000	-0.000000	-0.000000	0.000000

B_Matrix

0.196377	0	0	0
0.121377	0.117237	0	0
0.152896	0.0714537	0.15435	0
0.164305	0.0495302	0.00952933	0.022087

C_Matrix

1.02208	1.42797	0.358112	-2.34311
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0.474762	1.74809	0.3798	-2.37985
1.82844	0.00808892	1.36039	-2.79386
0.58955	1.33452	0.35718	-1.92405

G_Matrix

0.0385639	0.0238356	0.0300253	0.0322657
0.0238356	0.0284769	0.0269351	0.0257496
0.0300253	0.0269351	0.0523067	0.0301316
0.0322657	0.0257496	0.0301316	0.0300281

Season : 11

A_Matrix

-0.000000	-0.000000	-0.000000	0.000000
0.000000	0.000000	-0.000000	-0.000000
-0.000000	0.000000	-0.000001	0.000000
-0.000000	-0.000000	-0.000000	0.000000

B_Matrix

0.154498	0	0	0
0.133507	0.126207	0	0
0.149137	0.111312	0.199762	0
0.153733	0.0615936	0.0215073	0.0245934

C_Matrix

1.53006	0.694829	0.184828	-2.00602
0.254645	0.490447	0.244753	-0.506946
0.636679	-0.364887	1.03079	-0.670124
0.567432	0.352455	0.215191	-0.671244

G_Matrix

0.0238695	0.0206265	0.0230413	0.0237514
0.0206265	0.0337523	0.0339591	0.028298
0.0230413	0.0339591	0.074537	0.0340797
0.0237514	0.028298	0.0340797	0.0284951

Season : 12

A_Matrix

-0.000000	-0.000000	-0.000000	0.000000
-0.000000	-0.000000	-0.000001	0.000000
-0.000000	-0.000000	-0.000000	0.000000
-0.000000	-0.000000	-0.000001	0.000000

B_Matrix

0.173759	0	0	0
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0.206504	0.173242	0	0
0.259491	0.175559	0.18789	0
0.206645	0.0929467	0.0165958	0.0321004

C_Matrix

1.09253	0.991814	0.208764	-1.78649
1.11559	1.62944	0.462521	-2.57501
0.528264	1.12795	1.19301	-2.099
0.781725	1.0232	0.369226	-1.63444

G_Matrix

0.0301922	0.0358819	0.0450889	0.0359064
0.0358819	0.0726567	0.0840001	0.0587752
0.0450889	0.0840001	0.133459	0.0730583
0.0359064	0.0587752	0.0730583	0.052647

Group : 2

Keystations (6) : 11 12 13 14 15 16

Season : 1

A_Matrix

-0.000000	-0.000000	0.000000	-0.000000	0.000000	0.000000
-0.000000	-0.000000	0.000000	-0.000000	0.000000	0.000000
-0.000001	-0.000001	0.000002	-0.000000	0.000001	0.000000
-0.000001	-0.000001	0.000001	0.000000	0.000001	0.000000
-0.000000	-0.000000	0.000000	-0.000000	0.000001	0.000000
-0.000000	-0.000001	0.000000	-0.000000	0.000001	0.000000

B_Matrix

0.285005	0	0	0	0	0
0.147273	0.27085	0	0	0	0
0.20126	0.164535	0.415564	0	0	0
0.109297	0.186816	0.187282	0.340697	0	0
0.0578085	0.0919089	0.0436934	0.0166099	0.105877	0
0.154485	0.130975	0.0888181	0.083933	0.0169512	0.0682913

C_Matrix

0.847036	-0.139999	0.0169278	-5.119e-006	0.0499056	0.208286
-0.164877	0.492869	0.00705454	-3.66774e-007	0.315733	0.0184223
-0.126584	-0.129972	0.366793	-4.69759e-006	0.611799	0.434272
-0.0293906	0.332623	-0.0957983	-1.97631e-006	-0.16423	0.954438
0.0467824	0.106837	-0.038057	5.9042e-007	0.493149	-0.204799
0.0806382	0.0993473	-0.0335549	-3.75861e-006	0.127337	0.574945

G_Matrix

0.0812281	0.0419737	0.0573602	0.0311502	0.0164757	0.0440291
0.0419737	0.0950493	0.0742047	0.0666956	0.0334072	0.0582263
0.0573602	0.0742047	0.240271	0.130563	0.0449142	0.0895514
0.0311502	0.0666956	0.130563	0.197995	0.0373302	0.0865827
0.0164757	0.0334072	0.0449142	0.0373302	0.0251839	0.028038
0.0440291	0.0582263	0.0895514	0.0865827	0.028038	0.0609046

Season : 2

A_Matrix

0.000000	-0.000000	0.000000	-0.000001	0.000000	0.000000
0.000000	0.000000	-0.000001	-0.000000	0.000000	0.000000
-0.000000	-0.000001	0.000002	-0.000001	0.000000	0.000000
-0.000000	-0.000000	-0.000000	-0.000000	-0.000000	0.000000
0.000000	-0.000000	-0.000000	-0.000000	0.000000	0.000000
-0.000000	-0.000000	0.000000	-0.000001	0.000000	0.000000

B_Matrix

0.208608	0	0	0	0	0
0.0382309	0.130014	0	0	0	0
0.0986463	0.108202	0.436169	0	0	0
0.0443932	0.062832	0.0758254	0.179415	0	0
0.0196362	0.046147	0.018143	0.0264187	0.100145	0
0.0870833	0.0562514	0.0625358	0.052854	0.0303199	0.0555294

C_Matrix

0.525674	0.0310611	-0.0515085	-0.0540612	0.0659373	0.197631
0.0927287	0.538716	0.0192426	0.0312471	0.187425	-0.125084
-0.139031	-0.0131704	0.567466	-0.00831652	-0.545995	0.446387
0.0580618	-0.242813	-0.0438333	0.123865	0.0908805	0.678126
0.044274	0.0295561	-0.0462856	0.0572508	0.610288	-0.102927
0.114365	0.00689524	-0.0463633	0.0399899	0.0472178	0.454384

G_Matrix

0.0435174	0.00797528	0.0205784	0.00926079	0.00409628	0.0181663
0.00797528	0.0183654	0.0178392	0.00986626	0.00675048	0.0106428
0.0205784	0.0178392	0.211683	0.0442505	0.0148437	0.0419532
0.00926079	0.00986626	0.0442505	0.0438578	0.00988683	0.0216249
0.00409628	0.00675048	0.0148437	0.00988683	0.0135713	0.00987313
0.0181663	0.0106428	0.0419532	0.0216249	0.00987313	0.0214548

Season : 3

A_Matrix

-0.000000	-0.000001	-0.000000	-0.000001	0.000000	0.000000
0.000000	0.000000	0.000000	-0.000000	0.000000	-0.000000
0.000002	0.000001	0.000001	-0.000001	0.000001	-0.000001
-0.000000	-0.000001	-0.000001	-0.000001	-0.000001	0.000001
-0.000000	0.000000	-0.000000	-0.000000	-0.000001	0.000000
-0.000000	-0.000001	-0.000000	-0.000001	-0.000000	0.000001

B_Matrix

0.195419	0	0	0	0	0
0.0539538	0.171069	0	0	0	0
0.0647505	-0.0324771	0.661019	0	0	0
0.0287713	0.0661534	0.0271663	0.190175	0	0
0.0128477	0.0508977	0.0134093	0.0331178	0.154288	0
0.09405	0.0780211	0.0630443	0.0913526	0.0328096	0.112688

C_Matrix

0.619496	0.00333415	0.0118057	0.217456	0.0553685	-0.291704
0.198998	0.821329	0.0174582	0.077666	-0.0427117	-0.306024
-0.462329	0.215444	0.146556	0.478307	-1.06401	1.46074
0.43366	0.0531735	0.134351	0.622475	0.140597	-0.757733
0.290483	-0.0916578	0.0609616	0.0813061	1.16455	-0.582619
0.298901	0.138436	0.0623818	0.439362	0.197812	-0.386929

G_Matrix

0.0381888	0.0105436	0.0126535	0.00562248	0.00251068	0.0183792
0.0105436	0.0321755	-0.00206228	0.0128691	0.00940019	0.0184213
0.0126535	-0.00206228	0.442194	0.017672	0.00804268	0.0452293
0.00562248	0.0128691	0.017672	0.0421087	0.0103992	0.026953
0.00251068	0.00940019	0.00804268	0.0103992	0.027837	0.0141123
0.0183792	0.0184213	0.0452293	0.026953	0.0141123	0.0410277

Season : 4

A_Matrix

0.000000	-0.000000	0.000001	-0.000000	0.000000	-0.000000
0.000000	0.000000	0.000000	-0.000000	-0.000000	-0.000000
-0.000000	0.000001	-0.000002	-0.000000	-0.000003	0.000000
-0.000000	0.000000	-0.000000	0.000000	-0.000001	0.000000
0.000000	0.000000	-0.000001	0.000000	0.000000	-0.000000
-0.000000	0.000000	-0.000000	-0.000000	-0.000001	0.000000

B_Matrix

0.199906	0	0	0	0	0
0.0434595	0.15611	0	0	0	0
0.125496	0.0637498	0.645358	0	0	0
0.0517408	0.0502824	0.0530494	0.174034	0	0
0.0101648	0.0695764	0.0429212	0.017143	0.15085	0
0.0949124	0.0706154	0.0959296	0.0553493	0.0286384	0.112994

C_Matrix

0.456372	0.169639	0.0270982	0.173846	0.0719613	-0.297016
-0.0348421	0.647522	0.009666	-0.0626382	0.165722	0.0750104
0.214207	-0.130022	0.119922	0.0364644	0.1203	0.90666
0.134272	-0.177345	0.0595539	0.995467	0.123273	-0.38962
0.0579566	-0.0978169	0.0117883	-0.0598046	0.761261	0.0188213
0.0579642	0.0972999	0.0379637	0.0945716	0.212726	0.124871

G_Matrix

0.0399623	0.00868779	0.0250873	0.0103433	0.00203199	0.0189735
0.00868779	0.0262589	0.0154059	0.0100982	0.0113033	0.0151486
0.0250873	0.0154059	0.436301	0.0439346	0.0334106	0.0783218
0.0103433	0.0100982	0.0439346	0.0383074	0.0092848	0.0231832
0.00203199	0.0113033	0.0334106	0.0092848	0.0298362	0.0152643
0.0189735	0.0151486	0.0783218	0.0231832	0.0152643	0.0398487

Season : 5

A_Matrix

0.000000	-0.000000	-0.000000	-0.000000	0.000000	-0.000000
-0.000000	-0.000001	-0.000000	-0.000001	0.000001	0.000000
0.000000	-0.000002	0.000002	-0.000002	0.000001	0.000001
-0.000000	-0.000000	0.000000	-0.000000	0.000000	0.000000
0.000000	-0.000000	0.000000	-0.000000	0.000001	0.000000
-0.000000	-0.000001	0.000000	-0.000001	0.000000	0.000000

B_Matrix

0.159596	0	0	0	0	0
0.101062	0.167522	0	0	0	0
0.153032	0.0747625	0.570052	0	0	0
0.05178	0.0442827	0.0671975	0.158345	0	0
0.105814	0.0756747	0.0339011	0.0231095	0.160107	0
0.129943	0.07169	0.0955014	0.0431183	0.057536	0.106023

C_Matrix

0.685409	-0.076842	-0.0135456	0.0336975	-0.125412	0.0415845
0.00396154	0.717766	0.0280413	-0.148321	-0.139701	0.112067
-0.528005	0.689232	0.384568	0.469356	-0.743027	-0.280061
0.148998	0.147765	-0.0119671	0.712176	0.156593	-0.271624
-0.0556917	0.0836149	0.0118048	-0.0335969	0.474309	0.0363686
0.172447	0.192215	0.00116738	0.0776268	0.0232764	0.275925

G_Matrix

0.0254709	0.0161291	0.0244233	0.00826388	0.0168875	0.0207383
0.0161291	0.0382773	0.0279901	0.0126513	0.023371	0.025142
0.0244233	0.0279901	0.353968	0.0495408	0.0411759	0.0796859
0.00826388	0.0126513	0.0495408	0.0342309	0.0147675	0.0231481
0.0168875	0.023371	0.0411759	0.0147675	0.0442408	0.0326208
0.0207383	0.025142	0.0796859	0.0231481	0.0326208	0.0475555

Season : 6

A_Matrix

0.000000	-0.000000	0.000000	-0.000000	0.000001	-0.000000
0.000001	0.000001	0.000000	0.000001	0.000001	-0.000001
0.000001	-0.000000	0.000003	0.000001	0.000003	-0.000001
-0.000000	-0.000000	0.000000	0.000000	0.000000	0.000000
-0.000000	-0.000000	-0.000000	-0.000001	0.000001	0.000000
-0.000000	-0.000000	0.000000	-0.000001	0.000001	0.000000

B_Matrix

0.280219	0	0	0	0	0
0.19376	0.29	0	0	0	0
0.17493	0.159583	0.493805	0	0	0
0.0916778	0.0589661	0.0420819	0.188576	0	0
0.113837	0.0912975	0.00373795	0.066093	0.168548	0
0.194183	0.0937813	0.0610258	0.0839134	0.0383433	0.0764871

C_Matrix

0.740709	0.0601027	-0.0127117	0.0643772	-0.295936	-0.00685265
0.369643	0.800145	-0.0359232	0.241326	-0.436766	-0.175398
-0.0189178	0.0341422	0.09058	0.697201	-0.364466	-0.577036
0.27127	0.0339157	0.033778	0.486336	-0.0936392	-0.215272
0.124449	0.0720405	0.00407665	0.13609	0.0416237	-0.241706
0.391925	0.160908	-0.0189926	0.239589	-0.264417	-0.0780854

G_Matrix

0.0785225	0.0542952	0.0490187	0.0256898	0.0318993	0.0544137
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0.0542952	0.121643	0.0801735	0.0348637	0.0485334	0.0648215
0.0490187	0.0801735	0.299911	0.0462274	0.0363288	0.0790691
0.0256898	0.0348637	0.0462274	0.0492137	0.0284407	0.0417243
0.0318993	0.0485334	0.0363288	0.0284407	0.0540847	0.0429041
0.0544137	0.0648215	0.0790691	0.0417243	0.0429041	0.064588

Season : 7

A_Matrix

0.000000	-0.000000	0.000001	0.000000	0.000001	-0.000000
-0.000000	0.000000	-0.000001	-0.000001	-0.000001	0.000001
-0.000000	0.000000	0.000002	0.000001	-0.000002	-0.000000
0.000000	0.000000	0.000001	0.000002	0.000000	-0.000000
-0.000000	-0.000000	0.000000	-0.000000	0.000002	-0.000000
-0.000000	-0.000000	0.000000	-0.000000	0.000000	0.000000

B_Matrix

0.265269	0	0	0	0	0
0.196505	0.264928	0	0	0	0
0.160227	0.0622116	0.736595	0	0	0
0.269947	0.17689	0.155451	0.279643	0	0
0.0878972	0.134956	0.039934	0.0325675	0.148683	0
0.214783	0.123925	0.0737765	0.035938	0.0249915	0.0579129

C_Matrix

0.390164	-0.0973805	-0.0221249	0.469146	0.385594	-0.503603
0.37239	0.365767	0.0127926	0.416421	0.48238	-0.884575
-0.387133	-0.229884	-0.19923	0.20542	-0.670378	2.05003
0.417304	-0.101802	0.0585601	0.835012	0.204766	-0.203654
0.258418	0.178543	-0.0361912	0.286109	0.264855	-0.544883
0.342412	0.0110104	-0.00740742	0.518697	0.28329	-0.456308

G_Matrix

0.0703676	0.0521266	0.0425032	0.0716086	0.0233164	0.0569752
0.0521266	0.108801	0.0479669	0.099909	0.0530259	0.075037
0.0425032	0.0479669	0.572116	0.168762	0.0518945	0.0964669
0.0716086	0.099909	0.168762	0.206527	0.062915	0.101419
0.0233164	0.0530259	0.0518945	0.062915	0.0507011	0.0434356
0.0569752	0.075037	0.0964669	0.101419	0.0434356	0.072202

Season : 8

A_Matrix

-0.000000	-0.000000	0.000000	-0.000001	-0.000000	0.000000
-0.000000	0.000001	-0.000000	-0.000000	-0.000000	0.000000
-0.000000	0.000001	0.000004	-0.000002	-0.000003	0.000000
0.000000	0.000001	0.000000	0.000001	0.000001	-0.000000
-0.000000	-0.000000	0.000000	-0.000000	0.000001	0.000000
-0.000000	-0.000000	-0.000000	-0.000001	-0.000000	0.000000

B_Matrix

0.237447	0	0	0	0	0
0.117654	0.124301	0	0	0	0
0.0251154	0.000878296	0.460776	0	0	0
0.306972	0.163429	0.00353014	0.314666	0	0
0.139078	0.137868	0.0132099	0.000165088	0.100435	0
0.171747	0.0985993	0.0193738	0.00415752	0.00879976	0.0506578

C_Matrix

0.837814	0.377201	0.0465612	0.172505	-0.0572134	-0.897734
0.347186	0.062265	0.0130696	0.0441428	-0.0311445	-0.312111
0.135354	-0.092551	0.22483	0.722157	0.0387132	-0.932959
0.826371	0.252456	0.111369	0.52824	-0.32659	-0.893197
0.357567	0.081522	0.0257244	0.030719	0.128037	-0.364023
0.481888	0.164018	0.0106008	0.114589	-0.0841181	-0.380957

G_Matrix

0.0563812	0.0279367	0.00596357	0.0728897	0.0330236	0.0407808
0.0279367	0.0292933	0.00306411	0.056431	0.0335003	0.0324628
0.00596357	0.00306411	0.212946	0.00947986	0.00970089	0.0133271
0.0728897	0.056431	0.00947986	0.219968	0.0653232	0.0702121
0.0330236	0.0335003	0.00970089	0.0653232	0.048612	0.0386203
0.0407808	0.0324628	0.0133271	0.0702121	0.0386203	0.0422551

Season : 9

A_Matrix

0.000001	-0.000000	0.000001	-0.000000	0.000000	-0.000000
0.000000	0.000000	0.000001	-0.000000	-0.000000	0.000000
0.000000	0.000000	0.000002	-0.000000	-0.000001	0.000000
0.000001	-0.000000	0.000001	0.000001	0.000000	0.000000
0.000000	-0.000000	0.000001	-0.000001	0.000001	0.000000
0.000000	-0.000000	0.000001	-0.000000	-0.000000	0.000000

B_Matrix

0.193892	0	0	0	0	0
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0.142062	0.162094	0	0	0	0
0.187929	0.0846477	0.315797	0	0	0
0.226906	0.109477	0.0015968	0.157789	0	0
0.139317	0.167067	-0.0173124	0.0075406	0.106178	0
0.164551	0.0788464	0.0270704	0.0148108	0.0205552	0.0349233

C_Matrix

-0.335414	0.690554	-0.0471818	0.136273	-0.123464	-0.379109
-0.0990891	1.21683	-0.00647378	0.212881	-0.338529	-1.06975
-0.306851	0.59397	0.0330352	0.209438	0.21707	-0.698624
-0.464655	0.727712	-0.0536591	0.485781	-0.251305	-0.753889
0.0894981	1.03736	0.00456233	0.138789	0.00734894	-1.12235
-0.182701	0.877611	-0.0295626	0.147414	-0.123517	-0.674348

G_Matrix

0.0375941	0.0275448	0.036438	0.0439952	0.0270125	0.0319052
0.0275448	0.0464562	0.0404186	0.0499804	0.0468723	0.0361571
0.036438	0.0404186	0.14221	0.0524135	0.0348564	0.046147
0.0439952	0.0499804	0.0524135	0.0883713	0.0510641	0.0483498
0.0270125	0.0468723	0.0348564	0.0510641	0.0589509	0.037923
0.0319052	0.0361571	0.046147	0.0483498	0.037923	0.0358882

Season : 10

A_Matrix

-0.000000	-0.000000	-0.000001	-0.000000	-0.000001	0.000000
0.000000	0.000000	-0.000001	-0.000000	-0.000000	0.000000
-0.000001	-0.000000	0.000002	0.000001	-0.000000	0.000000
0.000000	0.000000	-0.000001	0.000002	-0.000001	-0.000000
0.000000	0.000000	-0.000001	0.000000	0.000001	-0.000000
0.000000	-0.000000	-0.000000	0.000000	-0.000000	0.000000

B_Matrix

0.231415	0	0	0	0	0
0.18348	0.192806	0	0	0	0
0.0858458	0.0396418	0.473277	0	0	0
0.166964	0.0390491	0.0485683	0.20094	0	0
0.0962816	0.129922	0.0346361	-0.00264773	0.124188	0
0.175676	0.0585303	0.0678499	0.0138436	0.0227715	0.0356419

C_Matrix

0.755708	0.63039	0.0197041	0.247076	-0.188985	-1.16575
-0.492869	1.12112	0.0271109	0.171024	-0.338696	-0.000672795

0.826359	0.209297	-0.678656	0.11915	0.390772	0.00750674
0.441224	0.704869	0.0660505	0.593249	-0.0460397	-1.38818
-0.473111	0.265123	-0.030752	0.136676	0.0250877	0.315268
0.213091	0.533318	-0.114337	0.161619	-0.134471	-0.281703

G_Matrix

0.0535528	0.0424599	0.019866	0.038638	0.022281	0.0406541
0.0424599	0.0708391	0.0233941	0.0381635	0.0427155	0.043518
0.019866	0.0233941	0.232933	0.0388674	0.0298082	0.0495131
0.038638	0.0381635	0.0388674	0.0721378	0.0222991	0.0376943
0.022281	0.0427155	0.0298082	0.0222991	0.0427792	0.0296601
0.0406541	0.043518	0.0495131	0.0376943	0.0296601	0.040872

Season : 11

A_Matrix

0.000000	-0.000000	-0.000000	-0.000000	0.000001	-0.000000
0.000000	0.000000	0.000000	0.000001	0.000000	-0.000000
0.000000	0.000000	-0.000001	0.000001	0.000001	-0.000000
0.009591	-0.001738	-0.022736	0.008301	0.022775	0.000608
0.000000	0.000000	-0.000000	0.000001	0.000001	-0.000000
0.000000	-0.000000	-0.000000	0.000000	0.000001	-0.000000

B_Matrix

0.153232	0	0	0	0	0
0.081007	0.153844	0	0	0	0
0.0477256	0.0127606	0.308689	0	0	0
4767	3245.48	2403.21	11667.1	0	0
0.0410478	0.0728789	0.0370756	0.0088064	0.10552	0
0.118075	0.0533969	0.0411878	0.0327145	0.0251705	0.0521391

C_Matrix

0.415481	0.225944	0.0365169	0.18394	-0.0204386	-0.492665
0.279419	0.473847	-0.111407	-0.0173875	-0.00543361	-0.318024
0.730571	-0.318439	1.34745	0.211189	0.777978	-1.98934
-12991.4	-4032.34	1687.79	38543.1	-16578.6	5307.71
0.289613	0.0722257	0.161461	0.0238898	0.383126	-0.657843
0.179116	0.0568042	0.116174	0.171603	0.0491121	-0.205721

G_Matrix

0.0234801	0.0124129	0.00731309	730.458	0.00628984	0.0180929
0.0124129	0.0302301	0.00582925	885.458	0.0145371	0.0177797

0.00731309	0.00582925	0.0977296	1010.77	0.0143339	0.0190308
730.458	885.458	1010.77	1.75155e+008	624.048	1216.83
0.00628984	0.0145371	0.0143339	624.048	0.0195829	0.0132094
0.0180929	0.0177797	0.0190308	1216.83	0.0132094	0.0229116

Season : 12

A_Matrix

0.000000	-0.000000	0.000000	0.000000	0.000000	-0.000000
0.000000	0.000000	0.000000	0.000000	0.000001	-0.000000
0.000001	0.000001	0.000001	0.000001	0.000001	-0.000001
0.015687	0.010801	-0.004570	0.026539	0.028536	-0.011774
0.000000	0.000000	0.000000	0.000000	0.000001	-0.000000
0.000000	-0.000000	0.000000	0.000000	0.000000	-0.000000

B_Matrix

0.165769	0	0	0	0	0
0.0834139	0.250452	0	0	0	0
0.144306	0.0970703	0.298996	0	0	0
6400.72	4228.54	3816.04	12363.3	0	0
0.0640092	0.112307	0.0430467	0.0226271	0.12529	0
0.144699	0.0897134	0.0855161	0.0757653	0.0312869	0.0533674

C_Matrix

0.845586	0.131461	0.00777954	2.88145e-006	0.288458	-0.593897
0.321292	0.677548	-0.0649811	7.13831e-006	0.509709	-0.943876
0.48584	-0.109588	0.949835	4.05525e-006	0.351754	-1.17103
16486.2	15548.9	5606.81	0.519911	4572.32	-50758.2
0.379453	-0.0466184	0.0468307	4.03325e-006	0.585432	-0.790496
0.514275	0.185854	0.0421857	4.16779e-006	0.336265	-0.574086

G_Matrix

0.0274794	0.0138274	0.0239214	1061.04	0.0106107	0.0239866
0.0138274	0.0696842	0.0363486	1592.96	0.0334668	0.0345388
0.0239214	0.0363486	0.119645	2475.11	0.0330094	0.0551583
1061.04	1592.96	2475.11	2.26264	1328.61	2568.58
0.0106107	0.0334668	0.0330094	1328.61	0.0347727	0.028653
0.0239866	0.0345388	0.0551583	2568.58	0.028653	0.0458665

These estimated parameters were used to generate 100 samples of monthly data each of 98 years long for the 10 sites. Part of the statistical analysis results of the generated data is shown below:

Model: Seasonal Disaggregation,(Statistical Analysis of Generated Data)

Site Number: 8

Stats	Season 1		Season 2		Season 3		Season 4		Season 5		Season 6	
	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen
Mean	2.55E+05	2.55E+05	2.14E+05	2.14E+05	1.77E+05	1.77E+05	1.62E+05	1.62E+05	1.57E+05	1.57E+05	2.19E+05	2.19E+05
StDev	9.06E+04	8.63E+04	4.78E+04	4.46E+04	3.62E+04	3.35E+04	2.75E+04	2.73E+04	2.80E+04	2.69E+04	6.38E+04	5.88E+04
CV	0.3556	0.3358	0.2236	0.2075	0.2042	0.188	0.1696	0.1679	0.1782	0.1708	0.2912	0.2671
Skew	1.191	0.9443	1.354	0.5748	1.425	0.5139	0.5625	0.5076	0.8878	0.4841	1.369	0.7417
Min	1.13E+05	1.07E+05	1.05E+05	1.25E+05	1.14E+05	1.09E+05	1.08E+05	1.06E+05	1.09E+05	1.01E+05	1.27E+05	1.09E+05
Max	5.84E+05	5.56E+05	4.07E+05	3.53E+05	3.09E+05	2.79E+05	2.46E+05	2.45E+05	2.45E+05	2.39E+05	4.47E+05	4.14E+05
acf(1)	0.1774	0.1439	0.4452	0.115	0.5758	0.09407	0.5258	0.07505	0.3037	0.08979	0.3578	0.06902
acf(2)	0.2127	0.02867	0.3428	0.0125	0.3529	0.01799	0.2203	0.002252	0.09943	0.016	0.1786	0.02124

Stats	Season 7		Season 8		Season 9		Season 10		Season 11		Season 12	
	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen
Mean	5.23E+05	5.30E+05	1.51E+06	1.54E+06	1.90E+06	1.93E+06	9.56E+05	9.64E+05	4.62E+05	4.64E+05	2.87E+05	2.88E+05
StDev	2.46E+05	2.63E+05	5.88E+05	6.99E+05	7.35E+05	8.87E+05	4.45E+05	4.57E+05	1.71E+05	1.67E+05	1.09E+05	1.03E+05
CV	0.4708	0.4908	0.3886	0.4481	0.3868	0.4542	0.4649	0.4688	0.3707	0.3571	0.3793	0.3566
Skew	1.007	1.381	0.4345	1.317	0.3724	1.3	1.201	1.319	1.046	1.025	1.372	1.054
Min	1.56E+05	1.47E+05	3.44E+05	4.79E+05	4.47E+05	6.02E+05	2.86E+05	2.91E+05	1.95E+05	1.83E+05	1.17E+05	1.15E+05
Max	1.38E+06	1.57E+06	3.42E+06	4.26E+06	3.71E+06	5.30E+06	2.47E+06	2.74E+06	1.02E+06	1.06E+06	7.29E+05	6.61E+05
acf(1)	0.07579	0.05275	0.2495	0.1621	0.1226	0.193	-0.00155	0.1898	0.1201	0.1694	0.06114	0.08354
acf(2)	0.07499	0.005327	-0.05038	0.04425	0.04898	0.04115	0.08138	0.04902	0.1682	0.04849	0.3109	0.005685

Site Number: 16

Stats	Season 1		Season 2		Season 3		Season 4		Season 5		Season 6	
	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen
Mean	1.83E+05	1.82E+05	1.56E+05	1.56E+05	1.17E+05	1.17E+05	1.20E+05	1.19E+05	1.43E+05	1.42E+05	2.92E+05	2.92E+05
StDev	7.88E+04	7.50E+04	4.61E+04	4.53E+04	3.67E+04	3.82E+04	3.17E+04	3.19E+04	5.13E+04	4.31E+04	1.06E+05	1.07E+05
CV	0.4301	0.4101	0.2951	0.2897	0.3126	0.3231	0.2654	0.2663	0.3583	0.302	0.3621	0.3647
Skew	1.293	1.17	0.7312	0.8623	0.5711	0.9595	0.5839	0.7619	2.335	0.941	0.9584	1.003
Min	5.49E+04	6.20E+04	5.74E+04	7.52E+04	4.60E+04	5.11E+04	6.15E+04	5.97E+04	6.16E+04	6.63E+04	1.18E+05	1.13E+05
Max	5.06E+05	4.57E+05	2.83E+05	3.10E+05	2.25E+05	2.54E+05	2.09E+05	2.25E+05	4.10E+05	2.93E+05	7.03E+05	6.70E+05
acf(1)	0.4071	0.1962	0.3239	0.1574	0.3953	0.09348	0.3352	0.07235	0.1401	0.07302	0.2517	0.09619
acf(2)	0.3724	0.06119	0.2887	0.04546	0.228	0.02202	0.2902	0.01816	0.0185	0.01632	0.07323	0.02852

Stats	Season 7		Season 8		Season 9		Season 10		Season 11		Season 12	
	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen	Hist.	Gen
Mean	4.86E+05	4.88E+05	1.13E+06	1.15E+06	1.45E+06	1.47E+06	7.57E+05	7.62E+05	3.72E+05	3.74E+05	2.09E+05	2.08E+05
StDev	2.03E+05	2.07E+05	4.29E+05	5.11E+05	6.15E+05	7.39E+05	3.67E+05	3.77E+05	1.34E+05	1.39E+05	8.68E+04	7.71E+04
CV	0.4174	0.421	0.38	0.4423	0.4242	0.4965	0.4844	0.4899	0.3586	0.3701	0.415	0.3683
Skew	1.122	1.217	0.2865	1.238	0.5391	1.471	1.277	1.398	0.7773	1.081	2.022	1.038
Min	1.84E+05	1.65E+05	2.44E+05	3.62E+05	2.85E+05	3.92E+05	2.04E+05	2.14E+05	1.51E+05	1.45E+05	8.70E+04	7.99E+04
Max	1.23E+06	1.28E+06	2.31E+06	3.07E+06	3.16E+06	4.51E+06	2.12E+06	2.27E+06	7.84E+05	8.85E+05	6.32E+05	4.83E+05
acf(1)	0.08158	0.1085	0.1879	0.157	0.1286	0.1813	0.05257	0.1676	0.2001	0.1709	0.06152	0.1188
acf(2)	0.09421	0.02964	0.02447	0.04916	0.05923	0.0495	0.07317	0.05436	0.1285	0.0532	0.2599	0.03823

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APPENDIX A: PARAMETER ESTIMATION AND GENERATION

A.1 Transformation

A.1.1 Tests of Normality

Two normality tests are used in SAMS, namely the skewness test of normality (Snedecor and Cochran, 1980) and Filliben probability plot correlation test (Filliben, 1975) both applied at the 10% significance level. Both tests can be applied on an annual or seasonal basis.

In the skewness test of normality we assume a sample $\{X_t\}_{t=1}^N \sim iid N(\mu_X, \sigma_X^2)$. Then the estimated sample skewness from Eq. (3.3) g is asymptotically distributed as $N(0, \sigma^2 = 6/N)$. The null hypothesis $H_0: g = 0$ vs $H_1: g \neq 0$ is rejected at the α significance level if $\text{abs}(g) > z_{1-\alpha/2} \sqrt{6/N}$, where z_q is the q th quantile from the standard normal distribution. According to Snedecor and Cochran (1980) the above probability limits are accurate for sample sizes greater than 150, for smaller sample sizes tabulated test statistics are given for example in Salas et al. (1980).

For a random sample $X_1, X_2, \dots, X_N$ of size N the Filliben probability plot correlation coefficient test of normality is applied on the cross correlation coefficient $R_0(X_{i:N}, M_{i:N})$ where the sample correlation coefficient is calculated by Eq. (3.4), $X_{i:N}$ is the i th sample order statistic and $M_{i:N}$ is the i th order statistic median from a standard normal distribution. $M_{i:N}$ is estimated as $F^{-1}(u_{i:N})$ where F^{-1} is the inverse of the standard normal cumulative distribution function and $u_{i:N}$ is the order statistic median from the uniform $U(0; 1)$ distribution estimated as $u_{1:N} = (1-2^{-1/N})$, $u_{i:N} = (i - 0.3175)/(N + 0.365)$ for $i = 2, \dots, N-1$, and $u_{N:N} = 2^{-1/N}$. The null hypothesis $H_0: r_0 = 1$ vs $H_1: r_0 < 1$ is rejected at the α significance level if $r_0 < \rho_\alpha(N)$ where $\rho_\alpha(N)$ is a tabulated test statistic given in Filliben (1975) and Vogel (1986) for the above plotting position. Johnson and Wichern (2002, page 182) give tabulated test statistics for the case when $u_{i:N}$ is estimated based on the Hazen plotting position.

A.1.2 Automatic Transformation

The user can select to have SAMS select the best transformation or to have SAMS suggest a Logarithmic, Power and Gamma transformation. The parameters of the transformations are estimated in the following way when “Auto” transformation button is selected:

Logarithmic: The location parameter a of Eq. (4.1) is estimated based on a method suggested by Boswell et al. (1979), with $a = (x_{\min} x_{\max} - x_{N/2:N}^2) / (x_{\min} + x_{\max} - 2x_{N/2:N})$, where $x_{N/2:N}$ is the

median of the sample series.

Gamma: The Wilson-Hilferty transformation (Loucks et al., 1981), is used for transforming a Gamma variate to a normal variate.

Power: The parameters of the Power transformation is Eq. (4.3) are estimated by an iterative process aimed at maximizing the Filliben correlation coefficient test statistic.

When the “Best Transf” button is pressed then SAMS chooses the best transformation among Normal, Logarithmic with $a = 0$ (LN-2), Logarithmic with a estimated as above (LN-3), Gamma, and if the sample skewness is negative the Power transformation is also used. The transformation resulting in the highest adjusted Filliben correlation coefficient test statistic is selected as the best one. The Filliben test statistic is slightly penalized for the LN-3, since the simpler LN-2 or Normal should be preferred if the test statistics are similar. In addition, the Gamma and the Power are slightly penalized over the LN-3. Due to this penalization, the distribution with the highest Filliben test statistic may not be selected as the best one.

A.2 Parameter Estimation of Univariate Models

A.2.1 Univariate ARMA(p,q)

The method of moments (MOM) and Least Squares (LS) method can be used for estimation of the parameters of the ARMA(p,q) model in chapter 4, Eq. (4.6). The MOM method is equivalent to Yule-Walker estimation in Brockwell and Davis (1996). For example, the moment estimators for the ARMA (1,0) , ARMA (1,1) and ARMA (2,1) models are given as:

- ARMA (1,0) model:

$$Y_t = \phi_1 Y_{t-1} + \varepsilon_t \quad (\text{A.1})$$

$$\hat{\phi}_1 = r_1 \quad (\text{A.2})$$

$$\hat{\sigma}^2(\varepsilon) = s^2(1 - \hat{\phi}_1^2) \quad (\text{A.3})$$

- ARMA (1,1) model:

$$Y_t = \phi_1 Y_{t-1} + \varepsilon_t - \theta_1 \varepsilon_{t-1} \quad (\text{A.4})$$

$$\hat{\phi}_1 = \frac{r_2}{r_1} \quad (\text{A.5})$$

$$\hat{\theta}_1 = \hat{\phi}_1 + \frac{1 - \hat{\phi}_1 r_1}{\hat{\phi}_1 - r_1} - \frac{1}{\hat{\theta}_1} \quad (\text{A.6})$$

$$\hat{\sigma}^2(\varepsilon) = s^2 \frac{\hat{\phi}_1 - r_1}{\hat{\theta}_1} \quad (\text{A.7})$$

where $\hat{\theta}_1$ is estimated by solving Eq. (A.6).

- ARMA (2,1) model:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t - \theta_1 \varepsilon_{t-1} \quad (\text{A.8})$$

$$\hat{\phi}_1 = \frac{r_2 r_1 - r_3}{r_1^2 - r_2} \quad (\text{A.9})$$

$$\hat{\phi}_2 = \frac{r_3 - \hat{\phi}_1 r_2}{r_1} \quad (\text{A.10})$$

$$\hat{\theta}_1 = \hat{\phi}_1 + \frac{1 - \hat{\phi}_1 r_1 - \hat{\phi}_2 r_2}{\hat{\phi}_1 - r_1 + \hat{\phi}_2 r_1} - \frac{\hat{\phi}_1 - r_1 + \hat{\phi}_2 r_1}{(\hat{\phi}_1 - r_1 + \hat{\phi}_2 r_1) \hat{\theta}_1} \quad (\text{A.11})$$

$$\hat{\sigma}^2(\varepsilon) = s^2 \frac{\hat{\phi}_1 + \hat{\phi}_2 r_1 - r_1}{\hat{\theta}_1} \quad (\text{A.12})$$

where s^2 is the variance of Y_t and $r_k = m_k / s^2$ is the estimate of the lag- k autocorrelation coefficient of Y_t which is defined as $R_k = E[Y_t Y_{t-k}] / E[Y_t Y_t]$. Similarly m_k is the estimate of the lag- k autocovariance coefficient of Y_t with $M_k = E[Y_t Y_{t-k}]$. In the foregoing model it is assumed that the mean has been removed or $E[Y_t] = 0$. Note also that $s^2 = m_0$.

The Least Squares (LS) method is generally a more efficient parameter estimation method. In this method, the parameters ϕ 's and θ 's are estimated by minimizing the sum of squares of the residuals defined by

$$F = \sum_{t=1}^N \varepsilon_t^2 \quad (\text{A.13})$$

where N is the number of years of data. For the ARMA(p,q) model, the residuals are defined as

$$\varepsilon_t = Y_t - \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} \quad (\text{A.14})$$

Once the ϕ 's and θ 's are determined, then the noise variance $\sigma^2(\varepsilon)$ is determined by $(1/N) \sum_{t=1}^N \varepsilon_t^2$.

The minimization of the sum of squares of Eq. (A.13) may be obtained by a numerical scheme. In SAMS first a high order AR(p) model is fitted to the data to get initial estimate of the noise terms ε_t .

Then iteratively a regression model is fitted to the data and the parameters ϕ 's and θ 's are re-

estimated and the residuals are re-calculated until the sum of the squares of the residuals has converged to a minimum value.

To generate synthetic series from an ARMA model, Eq. (4.6) can be used. The white noise process is generated by first generating a standard uncorrelated normal random variable z_t and then calculating ε_t as

$$\varepsilon_t = \sigma(\varepsilon)z_t \quad (\text{A.15})$$

For generation of the correlated series Y_t , a warm-up procedure is followed. In this procedure, values of Y_t prior to $t = 1$ are assumed to be equal to the mean of the process (which is zero in this case). Thus, $Y_1, Y_2, \dots, Y_{N+L}$ are generated using Eq. (4.6) by generating $\varepsilon_{1-q}, \varepsilon_{2-q}, \varepsilon_{3-q}, \dots$ from Eq. (A.15) where N is the required length to be generated and L is the warm-up length required to remove the effect of the initial assumptions of Y_t . L is arbitrarily chosen as 50 in SAMS. The advantage of the warm up procedure is that it can be used for low order and high order stationary and periodic models while exact generation procedures available in the literature apply only for stationary ARMA models or the low order periodic models.

A.2.2 Univariate GAR(1)

The stationary GAR(1) process of Eq. (4.7) has four parameters $\{\phi, \lambda, \alpha, \beta\}$. It may be shown that the relationships between the model parameters and the population moments of the underlying variable X_t are:

$$\mu = \lambda + \frac{\beta}{\alpha} \quad (\text{A.16})$$

$$\sigma^2 = \frac{\beta}{\alpha^2} \quad (\text{A.17})$$

$$\gamma = \frac{2}{\sqrt{\beta}} \quad (\text{A.18})$$

$$\rho_1 = \phi \quad (\text{A.19})$$

where μ, σ^2, γ and ρ_1 are the mean, variance, skewness coefficient, and the lag-one autocorrelation coefficient, respectively.

Estimation of the parameters of the GAR(1) model is based on results by Kendall (1968), Wallis and O'Connell (1972), and Matalas (1966) and based on extensive simulation experiments conducted by Fernandez and Salas (1990). These studies suggest the following estimation procedure for the four parameters $\{\phi, \lambda, \alpha, \beta\}$. First the sample moments are corrected to ensure unbiased

parameter estimates:

$$\hat{\sigma}^2 = s^2 \frac{N-1}{N-K} \quad (\text{A.20})$$

$$\hat{\rho}_1 = \frac{r_1 N + 1}{N-4} \quad (\text{A.21})$$

$$K = \frac{N(1-\hat{\rho}_1^2) - 2\hat{\rho}_1(1-\hat{\rho}_1^N)}{N(1-\hat{\rho}_1)^2} \quad (\text{A.22})$$

in which r_1 is the lag-1 sample autocorrelation coefficient and s^2 is the sample variance. In addition,

$$\hat{\gamma} = \frac{\hat{\gamma}_0}{1 - 3.12\hat{\rho}_1^{3.7}N^{-0.49}} \quad (\text{A.23})$$

where $\hat{\gamma}_0$ is the skewness coefficient suggested by Bobee and Robitaille (1975) as

$$\hat{\gamma}_0 = \frac{L \cdot g}{\sqrt{N}} \left[A + B \frac{L^2 g^2}{N} \right] \quad (\text{A.24})$$

in which g is the sample skewness coefficient and the constants A , B , and L are given by

$$A = 1 + \frac{6.51}{N} + \frac{20.2}{N^2} \quad (\text{A.25})$$

$$B = \frac{1.48}{N} + \frac{6.77}{N^2} \quad (\text{A.26})$$

and

$$L = \frac{N-2}{\sqrt{N-1}} \quad (\text{A.27})$$

respectively. Furthermore, the mean is estimated by the usual sample mean $\bar{x}$. Therefore, substituting the population statistics μ , σ^2 , γ and ρ_1 in Eqs. (A.16) through (A.19) by the corresponding estimates $\bar{x}$, $\hat{\sigma}^2$, $\hat{\gamma}$, and $\hat{\rho}_1$ as above suggested and solving the equations simultaneously give the MOM estimates of the GAR(1) model parameters. For more details, the interested reader is referred to Fernandez and Salas (1990).

To generate synthetic series from a GAR(1) model, Eq. (4.7) is used with the noise process generated by Eq. (4.9). A similar warm-up procedure is used as for the ARMA model.

A.2.3 Univariate SM

The MOM method along with LS smoothing of the sample correlogram (the autocorrelation function) is used for parameter estimation of the SM model in Eq. (4.10). For detailed description of

parameter estimation of the SM model refer to Sveinsson et al. (2003) and (2005). It may be shown that the relationships between the model parameters $\{\mu_Y, \sigma_Y^2, \sigma_M^2, p\}$ and the population moments of the underlying variable X_t in Eq. (4.10) are:

$$\mu_X = \mu_Y \quad (\text{A.28})$$

$$\sigma_X^2 = \sigma_Y^2 + \sigma_M^2 \quad (\text{A.29})$$

$$\rho_k(X) = \frac{\sigma_M^2 (1-p)^k}{\sigma_Y^2 + \sigma_M^2}, k = 1, 2, \dots \quad (\text{A.30})$$

where μ_X , σ_X^2 and $\rho_k(X)$ are the mean, variance, and the lag- k autocorrelation coefficient, respectively. The parameter estimates in terms of $\hat{\mu}_X = \bar{x}$, $\hat{\sigma}_X^2$, $\hat{\rho}_1(X)$ and $\hat{\rho}_2(X)$ are

$$\hat{p} = 1 - \frac{\hat{\rho}_2(X)}{\hat{\rho}_1(X)} \quad (\text{A.31})$$

$$\hat{\mu}_Y = \hat{\mu}_X \quad (\text{A.32})$$

$$\hat{\sigma}_M^2 = \hat{\sigma}_X^2 \frac{\hat{\rho}_1(X)}{(1 - \hat{p})} \quad (\text{A.33})$$

$$\hat{\sigma}_Y^2 = \hat{\sigma}_X^2 - \hat{\sigma}_M^2 \quad (\text{A.34})$$

The parameters are feasible if $\hat{\rho}_1(X) > \hat{\rho}_2(X) > \hat{\rho}_1^2(X)$. It is an option in SAMS to estimate the parameters given the value of the parameter p , in which case Eqs. (A.32)-(A.34) are used for estimation of the parameters. Because of sample variability of the sample correlogram, infeasible parameter estimates may result. To prevent this in SAMS the exact form of the model correlogram in Eq. (A.30) is fitted to the sample correlogram using LS. The modeller can choose up to which lag the sample correlogram should be fitted.

For generation of synthetic time series of the SM model, Eq. (4.10) is used with the noise level process generated by Eq. (4.11). A similar warm-up procedure is used as for the ARMA model.

A.2.4 Univariate Seasonal PARMA(p,q)

The MOM and LS methods may be used in parameter estimation of low order PARMA(p, q) models. In SAMS the MOM estimates are available for the PARMA(p,1) model. For example, the moment estimators for the PARMA (1,1) and PARMA (2, 1) models are shown below (Salas et al, 1982):

- PARMA (1,1) model:

$$Y_{\nu,\tau} = \phi_{1,\tau} Y_{\nu,\tau-1} + \varepsilon_{\nu,\tau} - \theta_{1,\tau} \varepsilon_{\nu,\tau-1} \quad (\text{A.35})$$

$$\hat{\phi}_{1,\tau} = \frac{m_{2,\tau}}{m_{1,\tau-1}} \quad (\text{A.36})$$

$$\hat{\theta}_{1,\tau} = \hat{\phi}_{1,\tau} + \frac{s_{\tau}^2 - \hat{\phi}_{1,\tau} m_{1,\tau}}{\hat{\phi}_{1,\tau} s_{\tau-1}^2 - m_{1,\tau}} - \frac{\hat{\phi}_{1,\tau+1} s_{\tau}^2 - m_{1,\tau+1}}{(\hat{\phi}_{1,\tau} s_{\tau-1}^2 - m_{1,\tau}) \hat{\theta}_{1,\tau+1}} \quad (\text{A.37})$$

$$\hat{\sigma}_{\tau}^2(\varepsilon) = \frac{\hat{\phi}_{1,\tau+1} s_{\tau-1}^2 - m_{1,\tau+1}}{\hat{\theta}_{1,\tau+1}} \quad (\text{A.38})$$

- PARMA (2,1) model:

$$Y_{\nu,\tau} = \phi_{1,\tau} Y_{\nu,\tau-1} + \phi_{2,\tau} Y_{\nu,\tau-2} + \varepsilon_{\nu,\tau} - \theta_{1,\tau} \varepsilon_{\nu,\tau-1} \quad (\text{A.39})$$

$$\hat{\phi}_{1,\tau} = \frac{m_{2,\tau} m_{1,\tau-2} - s_{\tau-2}^2 m_{3,\tau}}{m_{1,\tau-1} m_{1,\tau-2} - s_{\tau-2}^2 m_{2,\tau-1}} \quad (\text{A.40})$$

$$\hat{\phi}_{2,\tau} = \frac{m_{3,\tau} - \hat{\phi}_{1,\tau} m_{2,\tau-1}}{m_{1,\tau-2}} \quad (\text{A.41})$$

$$\hat{\theta}_{1,\tau} = \hat{\phi}_{1,\tau} + \frac{s_{\tau}^2 - \hat{\phi}_{1,\tau} m_{1,\tau} - \hat{\phi}_{2,\tau} m_{2,\tau}}{\hat{\phi}_{1,\tau} s_{\tau-1}^2 - m_{1,\tau} + \hat{\phi}_{2,\tau} m_{1,\tau-1}} - \frac{\hat{\phi}_{1,\tau+1} s_{\tau}^2 - m_{1,\tau+1} + \hat{\phi}_{2,\tau+1} m_{1,\tau}}{(\hat{\phi}_{1,\tau} s_{\tau-1}^2 - m_{1,\tau} + \hat{\phi}_{2,\tau} m_{1,\tau-1}) \hat{\theta}_{1,\tau+1}} \quad (\text{A.42})$$

$$\hat{\sigma}_{\tau}^2(\varepsilon) = \frac{\hat{\phi}_{1,\tau+1} s_{\tau}^2 + \hat{\phi}_{2,\tau+1} m_{1,\tau} - m_{1,\tau+1}}{\hat{\theta}_{1,\tau+1}} \quad (\text{A.43})$$

wheres s_{τ}^2 is the seasonal variance and $m_{k,\tau}$ is the estimate of the lag- k season-to-season autocovariance coefficient of $Y_{\nu,\tau}$ which is defined as $M_{k,\tau} = E[Y_{\nu,\tau} Y_{\nu,\tau+k}]$, where it is assumed $E[Y_{\nu,\tau}] = 0$. Note also that $s_{\tau}^2 = m_{0,\tau}$.

In a similar manner as for the ARMA(p,q) model, the Least Squares (LS) method can be used to estimate the model parameters of PARMA(p,q) models. In this case, the parameters ϕ 's and θ 's are estimated by minimizing the sum of squares of the residuals defined by

$$F = \sum_{\nu=1}^N \sum_{\tau=1}^{\omega} \varepsilon_{\nu,\tau}^2 \quad (\text{A.44})$$

where ω is the number of seasons and N is the number of years of data. For the PARMA(p,q) model, the residuals are defined as

$$\varepsilon_{v,\tau} = Y_{v,\tau} - \sum_{i=1}^p \phi_{i,\tau} Y_{v,\tau-i} + \sum_{j=1}^q \theta_{j,\tau} \varepsilon_{v,\tau-j} \quad (\text{A.45})$$

Once the ϕ 's and θ 's are determined the seasonal noise variance $\sigma_\tau^2(\varepsilon)$ can be estimated by

$$(1/N) \sum_{v=1}^N \varepsilon_{v,\tau}^2.$$

Generation of data from PARMA(p,q) models is carried out in a similar manner as for ARMA(p,q) models. The warm up length procedure is used to generate seasonal sequences of the $Y_{v,\tau}$ process by assuming that values of $Y_{v,\tau}$ prior to season 1 of year 1 are equal to zero and generating uncorrelated random sequences of $\varepsilon_{v,\tau}$ as needed in a similar manner as for the ARMA (p,q) model. The warm-up period is taken as 50 years.

A.3 Parameter Estimation of Multivariate Models

A.3.1 Multivariate MAR(p)

The MOM method is used for parameter estimation of the MAR(p) model. It can be shown that the MOM equations of the MAR(p) model in Eq. (4.13) are given by:

$$\mathbf{M}_0 = \mathbf{G} + \sum_{i=1}^p \Phi_i \mathbf{M}_i^T \quad (\text{A.46})$$

$$\mathbf{M}_k = \sum_{i=1}^p \Phi_i \mathbf{M}_{k-i}, \quad k \geq 1 \quad (\text{A.47})$$

where $\mathbf{M}_k$ is the lag- k cross covariance matrix of $\mathbf{Y}_t$ defined as:

$$\mathbf{M}_k = E[\mathbf{Y}_t \mathbf{Y}_{t-k}^T] \quad (\text{A.48})$$

in which the superscript T indicates a matrix transpose and $E[\mathbf{Y}_t] = \mathbf{0}$. In finding the MOM estimates, Eq. (A.47) for $k = 1, \dots, p$, is solved simultaneously for the parameter matrixes Φ_i , $i = 1, \dots, p$, by substituting in Eq. (A.47) the population covariance matrixes $\mathbf{M}_k$, $k = 1, 2, \dots, p$, by the sample covariance matrixes $\mathbf{m}_k$, $k = 1, 2, \dots, p$. Then Eq. (A.46) is used to estimate the variance-covariance matrix of the residuals $\mathbf{G}$. For example, the moment estimators of the MAR(1) model are:

$$\hat{\Phi}_1 = \frac{\mathbf{m}_1}{\mathbf{m}_0} \quad (\text{A.49})$$

$$\hat{\mathbf{G}} = \mathbf{m}_0 - \mathbf{m}_1 \mathbf{m}_0^{-1} \mathbf{m}_1^T \quad (\text{A.50})$$

in which superscript -1 indicates a matrix inverse.

After estimating Φ_i , $i = 1, \dots, p$, and $\mathbf{G}$ as indicated above, $\mathbf{B}$ of Eq. (4.14) can be determined from

$$\hat{\mathbf{G}} = \hat{\mathbf{B}}\hat{\mathbf{B}}^T \quad (\text{A.50})$$

The above matrix equation can have more than one solution. However, a unique solution can be obtained by assuming that $\mathbf{B}$ is a lower triangular matrix. This solution, however, requires that $\mathbf{G}$ be a positive definite matrix.

Generation of synthetic series for the MAR(p) model is carried out using Eq. (4.13) with the spatially correlated noise generated by Eq. (4.14). The warm-up period is defined in the same way as for the ARMA model.

A.3.2 Multivariate CARMA(p,q)

The parameter matrixes of the CARMA(p,q) in Eq. (4.15) are diagonal. Thus, as described in section 4.3.2 the estimation of parameters of the CARMA model is done by decoupling it into univariate ARMA models:

$$Y_t^{(k)} = \sum_{i=1}^p \phi_i^{(k)} Y_{t-i}^{(k)} + \varepsilon_t^{(k)} - \sum_{j=1}^q \theta_j^{(k)} \varepsilon_{t-j}^{(k)} \quad (\text{A.51})$$

where the superscript (k) indicates the k th site and as such the parameters shown indicate the kk diagonal element in the diagonal parameter matrixes in Eq. (4.15). The best univariate ARMA model is identified for each site and the parameters are estimated at each site using MOM or LS estimation methods. After having estimated the diagonal parameter matrixes $\Phi_1, \Phi_2, \dots, \Phi_p$ and $\Theta_1, \Theta_2, \dots, \Theta_q$, what remains is estimation of the noise variance-covariance matrix $\mathbf{G}$. The procedure is simple, but a necessary condition is that the CARMA(p,q) is causal. This is equivalent to requiring each of the estimated univariate ARMA(p,q) models to be causal (often a common requirement in estimation procedures for ARMA models). Causality implies that $\mathbf{Y}_t$ in Eq. (4.15) can be written out as an infinite moving average model (Brockwell and Davis, 1996):

$$\mathbf{Y}_t = \sum_{j=0}^{\infty} \Psi_j \varepsilon_{t-j} \quad (\text{A.52})$$

where $E[\mathbf{Y}_t] = \mathbf{0}$ and Ψ_j are matrixes with absolutely summable elements given by

$$\begin{aligned}\Psi_0 &= \mathbf{I} \\ \Psi_j &= -\Theta_j + \sum_{i=1}^p \Phi_i \Psi_{j-i}^{T}\end{aligned}\tag{A.53}$$

where $\Psi_j = \mathbf{0}$ for $j < 0$, $\Theta_j = \mathbf{0}$ for $j > q$ and $\mathbf{I}$ is the identity matrix. For the special case when $p = 1$ and $q = 0$ then $\Psi_j = \Phi_1^j$, for $j = 1, 2, \dots$. Multiplying each side of Eq. (A.52) by its transpose and taking expectations gives

$$\mathbf{M}_0 = \sum_{j=0}^{\infty} \Psi_j \mathbf{G} \Psi_j^T \tag{A.54}$$

Since Ψ_j , $j = 0, 1, \dots$, are diagonal matrixes the i th row and j th column element of $\mathbf{G}$ is

$$G^{ij} = \frac{M_0^{ij}}{\sum_{k=0}^{\infty} \psi_k^{ii} \psi_k^{jj}} \tag{A.55}$$

where G^{ij} , M_0^{ij} , ψ_k^{ij} are the i th row and j th column element of $\mathbf{G}$, $\mathbf{M}_0$ and Ψ_k , respectively. The elements of Ψ_j decay rather quickly with increasing j , thus the sum in Eq. (A.55) can usually be truncated at a fairly low value of k . An estimate of the $\mathbf{G}$ matrix is obtained by replacing population statistics and parameters in Eq. (A.55) by their corresponding estimates. The above procedure for estimation of the noise variance-covariance matrix $\mathbf{G}$ utilizing only estimated parameter matrixes and the lag 0 covariance matrix of $\mathbf{Y}_t$ ensures that the estimate of $\mathbf{G}$ is consistent with the estimates of the diagonal parameter matrixes.

Generation of synthetic series for the CARMA(p,q) model is carried out using Eq. (4.15) with the spatially correlated noise generated in the same way as for the MAR(p) model. The warm-up period is defined in the same way as for the ARMA model.

A.3.3 Multivariate CSM – CARMA(p,q)

The estimation of the CSM – CARMA(p,q) model is done by decoupling the model first into its CSM and CARMA(p,q) counterparts (refer to Eq. (4.16)). The parameter of the CSM and CARMA models are then estimated separately, where further decoupling takes place into univariate SM models and univariate ARMA(p,q) models. This modeling option can also be used to estimate a CSM model only or a CARMA(p,q) model only.

First it is demonstrated how the CSM part of the model is estimated. The CSM part of the model in Eq. (4.16) has the following properties

1. The lag k covariance function of $\mathbf{X}_t$ of the CSM model is given by

$$\mathbf{M}_k(\mathbf{X}) = \begin{cases} \mathbf{G}_Y + \mathbf{G}_M & \text{if } k = 0 \\ (1-p)^k \mathbf{G}_M & \text{for } k = 1, 2, \dots \end{cases} \tag{A.56}$$

where $\mathbf{G}_Y$ and $\mathbf{G}_M$ are the variance-covariance matrixes (lag 0 covariance matrixes) of $\mathbf{Y}$ and $\mathbf{M}$, respectively.

2. The sequences $\{Y_t^{(1)}\}, \{Y_t^{(2)}\}, \dots, \{Y_t^{(n_1)}\}$ are correlated in space at lag 0 only, and independent in time, with $\{\mathbf{Y}_t\} \sim iid \text{MVN}(\mathbf{0}, \mathbf{G}_Y)$.
3. The sequences $\{M_i^{(1)}\}, \{M_i^{(2)}\}, \dots, \{M_i^{(n_1)}\}$ are correlated in space only at lag zero. That is, $\{\mathbf{M}_i\} \sim iid \text{MVN}(\mathbf{0}, \mathbf{G}_M)$. It can be shown (Sveinsson and Salas, 2006) that a necessary and sufficient condition for $\{\mathbf{Z}_t\}$ to be stationary in the covariance is that $N_1, N_2, \dots$ is a common sequence for all sites. In that case the covariance function of $\mathbf{Z}_t$ at lag k is:

$$\mathbf{M}_k(\mathbf{Z}) = (1 - p)^k \mathbf{G}_M \quad k = 0, 1, \dots \quad (\text{A.57})$$

The condition that $\{N_t\}_{t=1}^\infty$ is a common sequence for all sites may also be supported in practice, if the shifts in the means are thought of being caused by changes in natural processes, such as changes in climate. In such cases it should be expected that time series of the same hydrologic variable within a geographic region would all exhibit shifts at the same times. Thus, in general the CSM model should not be applied for multivariate analysis of time series if it is clear that shifts in different time series do not coincide in time. Such cases can come up if a shift in a time series is caused by a construction of a dam or other man made constructions, where the construction does not affect the other time series being analyzed. Note that if $\mathbf{M}_t$ is assumed uncorrelated in space then the condition for stationarity that $\{N_t\}_{t=1}^\infty$ is a common sequence for all sites is not necessary any more (that option though is not available in SAMS).

The CSM is decoupled into univariate SM models and the parameters are estimated at each site using the procedures for the SM models. If the common p is not known, then $p^{(i)}$ is first estimated at each site i (Sveinsson and Salas, 2006). The common p can then be estimated as a weighted average of the $\hat{p}^{(i)}$ s

$$\hat{p} = \frac{1}{n_1^{(1)} + n_1^{(2)} + \dots + n_1^{(n_1)}} \sum_{i=1}^{n_1} n_1^{(i)} \hat{p}^{(i)} \quad (\text{A.58})$$

Given $\hat{p}$ the parameters of the univariate SM-1 models are reestimated. What remains is estimating the non-diagonal elements of $\mathbf{G}_Y$ and $\mathbf{G}_M$ (note the diagonal elements, i.e. the variances, have already been estimated in the univariate models). Using Eq. (A.56) $\mathbf{G}_M$ is

estimated from

$$\hat{\mathbf{G}}_{\mathbf{M}} = \frac{\mathbf{m}_1(\mathbf{X})}{1 - \hat{\rho}} \quad (\text{A.57})$$

where if necessary $\hat{\mathbf{G}}_{\mathbf{M}}$ is made symmetric by replacing $\hat{g}_{\mathbf{M}}^{ij}$ and $\hat{g}_{\mathbf{M}}^{ji}$ with their respective averages. Then $\mathbf{G}_{\mathbf{M}}$ is estimated from (Eq. (A.56))

$$\hat{\mathbf{G}}_{\mathbf{Y}} = \mathbf{m}_0(\mathbf{X}) - \hat{\mathbf{G}}_{\mathbf{M}} \quad (\text{A.58})$$

where as before $\mathbf{m}_k(\mathbf{X})$ is the sample estimate of the lag- k covariance matrix $\mathbf{M}_k(\mathbf{X})$ as defined in Eq. (A.48).

Estimation of the CARMA part of the model in Eq. (4.16) is done by decoupling it into univariate ARMA(p_i, q_i), $i = n_1 + 1, n_1 + 2, \dots, n$ models and fitting the best ARMA model for each site using the parameter estimation procedure for the multivariate CARMA model. For estimation of the variance-covariance matrix of the noise ($\mathbf{G}$) of the CARMA modelled $\mathbf{Y}_t$, the procedures of the CARMA models are used, where each of the elements of $\mathbf{Y}_t$ corresponding to the CSM process is looked at as being modelled by an ARMA(0,0) model. The upper left $n_1 \times n_1$ part of the $n \times n$ estimated $\mathbf{G}$ matrix is replaced by $\hat{\mathbf{G}}_{\mathbf{Y}}$ in Eq. (A.58).

For generation of synthetic time series of the CSM-CARMA model, Eq. (4.16) is used with the noise level process generated by Eq. (4.11). A similar warm-up procedure is used as for the ARMA model.

A.3.4 Multivariate Seasonal MPAR (p)

The parameters of the multivariate seasonal MPAR(p) model in Eq. (4.17) are estimated by the MOM by substituting the sample moments into the moment equations in a similar manner as for the MAR(p) model. The moment equations of the MPAR(p) model may be shown to be:

$$\mathbf{M}_{0,\tau} = \mathbf{G}_{\tau} + \sum_{i=1}^p \Phi_{i,\tau} \mathbf{M}_{i,\tau}^T \quad (\text{A.59})$$

$$\mathbf{M}_{k,\tau} = \sum_{i=1}^p \Phi_{i,\tau} \mathbf{M}_{k-i,\tau-i} \quad , \text{ for } \tau - i \geq 0 \text{ and } k \geq 1 \quad (\text{A.60a})$$

$$\mathbf{M}_{k,\tau} = \sum_{i=1}^p \Phi_{i,\tau} \mathbf{M}_{i-k,\tau-k}^T \quad , \text{ for } \tau - i < 0 \text{ and } k \geq 1 \quad (\text{A.60b})$$

where $\mathbf{M}_{k,\tau}$ is the lag- k cross covariance matrix of $\mathbf{Y}_{v,\tau}$ defined as:

$$\mathbf{M}_{k,\tau} = E[\mathbf{Y}_{v,\tau} \mathbf{Y}_{v,\tau-k}^T] = \{E[\mathbf{Y}_{v,\tau-k} \mathbf{Y}_{v,\tau}^T]\}^T = \mathbf{M}_{-k,\tau-k}^T \quad (\text{A.62})$$

in which the superscript T indicates a matrix transpose and $E[\mathbf{Y}_{v,\tau}] = \mathbf{0}$. In a similar manner as for the MAR(p) model, the MOM estimates can be found by solving Eq. (A.60) for $k = 1, 2, \dots, p$ simultaneously for Φ 's by substituting the population covariance matrixes $\mathbf{M}_{k,\tau}$, $k = 1, \dots, p$ by the

corresponding sample covariance matrixes. Then Eq. (A.59) is used to estimate the variance-covariance matrix of the residuals $\mathbf{G}_\tau$.

For generation of synthetic time series similar procedures as for the MAR(p) and PARMA(p,q) models are used. As for the MAR(p) model the generation process of the noise is simplified by using a lower triangular matrix $\mathbf{B}_\tau$ similar as in Eq. (4.14) for the MAR(p) model, i.e. $\mathbf{G}_\tau = \mathbf{B}_\tau \mathbf{B}_\tau^T$. As for other models a warm-up period is used to remove the effects of initial conditions of the generation process.

A.4 Parameter Estimation of Disaggregation Models

A.4.1 Valencia and Schaake Spatial Disaggregation

The model parameter matrixes $\mathbf{A}$ and $\mathbf{B}$ of the VS model in Eq. (4.18) can be estimated by using MOM (Valencia and Schaake, 1973):

$$\mathbf{A} = \mathbf{M}_0(\mathbf{YX})\mathbf{M}_0^{-1}(\mathbf{X}) \quad (\text{A.63})$$

$$\mathbf{BB}^T = \mathbf{M}_0(\mathbf{Y}) - \mathbf{A}\mathbf{M}_0(\mathbf{X})\mathbf{A}^{-1} \quad (\text{A.64})$$

where $\mathbf{G} = \mathbf{BB}^T$ is the noise variance-covariance matrix ($\mathbf{B}$ is the Cholesky decomposition of $\mathbf{G}$), and $\mathbf{M}_k(\mathbf{Y}) = E[\mathbf{Y}_\nu \mathbf{Y}_{\nu-k}^T]$ and $\mathbf{M}_k(\mathbf{YX}) = E[\mathbf{Y}_\nu \mathbf{X}_{\nu-k}^T]$. The VS model is not available for spatial disaggregation of seasonal data in SAMS, since the MR model is thought to be better suited.

A.4.2 Mejia and Rousselle Spatial Disaggregation

The model parameter matrixes $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ of the MR model in Eq. (4.19) can be estimated by using MOM as:

$$\mathbf{A} = [\mathbf{M}_0(\mathbf{YX}) - \mathbf{M}_1(\mathbf{Y})\mathbf{M}_0^{-1}(\mathbf{Y})\mathbf{M}_1^T(\mathbf{XY})][\mathbf{M}_0(\mathbf{X}) - \mathbf{M}_1(\mathbf{XY})\mathbf{M}_0^{-1}(\mathbf{Y})\mathbf{M}_1^T(\mathbf{XY})]^{-1} \quad (\text{A.65})$$

$$\mathbf{C} = [\mathbf{M}_1(\mathbf{Y}) - \mathbf{A}\mathbf{M}_1(\mathbf{XY})]\mathbf{M}_0^{-1}(\mathbf{Y}) \quad (\text{A.66})$$

$$\mathbf{BB}^T = \mathbf{M}_0(\mathbf{Y}) - \mathbf{A}\mathbf{M}_0(\mathbf{XY}) - \mathbf{C}\mathbf{M}_1^T(\mathbf{Y}) \quad (\text{A.67})$$

Equations (A.65) through (A.67) can be used to obtain estimates of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ by substituting the population covariance matrixes by their corresponding sample estimates. Lane (1981) showed that some problems exist if one uses the above equations to estimate the parameters. Specifically, the problem is in using $\mathbf{M}_1(\mathbf{XY})$, since the model structure does not preserve this particular lag-1 dependence between $\mathbf{X}$ and $\mathbf{Y}$. Lane verified this and showed that the generated moments are affected and some key moments are not preserved. As a result, he suggested that, instead of using a sample estimate of $\mathbf{M}_1(\mathbf{XY})$, one should use the model $\mathbf{M}_1(\mathbf{XY})$ that would result from the model

structure (for further details, the reader is referred to Lane and Frevert, 1990). In the final analysis, the suggested equation is

$$\mathbf{M}_1^*(\mathbf{XY}) = \mathbf{M}_1(\mathbf{X}) \mathbf{M}_0^{-1}(\mathbf{X}) \mathbf{M}_0(\mathbf{XY}) \quad (\text{A.68})$$

For consistency $\mathbf{M}_1(\mathbf{Y})$ also needs to be adjusted

$$\mathbf{M}_1^*(\mathbf{Y}) = \mathbf{M}_1(\mathbf{Y}) + \mathbf{M}_0(\mathbf{YX}) \mathbf{M}_0^{-1}(\mathbf{X}) [\mathbf{M}_1^*(\mathbf{XY}) - \mathbf{M}_1(\mathbf{XY})] \quad (\text{A.69})$$

Equations (A.68) and (A.69) should be used for calculating $\mathbf{M}_1(\mathbf{XY})$ and $\mathbf{M}_1(\mathbf{Y})$, and these calculated values should be used in Eqs. (A.65) through (A.67) for estimating the model parameters.

The reader is referred to Lane and Frevert (1990) for more in depth details about these adjustments.

A.4.3 Mejia and Rousselle Spatial Disaggregation of Seasonal Data

The model parameter matrixes $\mathbf{A}_\tau$, $\mathbf{B}_\tau$, and $\mathbf{C}_\tau$ of the MR model in Eq. (4.21) can be estimated in a similar way as for the spatial disaggregation of annual data above by using MOM. The MOM equations are similar as for the annual MR model:

$$\begin{aligned} \mathbf{A}_\tau &= [\mathbf{M}_{0,\tau}(\mathbf{YX}) - \mathbf{M}_{1,\tau}(\mathbf{Y}) \mathbf{M}_{0,\tau-1}^{-1}(\mathbf{Y}) \mathbf{M}_{1,\tau}^T(\mathbf{XY})] \\ &\quad [\mathbf{M}_{0,\tau}(\mathbf{X}) - \mathbf{M}_{1,\tau}(\mathbf{XY}) \mathbf{M}_{0,\tau-1}^{-1}(\mathbf{Y}) \mathbf{M}_{1,\tau}^T(\mathbf{XY})]^{-1} \end{aligned} \quad (\text{A.70})$$

$$\mathbf{C}_\tau = [\mathbf{M}_{1,\tau}(\mathbf{Y}) - \mathbf{A}_\tau \mathbf{M}_{1,\tau}(\mathbf{XY})] \mathbf{M}_{0,\tau-1}^{-1}(\mathbf{Y}) \quad (\text{A.71})$$

$$\mathbf{B}_\tau \mathbf{B}_\tau^T = \mathbf{M}_{0,\tau}(\mathbf{Y}) - \mathbf{A}_\tau \mathbf{M}_{0,\tau}(\mathbf{XY}) - \mathbf{C}_\tau \mathbf{M}_{1,\tau}^T(\mathbf{Y}) \quad (\text{A.72})$$

where $\mathbf{M}_{k,\tau}(\mathbf{Y}) = E[\mathbf{Y}_{\nu,\tau} \mathbf{Y}_{\nu,\tau-k}^T]$ and $\mathbf{M}_{k,\tau}(\mathbf{YX}) = E[\mathbf{Y}_{\nu,\tau} \mathbf{X}_{\nu,\tau-k}^T]$. Since the model structure of Eq. (4.21) does not preserve the dependence structure between $\mathbf{X}_{\nu,\tau}$ and $\mathbf{Y}_{\nu,\tau-1}$ for any season, same type of adjustment procedures as for the annual MR model have to be applied for each season for estimation of $\mathbf{M}_{1,\tau}(\mathbf{Y})$ and $\mathbf{M}_{1,\tau}(\mathbf{XY})$. Thus for each season the following corrected model covariances are used:

$$\mathbf{M}_{1,\tau}^*(\mathbf{XY}) = \mathbf{M}_{1,\tau}(\mathbf{X}) \mathbf{M}_{0,\tau-1}^{-1}(\mathbf{X}) \mathbf{M}_{0,\tau-1}(\mathbf{XY}) \quad (\text{A.73})$$

$$\mathbf{M}_{1,\tau}^*(\mathbf{Y}) = \mathbf{M}_{1,\tau}(\mathbf{Y}) + \mathbf{M}_{0,\tau}(\mathbf{YX}) \mathbf{M}_{0,\tau}^{-1}(\mathbf{X}) [\mathbf{M}_{1,\tau}^*(\mathbf{XY}) - \mathbf{M}_{1,\tau}(\mathbf{XY})] \quad (\text{A.74})$$

The above corrected model covariances need to be substituted into the MOM equations, and then the estimates of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are obtained by substituting the population covariance matrixes in the MOM equations by their corresponding sample estimates.

A.4.4 Lane Temporal Disaggregation

The model parameter matrixes $\mathbf{A}_\tau$, $\mathbf{B}_\tau$, and $\mathbf{C}_\tau$ of the temporal Lane model in Eq. (4.22) can

be estimated by using the MOM as (Lane and Frevert, 1990). To avoid confusion we have $\mathbf{X}$ denote the annual flows at the N stations and $\mathbf{Y}$ the seasonal flows at the same stations.

$$\mathbf{A}_\tau = [\mathbf{M}_{0,\tau}(\mathbf{YX}) - \mathbf{M}_{1,\tau}(\mathbf{Y})\mathbf{M}_{0,\tau-1}^{-1}(\mathbf{Y})\mathbf{M}_{1,\tau}^T(\mathbf{XY})] \\ [\mathbf{M}_0(\mathbf{X}) - \mathbf{M}_{1,\tau}(\mathbf{XY})\mathbf{M}_{0,\tau-1}^{-1}(\mathbf{Y})\mathbf{M}_{1,\tau}^T(\mathbf{XY})]^{-1} \quad (\text{A.75})$$

$$\mathbf{C}_\tau = [\mathbf{M}_{1,\tau}(\mathbf{Y}) - \mathbf{A}_\tau\mathbf{M}_{1,\tau}(\mathbf{XY})]\mathbf{M}_{0,\tau-1}^{-1}(\mathbf{Y}) \quad (\text{A.76})$$

$$\mathbf{B}_\tau\mathbf{B}_\tau^T = \mathbf{M}_{0,\tau}(\mathbf{Y}) - \mathbf{A}_\tau\mathbf{M}_{0,\tau}(\mathbf{XY}) - \mathbf{C}_\tau\mathbf{M}_{1,\tau}^T(\mathbf{Y}) \quad (\text{A.77})$$

where $\mathbf{M}_k(\mathbf{X}) = E[\mathbf{X}_\nu\mathbf{X}_{\nu-k}^T]$, $\mathbf{M}_{k,\tau}(\mathbf{Y}) = E[\mathbf{Y}_{\nu,\tau}\mathbf{Y}_{\nu,\tau-k}^T]$, $\mathbf{M}_{k,\tau}(\mathbf{XY}) = E[\mathbf{X}_\nu\mathbf{Y}_{\nu,\tau-k}^T]$ and $\mathbf{M}_{k,\tau}(\mathbf{YX}) = E[\mathbf{Y}_{\nu,\tau}\mathbf{X}_{\nu-k}^T]$. Since the model structure of Eq. (4.22) does preserve the dependence structure between $\mathbf{X}_\nu$ and $\mathbf{Y}_{\nu,\tau-1}$ (i.e. $\mathbf{M}_{1,\tau}(\mathbf{XY})$) for all seasons except the first one, adjustment procedures as for the MR models need only to be applied for the first season in estimation of $\mathbf{M}_{1,\tau}(\mathbf{Y})$ and $\mathbf{M}_{1,\tau}(\mathbf{XY})$. Thus only for the first season need the following corrected model covariances to be used:

$$\mathbf{M}_{1,\tau}^*(\mathbf{XY}) = \mathbf{M}_1(\mathbf{X})\mathbf{M}_0^{-1}(\mathbf{X})\mathbf{M}_{0,\tau-1}(\mathbf{XY}) \quad (\text{A.78})$$

$$\mathbf{M}_{1,\tau}^*(\mathbf{Y}) = \mathbf{M}_{1,\tau}(\mathbf{Y}) + \mathbf{M}_{0,\tau}(\mathbf{YX})\mathbf{M}_0^{-1}(\mathbf{X})[\mathbf{M}_{1,\tau}^*(\mathbf{XY}) - \mathbf{M}_{1,\tau}(\mathbf{XY})] \quad (\text{A.79})$$

The MOM parameter matrixes are then estimated by substituting the population moments by their corresponding sample estimates.

A.4.5 Grygier and Stedinger Temporal Disaggregation

The parameter matrixes of the contemporaneous Grygier and Stedinger disaggregation model in Eq. (4.23) are diagonal. Similar as for other contemporaneous models the parameters of the diagonal $\mathbf{A}_\tau$, $\mathbf{C}_\tau$, and $\mathbf{D}_\tau$ matrixes are estimated by decoupling the model into univariate models for each station and each season and estimating the parameters using the Least Squares method (LS). What remains is estimation of $\mathbf{G}_\tau = \mathbf{B}_\tau\mathbf{B}_\tau^T$, the variance-covariance matrix of the noise for each season. The procedure for estimating the noise variance-covariance matrixes is rigorous, and in the case when adjustments need to be made to $\mathbf{G}_\tau$ to make it positive definite, then these adjustments are accounted for in the estimated $\mathbf{G}_\tau$ for the following seasons. For detailed information on the estimation of parameters refer to Grygier and Stedinger (1990). In the following equations we use that the transpose of a diagonal matrix is the matrix itself. To avoid confusion we have $\mathbf{X}$ denote the annual flows at the N stations and $\mathbf{Y}$ the seasonal flows at the same stations. For all seasons below

the population covariance matrixes $\mathbf{M}_0(\mathbf{X})$ and $\mathbf{M}_{0,\tau}(\mathbf{Y})$ are estimated by the sample covariance matrixes $\mathbf{m}_0(\mathbf{X})$ $\mathbf{m}_{0,\tau}(\mathbf{Y})$.

Season $\tau = 1$:

$$\mathbf{M}_{0,1}(\mathbf{YX}) = \mathbf{A}_1 \mathbf{M}_0(\mathbf{X}) \quad (\text{A.80})$$

$$\mathbf{B}_1 \mathbf{B}_1^T = \mathbf{M}_{0,1}(\mathbf{Y}) - \mathbf{A}_1 \mathbf{M}_0(\mathbf{X}) \mathbf{A}_1 \quad (\text{A.81})$$

Season $\tau = 2$: Let

$$\mathbf{M}_{1,2}(\Lambda \mathbf{Y}) = \mathbf{W}_1 \mathbf{M}_{0,1}(\mathbf{Y}) \quad (\text{A.82})$$

$$\mathbf{M}_{0,2}(\Lambda \mathbf{X}) = \mathbf{W}_1 \mathbf{M}_{0,1}(\mathbf{YX}) \quad (\text{A.83})$$

$$\mathbf{M}_{0,2}(\Lambda) = \mathbf{W}_1 \mathbf{M}_{0,1}(\mathbf{Y}) \mathbf{W}_1 \quad (\text{A.84})$$

$$\mathbf{M}_{0,2}(\mathbf{YX}) = \mathbf{A}_2 \mathbf{M}_0(\mathbf{X}) + \mathbf{D}_2 \mathbf{M}_{0,2}(\Lambda \mathbf{X}) \quad (\text{A.85})$$

then

$$\begin{aligned} \mathbf{B}_2 \mathbf{B}_2^T &= \mathbf{M}_{0,2}(\mathbf{Y}) - \mathbf{A}_2 \mathbf{M}_0(\mathbf{X}) \mathbf{A}_2 - \mathbf{D}_2 \mathbf{M}_{0,2}(\Lambda) \mathbf{D}_2 \\ &\quad - \mathbf{D}_2 \mathbf{M}_{0,2}(\Lambda \mathbf{X}) \mathbf{A}_2 - \mathbf{A}_2 \mathbf{M}_{0,2}^T(\Lambda \mathbf{X}) \mathbf{D}_2 \end{aligned} \quad (\text{A.86})$$

Season $\tau > 2$: Let

$$\mathbf{M}_{0,\tau}(\Lambda \mathbf{Y}) = \mathbf{M}_{1,\tau-1}(\Lambda \mathbf{Y}) \mathbf{C}_{\tau-1} + \mathbf{M}_{0,\tau-1}(\Lambda \mathbf{X}) \mathbf{A}_{\tau-1} + \mathbf{M}_{0,\tau-1}(\Lambda) \mathbf{D}_{\tau-1} \quad (\text{A.87})$$

$$\mathbf{M}_{1,\tau}(\Lambda \mathbf{Y}) = \mathbf{W}_{\tau-1} \mathbf{M}_{0,\tau-1}(\mathbf{Y}) + \mathbf{M}_{0,\tau}(\Lambda \mathbf{Y}) \quad (\text{A.88})$$

$$\mathbf{M}_{0,\tau}(\Lambda \mathbf{X}) = \mathbf{M}_{0,\tau-1}(\Lambda \mathbf{X}) + \mathbf{W}_{\tau-1} \mathbf{M}_{0,\tau-1}(\mathbf{YX}) \quad (\text{A.89})$$

$$\begin{aligned} \mathbf{M}_{0,\tau}(\Lambda) &= \mathbf{M}_{0,\tau-1}(\Lambda) + \mathbf{W}_{\tau-1} \mathbf{M}_{0,\tau-1}(\mathbf{Y}) \mathbf{W}_{\tau-1} \\ &\quad + \mathbf{M}_{0,\tau}(\Lambda \mathbf{Y}) \mathbf{W}_{\tau-1} + \mathbf{W}_{\tau-1} \mathbf{M}_{0,\tau}^T(\Lambda \mathbf{Y}) \end{aligned} \quad (\text{A.90})$$

$$\mathbf{M}_{0,\tau-1}(\mathbf{YX}) = \mathbf{A}_{\tau} \mathbf{M}_0(\mathbf{X}) + \mathbf{D}_{\tau} \mathbf{M}_{0,\tau}(\Lambda \mathbf{X}) + \mathbf{C}_{\tau} \mathbf{M}_{0,\tau-1}(\mathbf{YX}) \quad (\text{A.91})$$

then

$$\begin{aligned} \mathbf{B}_{\tau} \mathbf{B}_{\tau}^T &= \mathbf{M}_{0,\tau}(\mathbf{Y}) - \mathbf{A}_{\tau} \mathbf{M}_0(\mathbf{X}) \mathbf{A}_{\tau} - \mathbf{C}_{\tau} \mathbf{M}_{0,\tau-1}(\mathbf{Y}) \mathbf{C}_{\tau} - \mathbf{D}_{\tau} \mathbf{M}_{0,\tau}(\Lambda) \mathbf{D}_{\tau} \\ &\quad - \mathbf{D}_{\tau} \mathbf{M}_{0,\tau}(\Lambda \mathbf{X}) \mathbf{A}_{\tau} - \mathbf{A}_{\tau} \mathbf{M}_{0,\tau}^T(\Lambda \mathbf{X}) \mathbf{D}_{\tau} \\ &\quad - \mathbf{D}_{\tau} \mathbf{M}_{1,\tau}(\Lambda \mathbf{Y}) \mathbf{C}_{\tau} - \mathbf{C}_{\tau} \mathbf{M}_{1,\tau}^T(\Lambda \mathbf{Y}) \mathbf{D}_{\tau} \\ &\quad - \mathbf{C}_{\tau} \mathbf{M}_{0,\tau-1}(\mathbf{YX}) \mathbf{A}_{\tau} - \mathbf{A}_{\tau} \mathbf{M}_{0,\tau-1}^T(\mathbf{YX}) \mathbf{C}_{\tau} \end{aligned} \quad (\text{A.92})$$

If adjustments are needed for any season to make $\mathbf{G}_{\tau} = \mathbf{B}_{\tau} \mathbf{B}_{\tau}^T$ positive definite then the following

adjusted estimate is used for $\mathbf{M}_{0,\tau-1}(\mathbf{Y})$ for the next season:

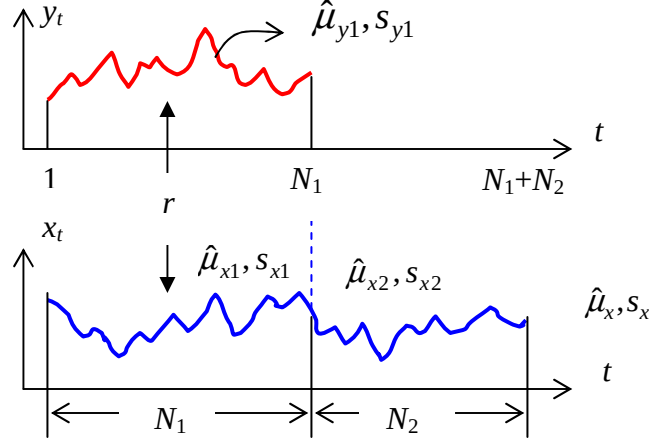
$$\mathbf{m}_{0,\tau-1}^*(\mathbf{Y}) = \mathbf{m}_{0,\tau-1}(\mathbf{Y}) + \hat{\mathbf{B}}_{\tau-1} \hat{\mathbf{B}}_{\tau-1}^T - \hat{\mathbf{G}}_{\tau-1} \quad (\text{A.93})$$

in Eqs. (A.82), (A.88), (A.90) and (A.92).

A.5 Unequal Record Lengths

The models that can deal with unequal record lengths are listed in section 4.5. When working with different length records difficulties can arise in the use of multivariate procedures that require the records to be of same lengths. There are several options to overcome this difficulty, the traditional ones being to either extend the shorter records or to work with the common period of the records. Record extension is usually the way to go, but can be a tedious task that has to be done with a special care. Correctly done, record extension will account for changes in the mean, variance, and autocorrelation over time. If record extension is considered to large of a task, then decisions need to be taken whether only to use the common period of records (sometimes referred to as complete-case methods) or to use all available data (sometimes referred to as available case methods). Using only the common period of record has the advantages of being simple and that univariate statistics across records can be compared since they are estimated from a common sample base. The disadvantages stem from potential loss of information in discarding the uncommon sample base. The advantage of using all available data is simply that all available information is being used, while the disadvantages are that the sample base changes for variable to variable yielding problems in comparability of statistics across variables.

The approach used in SAMS is the one of using all available data in such a way that the overall mean and the variance of each record will be preserved. To further visualize what happens in such an approach, the figure below shows the case of two different length records x_t and y_t :



where

$\hat{\mu}_{y1}$ = mean of the short y_t record of length N_1 .

s_{y1} = standard deviation of the short y_t record of length N_1 .

$\hat{\mu}_{x1}$ = mean of x_t based on the record of length N_1

$\hat{\mu}_{x2}$ = mean of x_t based on the record of length N_2

$\hat{\mu}_x$ = mean of the whole record, x_t .

s_{x1} = standard deviation of x_t based on the record of length N_1

s_{x2} = standard deviation of x_t based on the record of length N_2

s_x = standard deviation of the whole record, x_t .

r = correlation coefficient between the concurrent records of x_t and y_t

For joint modeling of the above data the statistics to be preserved are the overall mean and the standard deviation ($\hat{\mu}_{y1}, s_{y1}$) of the shorter record y_t , and the overall mean and the standard deviation ($\hat{\mu}_x, s_x$) of the longer record x_t . In addition, we would like to preserve the correlation coefficient r or the covariance coefficient m between the concurrent records of x_t and y_t . It should be fairly obvious that for this scenario we can not preserve both the correlation coefficient r and the covariance m of the concurrent records, since

$$m = r s_{x1} s_{y1} \quad (\text{A.94})$$

where s_{x1} is the standard deviation of x_t based on the record of length N_1 , which is not preserved. If r is preserved then the covariance that will be preserved is given by:

$$m^* = r s_x s_{y1} = m \frac{s_x}{s_{x1}} \quad (\text{A.95})$$

or opposite if m is preserved then the preserved correlation is

$$r^* = \frac{m}{S_x S_{y1}} = r \frac{S_{x1}}{S_x} \quad (\text{A.96})$$

As stated above the modeling approach is designed to preserve the long term mean and variances of each site being modeled whether or not the different sites have equal record lengths. As a consequence the actual historical ratio of mean flows or variances of flows between two sites is not necessarily preserved. That is the physically consistent relationship between the two sites of the ratio of mean flows and standard deviations is

$$\hat{\mu}_{x1}/\hat{\mu}_{y1}, \quad \hat{\sigma}_{x1}/\hat{\sigma}_{y1}$$

while the preserved relationship will be

$$\hat{\mu}_x/\hat{\mu}_{y1}, \quad \hat{\sigma}_x/\hat{\sigma}_{y1}$$

Thus if there are differences in the mean and the variances of the series x_t between the two flow periods N_1 and N_2 , then there will be some distortion in the ratio of the flows and the ratio of the variability of the flows at the two sites from what is expected.

A.5.1 Sample Covariance Matrixes

Adjusted procedures are used in estimation of a covariance matrix for a group of sites with unequal record lengths. These covariance matrixes are then used in the parameter estimation procedures of the models presented in this appendix. The goal is to use a covariance estimator that utilizes the best information from the data available, such that the overall variances at each site are preserved and the correlation or covariance between concurrent records at any two sites is preserved.

Correlation Preserved

When the correlation coefficients are to be preserved and adjusted covariance according to Eq. (A.95) then the lag zero variance-covariance matrix of the mean subtracted data set $\mathbf{X}$ representing sites with different record lengths is estimated from

$$\mathbf{m}_0(\mathbf{X}) = \sqrt{\mathbf{v}_X} \mathbf{r}_0(\mathbf{X}) \sqrt{\mathbf{v}_X^T} \quad (\text{A.97})$$

where $\mathbf{v}_X$ is a diagonal matrix with the i th diagonal value being the estimated variance from the full record at site i , and $\mathbf{r}_0(\mathbf{X})$ is the estimated correlation matrix with the i th row, j th column element being estimated as the correlation coefficient computed from the concurrent record at sites i and j . Thus the estimated covariance matrix represents the at-site variances as we wish them to be preserved, and the corresponding covariances needed to preserve the correlation coefficient of the concurrent record between any two sites (refer to Eg. (A.95)). If there is a need to estimate lagged

covariance's, then the corresponding lagged correlation matrix is used. I.e.

$$\mathbf{m}_k(\mathbf{X}) = \text{Cov}(\mathbf{X}_t, \mathbf{X}_{t-k}) = \sqrt{\mathbf{v}_X} \mathbf{r}_k(\mathbf{X}) \sqrt{\mathbf{v}_X^T} \quad (\text{A.97})$$

gives an estimate of the lag- k variance-covariance matrix of $\mathbf{X}$. The covariance matrix between two different data arrays such as $\mathbf{X}$ and $\mathbf{Y}$ is denoted by $\mathbf{m}_k(\mathbf{XY})$ as before.

Covariance Preserved

When the covariance is to be preserved and adjusted correlation according to Eq. (A.96) then each element of the lag- k covariance matrix between $\mathbf{X}$ and $\mathbf{Y}$, $\mathbf{m}_k(\mathbf{XY})$, is estimated as the covariance coefficient computed from the concurrent records of the corresponding sites as for the correlation matrix above.

A.6 Residual Variance-Covariance Non-Positive Definite

It can happen that the matrix $\mathbf{G} = \mathbf{BB}^T$ is not positive definite. Especially when using different record lengths it is more likely that variance-covariance matrixes are not positive definite, and thus adjustments are needed to make the matrixes positive definite. In the temporal disaggregation models by Lane, and by Grygier and Stedinger, as well as in the spatial disaggregation of seasonal data using the MR model (a condensed model), the estimated variance-covariance noise matrix of the previous season is used for estimation of the parameters of the current season. As such, frequent corrections to make matrixes positive definite can have an accumulated effect. To minimize the effects of such corrections on extreme quantiles, decomposition routines that only alter the off-diagonal values to make variance-covariance matrixes positive definite should be preferred. Thus the variance coefficients on the diagonal are not affected, and as such extreme quantiles are more likely to be reproduced. For the above disaggregation models and for the annual CSM-CARMA, decomposition routines are used where off-diagonal values are reduced to make variance-covariance matrixes positive definite. The result should be that the variance of the data will be preserved while the covariance between two different records may be preserved in a reduced form.

APPENDIX B: EXAMPLE OF MONTHLY INPUT FILE

This appendix contains a sample of a monthly input data file used in this manual that corresponds to 12 stations of monthly flows for the Colorado River basin. The data file name is Colorao_River.DAT. Printed below for illustration is data for only two stations (sites 1 and 20). Note that except the first block entitled “station” containing the stations’ names, all other items must be included in the data file.

Remarks:

1. Data values are in free format but they must be separated by at least one space.
2. The item titles including “tot_num_stats”, “Years”, “Seasonal”, “Station”, “Station_id”, and “Duration” depend on the case at hand.
3. The station names following the item title “Station_id” must be one word. If the name has more than one word, the words must be connected by underline “_” such as “AF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ”.
4. The “Station_id” term is optional. Note the if a data file does not include the “Station_id” term, the results in tables and graphs will not show the station’s identification.

station

```
1 AF0725_COLO_RIV_NEAR_GLENWOOD_SPRINGS_CO
2 AF0955_GAINS_ON_COLO_RIV_ABOVE_CAMEO_CO
3 AF1090_TAYLOR_RIV_BELOW_TAYLOR_PARK_RES_CO
4 AF1247_GAINS_ON_GUNNISON_RIV_ABOVE_BLUE_MESA_DAM
5 AF1278_GAINS_ON_GUNNISON_RIV_ABOVE_CRYSTAL_DAM_CO
6 AF1525_GAINS_ON_GUNNISON_RIV_ABOVE_GRAND_JUNCTION
7 AF1800_DOLORES_RIV_NEAR_CISCO_UT
8 AF1805_GAINS_ON_COLO_RIV_ABOVE_CISCO_UT
9 AF2112_GREEN_RIV_BELOW_FONTENELLE_RES_WY
10 AF2170_GAINS_ON_GREEN_RIV_ABOVE_GREEN_RIV_WY
11 AF2345_GAINS_ON_GREEN_RIV_ABOVE_GREENDALE_UT
12 AF2510_YAMPA_RIV_NEAR_MAYBELL_CO
13 AF2600_LITTLE_SNAKE_RIV_NEAR_LILLY_CO
14 AF3020_DUCHESNE_RIV_NEAR_RANDLETT_UT
15 AF3065_WHITE_RIV_NEAR_WATSON_UT
16 AF3150_GAINS_ON_GREEN_RIV_ABOVE_GREEN_RIV_UT
17 AF3285_SAN_RAFAEL_RIV_NEAR_GREEN_RIV_UT
18 AF3555_SAN_JUAN_RIV_NEAR_ARCHULETA_NM
19 AF3795_GAINS_ON_SAN_JUAN_RIV_ABOVE_BLUFF_UT
20 AF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ
21 AF38200_PARIA_RIV_AT_LEES_FERRY_AZ
22 AF40200_LITTLE_COLO_RIV_NEAR_CAMERON_AZ
23 AF40210_GAINS_ON_COLO_RIV_ABOVE_GRAND_CANYON
24 AF41500_VIRGIN_RIV_AT_LITTLEFIELD_AZ
25 AF42100_GAINS_ON_COLO_RIV_ABOVE_HOOVER_DAM
26 AF42250_GAINS_ON_COLO_RIV_ABOVE_DAVIS_DAM
27 AF42600_BILL_WILLIAMS_RIV_BELOW_ALAMO_DAM_AZ
28 AF42750_GAINS_ON_COLO_RIV_ABOVE_PARKER_DAM
29 AF42949_GAINS_TO_COLO_RIV_ABOVE_IMPERIAL_DAM
```

tot_num_stats 29

Years 98
Seasonal 12

Station 1
Station_id AF0725_COLO_RIV_NEAR_GLENWOOD_SPRINGS_CO

Duration	1906	2003									
66982	60131	37105	37525	38047	64812	166869	603358	809692	417092	193160	210126
108379	64733	49279	42194	50071	96240	196106	433066	1001772	718018	229194	116369
92511	59764	46132	52790	40479	62629	127924	244207	528043	237460	144038	69132
54734	46300	41728	47445	36981	53003	94156	365065	1492179	564560	199280	154107
95330	66070	47527	51775	40592	114650	192236	432027	495871	168640	103566	91501
61615	53782	40929	43131	41643	57967	107070	505588	720399	336010	140938	83611
88882	54486	40166	47237	43409	49562	84179	469364	1164973	617765	218221	108734
92922	53868	45820	45295	37443	43405	178781	452171	454694	223095	111851	88371
89738	63309	41991	45102	41898	62247	154771	770934	1132594	382642	186215	112069
101672	58314	38006	38568	37156	45815	136351	286563	584541	309362	109559	67986
71183	48013	42722	44486	39515	76730	142655	457586	730234	332998	197641	113043
96695	62460	48365	42783	40979	48255	172487	422734	1203686	620232	177627	99293
74129	68945	59410	50853	49058	79078	126500	570986	1198263	356694	132773	100034
89297	68111	52079	47061	40205	63580	156749	463641	345500	178533	114824	82416
65121	60461	51737	44661	39778	47030	74993	734672	1025073	404277	178127	106418
83106	67752	50766	49619	38960	76953	103032	639542	1176384	372525	194316	125361
75505	63162	65286	50157	46971	75605	102674	472774	700102	227573	126765	88307
59235	51476	49008	46922	41132	47049	92987	516926	930901	449785	203799	109083
111254	76335	52703	52309	48601	51715	149244	497805	814116	259635	99639	66221
84166	67813	45164	45183	40804	77956	166733	419751	470355	236521	125275	110903
91396	64705	48559	45402	39001	54752	197594	601811	888322	420167	165558	72277
65037	56936	45788	43431	38914	52940	135702	711125	711514	324628	183243	99459
95601	79431	62131	58637	46871	72912	129461	852689	814608	451908	154077	96451
77619	66884	42504	47234	41334	58493	135888	594891	945700	424620	235817	162094
110901	76219	54813	51483	47105	55027	266239	386314	604498	224591	222107	104215
79936	50877	40409	36151	35177	44368	94951	273464	398147	136312	84319	62930
51738	38483	30109	28846	30518	43989	144397	551959	676203	315056	129320	65406
58090	49708	40744	41048	33527	46979	73029	287102	948901	266211	107509	74499
62904	46937	41508	36875	33898	44832	106842	363804	184163	77343	72687	46369
38938	35332	32682	32878	29818	41314	76316	208557	719505	292846	120373	71490
57312	52782	34132	35643	34364	44126	214794	767749	615416	270357	186905	87169
65957	53581	39995	34848	36284	48119	97426	443047	378819	197450	91519	73901
67429	58744	42405	39685	39156	67985	168126	549480	949532	339706	132250	120903
70947	53927	53038	50442	40776	67632	148617	608106	460937	159126	88403	57218
55562	44186	35645	35867	36514	48635	92474	378629	416734	160366	70378	67193
72396	49412	42456	36434	37404	50491	84923	559432	512594	199903	105984	69744
75150	59316	47994	42607	40608	46403	170864	396936	752737	269106	103729	54415
50846	49200	43699	44075	41715	55982	191114	377768	651523	290818	123771	68651
54803	49054	43001	36510	38245	47533	80045	333531	588604	241426	81835	45259
46438	41902	36767	37327	36204	44520	68685	362894	566544	348464	209042	79442
69914	61124	46655	42544	41288	61161	194137	303346	504444	202783	104160	68984
66305	55748	44939	33343	42147	58331	112398	549128	675361	494113	183564	92961
80476	72139	50291	48266	48303	51670	158027	622306	549095	195700	106182	61610
55019	50876	46646	44624	40869	54644	124385	412204	766021	395249	131770	68049
69377	47693	42590	40070	40408	48636	117622	277940	614303	212420	84390	62783
54475	49261	49546	39207	43699	46766	91019	449733	740876	440071	167291	77705
71298	58153	54298	49412	42677	48439	185739	666580	1075525	321526	195481	103520
64890	55042	48392	58034	44877	57621	86564	284166	739652	243074	146609	65314
50632	55257	50101	49325	39339	44595	95489	239584	188348	119991	70003	53496
60569	48668	40662	37153	34292	43084	115940	303573	370450	178407	125269	55446
45293	49645	49765	44365	37678	55256	130022	610195	559630	152713	100601	48529
42833	46291	41707	44234	41397	48877	82779	405790	1124200	800408	235575	107050
80661	67628	58387	47085	47229	55450	85473	702806	620916	153944	87635	56025
46091	49144	48527	45269	39796	46741	85664	316390	648090	207743	117050	65122
99832	70173	47771	42496	40149	81815	204044	381424	657146	217525	95350	58241
51494	48161	41683	34846	37768	48018	67935	303658	442177	151143	116820	157794
164750	83858	55792	52436	58677	62034	354442	652543	728064	406813	146639	70234
67380	63937	48655	42399	45223	56668	101600	298438	263378	118472	122212	82759
50658	49824	38421	37055	32555	37770	73059	360298	415889	206374	111718	60328
48835	47326	48363	48033	39346	43406	111474	403793	843039	509240	235520	112587
101622	70219	60033	51315	42295	64353	101860	308340	250274	134723	87730	56271
66110	52708	41946	43206	37858	61896	113125	313901	532450	273237	111763	89660
74455	63001	50627	40685	42696	49542	83307	239313	714275	248537	165277	82753
67198	57297	53505	54369	43247	48579	152516	495605	477203	285049	124469	84804

99534	68529	53773	51667	48101	49606	89445	646924	728044	324818	140779	107343
107520	78589	56439	56801	53591	87036	194003	412919	813367	372231	140597	115143
86357	69319	61454	54262	50520	82030	129653	371889	658216	179920	101278	111506
95102	72427	62874	54335	47527	60767	93038	471001	727002	419052	169483	81365
72362	69298	55128	56077	49782	73795	128757	665951	656219	286117	122882	73215
75893	67402	52703	54350	52119	65065	96551	303915	661220	490561	153829	80089
62314	57959	52696	51309	53891	64242	109161	345056	441495	232808	132114	84661
72467	49566	38248	36344	35485	44764	92476	195933	250533	102959	82813	54815
62417	51366	50402	72755	44631	61093	113823	364554	826560	524972	204840	79048
69504	61360	58841	42795	45328	63979	106088	424516	748100	502968	199663	90468
79310	65104	61707	67622	64083	60419	87043	405019	718865	395812	147653	74086
62018	60106	53199	48714	36199	37216	68780	180660	346432	217544	90700	71609
61005	46826	54879	46150	38059	56801	70059	324073	634782	505031	231892	114411
103772	63707	59730	54701	47185	57651	70535	295392	945138	796122	336116	135273
100277	71109	77365	41342	64913	74365	114613	759155	1029067	643457	305878	163253
120208	106556	88230	67306	67346	89336	220625	630298	695074	376223	166450	82388
92031	95390	75647	54683	63393	103571	199973	514564	795578	514640	161439	115048
115259	96403	74765	44881	40610	47496	143451	365781	365417	198847	101350	46955
70432	65920	55366	55946	47057	69184	125677	372696	534614	332562	115617	63175
56538	42217	46468	43907	41094	67963	139104	325377	374434	228274	126582	66294
55371	60754	50190	37115	39109	42310	81947	210037	451000	285667	110453	69747
82310	48297	65646	42659	39682	44481	89779	332721	555775	330371	141464	106982
46174	80357	44404	43958	38438	51028	100890	350892	373490	247464	131607	87424
56700	70794	52588	42062	37737	40482	96171	503189	757024	504042	207955	93620
75485	62401	69346	48943	38264	56740	99169	358395	417848	205294	89234	75710
64275	54843	50394	44354	34743	53475	52437	216678	763853	732973	302524	97708
89076	73459	59770	54274	54964	63698	176645	607903	733404	351918	143289	86605
81153	91386	56034	68171	49978	68594	131076	597346	1003665	423311	200850	127746
123060	89557	74956	63650	43293	66319	113531	345083	426884	350099	176237	94519
67945	73004	71197	61824	66417	81649	91448	278322	590508	402119	179053	124181
86787	56637	66008	60078	42381	56636	107827	489284	470137	228092	115595	75113
48012	59550	65627	24466	35835	51058	80246	365363	419643	209847	107744	84498
55925	44124	53872	35838	24350	48853	67452	176288	174615	87802	77376	46494
42935	71020	43158	44687	38970	50672	79137	414097	629629	336948	141320	83834

Station 20
Station_id AF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ

Duration	1906	2003									
458528	401644	226871	244314	292534	678174	1204640	3635101	5014167	2950460	1605086	1503159
739807	503006	353312	356760	377349	789130	1465838	2702179	5967232	5103491	1920787	955414
608812	377467	268130	276192	379543	664762	1041224	1595614	2922360	1924283	1117477	598088
483627	395707	312145	378989	317458	763721	1120492	3349297	7203254	4109919	1880422	1526396
680646	489990	377548	289322	493565	1403871	1730475	3298793	3101705	1373125	866631	630999
616468	445769	345922	367374	482597	902111	951815	2924637	4124342	2353784	1016615	593647
1138005	442055	353040	346040	327040	538145	902409	3684152	6151097	3206236	1362372	631542
636272	533065	305040	354040	314035	523340	1829661	3270774	3144985	1984476	874869	701626
670353	538369	329845	369540	401135	876055	1593814	4685650	6296013	3116692	1405438	783864
964928	527355	334330	304135	397335	525840	1483873	2427137	3642473	2147795	853538	528870
557984	411050	343247	393997	424368	1391402	1802736	3736188	4752150	2633062	1931864	809499
1402793	495715	369118	260296	351858	506891	1545288	3763312	7772051	4940893	1618993	822053
510346	448052	402771	356292	373570	655997	901047	2760607	5393082	2288860	968227	691873
569910	496385	410089	287188	316951	653288	1414719	3231444	2597757	1537305	904498	531938
377402	404787	394092	406940	601645	685472	983984	5917499	6993901	3165233	1376497	620527
534995	596367	404572	414071	456636	943675	930238	4180109	8467230	2849389	1972571	953215
488368	417789	453490	351437	438928	907266	1185878	4699578	5761054	2159890	1148518	657391
336581	400845	399832	375213	340452	449461	1316359	3835398	5077612	3053685	1744686	1013539
747521	646295	423825	312563	506890	508913	1665561	3264099	3780821	1672023	720755	389827
388361	392567	275418	262125	403157	607575	1382195	2536635	2860901	2086524	1040652	1174710
1020530	608566	447131	359577	353544	643799	1634988	3546065	4075706	1998872	966236	459006
461696	334894	379348	337439	388832	605741	1269471	4135924	4064755	3135304	1321496	2116962
979882	739253	444153	469629	463036	754898	1025978	4580808	4271762	2241461	1048280	558717
625418	570090	344257	331823	346061	923749	1698112	4276261	5414640	2744488	2389754	1742400
964743	560310	437244	298790	485407	575246	1792671	2168481	3724824	1693721	1891015	691053
587559	423714	288560	263662	366639	429833	597640	1387684	2042727	1147598	671677	424426
536283	353322	252643	272930	557282	673831	1676128	4286246	4193514	2684941	1364498	693906
367644	378380	272887	273376	255953	501362	515700	1604249	4680018	1898287	818373	563832
440664	297779	333907	308075	303395	349072	557263	1480351	1018245	721126	532811	284828
212899	181355	228772	254933	274011	339574	685733	1585305	4708552	2255472	959192	594224
387726	319435	266192	264047	318400	459898	1400149	4032422	3360120	1709054	1262461	705479

376632	443083	317128	200331	414259	700570	1559558	3833665	2958383	1923464	838115	596566
505920	384592	390633	325637	354575	794138	1659082	3599128	5324845	2503358	1027381	1050775
618499	479804	411097	348487	300377	809304	1228538	2865278	2250280	1104801	629626	671962
358134	312958	284314	261837	301174	439068	735512	2442459	2212812	984226	522322	525463
731809	410102	364873	355941	430079	675567	1127132	5323093	4598652	2428433	1190375	683285
1813960	913232	576929	404450	395910	660985	2902862	3500486	4834784	2074078	938573	412011
358326	373655	369016	345094	344573	533607	1624861	2446508	3294191	2132619	1188466	613563
386115	442637	379167	284953	344393	515043	1060878	3622415	4760167	2526378	857685	332678
378318	378988	307526	330444	359434	430301	790464	3150282	3358331	2468347	1465735	494543
538329	434400	319918	348056	313504	506085	1141098	1970811	2755500	1432801	852859	449365
430011	472765	422598	265000	353367	656705	844051	3600478	3790869	2726585	1575216	778634
830411	577524	440646	375949	432004	624879	1728270	4032836	3915190	1662639	887163	372680
361999	399745	345940	326826	350930	692324	1377417	3474042	5116808	2809867	985997	420279
539733	475823	363538	346883	394729	631678	1270496	2239296	3782181	2027117	817606	428842
423062	355831	422695	307758	356588	416528	564533	2034805	3694508	2205007	1172271	532247
430044	451352	340253	490658	385654	435309	2329319	5569121	6201051	2317967	1255129	694186
376061	376582	374385	402474	365207	458845	554827	1285021	3910327	1662389	1032517	405366
318406	427066	342974	317925	341586	388722	666898	1753441	1396009	1255884	664718	494512
570813	355936	289658	254680	252729	590617	697977	1950795	2332135	1220313	920244	359573
225234	274490	335121	379784	279980	513692	993694	2814517	3534913	1151798	703754	298120
193813	304643	258166	295275	331116	508805	868604	2805792	6669099	4906010	2007877	1010603
756358	838468	502956	392045	536727	688965	1599996	4597690	4562509	1308184	677219	438820
333453	358554	368349	306407	313512	350118	463516	1380376	2826173	1448923	766845	316311
557316	517710	350962	289809	314720	749816	1720737	1977890	3222979	1361812	582813	328283
361418	348931	264952	244498	318919	368225	637529	1642974	2528584	956734	718990	856024
819598	547420	370764	334494	774737	545028	2532520	4119768	3849168	2550866	912852	412135
555007	448064	342970	201557	370712	575260	763590	1808387	1839152	933748	685572	735431
319363	342117	266011	267885	262479	343862	649129	2354779	2984535	1729449	915192	366401
301361	325117	363397	379725	369167	443493	1400634	3392487	5596742	3793601	1623391	877286
875445	570571	552485	455182	395360	981129	1333026	2523296	1934274	1053979	589839	357643
335665	349297	371154	289340	307306	576121	604735	1690771	3628249	2187199	951241	517396
351908	327692	238872	313145	337745	517660	639473	2123875	5021980	1742269	1468908	424710
443620	385800	320747	391523	352823	571807	1972984	3869865	3004303	2035789	892706	607745
675186	513856	383572	382737	361005	447329	615374	3630445	4189472	2096715	917019	1131553
649771	515681	407035	490428	491917	609586	1346671	2442170	4378219	2193103	898173	672638
661344	551215	479703	503247	467405	821308	823858	1927411	3758045	1164322	651283	570793
1117457	673421	453960	440638	460070	850577	1352785	4438702	5017892	2725430	995488	671589
463087	519916	440926	461515	405635	789606	941881	3337175	3326953	1524780	707333	366575
408958	509477	344922	427953	377807	626521	867004	2690100	4980524	3983484	1022959	525705
405373	448805	425556	398670	474045	549727	842291	2425697	2791610	1295121	741627	491105
442882	374018	283634	323678	293529	279558	362869	621039	948914	655405	568483	370874
324926	342911	315076	366742	305861	615011	1229315	2725495	4996726	2527981	792844	409148
347725	487162	398648	388798	359164	748474	1813403	3987890	5216554	2656609	1048268	418098
366547	408377	359686	477018	610450	643393	1423153	4334181	5335616	2224502	714267	608819
415724	481358	427048	392521	321594	365405	625969	1207244	2350327	1142295	520895	542125
645087	465497	407287	353944	322075	649796	980164	3005422	4261797	2997487	1549240	1080058
997492	726531	620587	395661	459797	896921	1130239	3529632	7749358	5119270	2123359	847309
1056364	707826	650196	436388	516388	855802	1439814	6051182	6696277	3864820	1957909	1063077
1042063	829281	644638	588807	590005	1126236	2928584	4877643	4709583	2281036	1097301	738505
914200	748568	638841	549620	744835	1089259	2171122	3843805	6019606	3406845	1334490	992365
1144081	999075	730182	526848	623570	948887	1875380	3672651	3171561	1549328	1035658	655526
490099	630954	431240	358635	413160	716818	1045200	2042327	2757869	1464669	866077	579061
478381	379938	344918	331736	369328	824828	1195159	1738912	1989335	1218965	819605	458147
378394	412665	300616	283874	286727	406718	623336	1272203	2650122	1431174	734448	546716
584293	452232	300982	302663	387347	488399	805069	2142072	3397023	1576149	966693	798055
366329	571913	355320	331233	423080	597821	1077164	2233709	2128362	1358845	893001	639570
390254	461410	317306	422016	406299	850921	1306387	4392963	5018064	2568377	1197118	773740
565586	473521	405719	394861	356573	667133	796880	2280260	2463400	1063026	680931	529090
535822	461242	392859	365597	451478	838692	711196	2276743	6260631	5275349	1721807	746118
665334	549699	472341	432199	506944	548217	1093148	3310201	3633660	2037048	780273	539180
574916	626952	506313	501689	446694	1051532	1459934	4542370	6138492	2439388	1511481	1225995
1045442	719272	555801	522285	476725	751272	1315036	3592304	3606179	2583291	1300234	734395
705799	758134	500151	486391	427290	670363	798775	2578717	4445246	2538308	1606115	1074032
597603	512581	410215	441605	431047	519651	1116217	2559824	2296158	1076228	636522	537722
465751	458174	402471	304030	321668	583675	901810	2742548	2294801	1166622	824135	482174
370904	383981	334526	301419	255292	374686	584548	811697	1124148	727763	438371	483371
361480	428525	297771	283654	279225	499211	644975	2002318	2954098	1215702	622129	674693

APPENDIX C: EXAMPLE OF ANNUAL INPUT FILE

This appendix contains a sample of an annual input data file used by SAMS corresponding to 98 stations of annual flows for the Colorado River basin. Printed below for illustration are data for only two stations (sites 1 and 20).

tot_num_stats 12

Years 98

Annual

```
Station      1
Station_id AF0725_COLO_RIV_NEAR_GLENWOOD_SPRINGS_CO
705000
3105000
1705000
3150000
1900000
2193000
2987000
1828000
3084000
1814000
2297000
3036000
2867000
1702000
2832000
2978000
2095000
2598000
2280000
1891000
2690000
2469000
2915000
2833000
2204000
1337000
2106000
2027000
1118000
1700000
2401000
1561000
2575000
1859000
1442000
1821000
2060000
1989000
1640000
1878000
1701000
```

2408000
2044000
2190000
1658000
2250000
2873000
1894000
1056000
1414000
1884000
3021000
2063000
1716000
1996000
1501000
2836000
1311000
1474000
2491000
1329000
1738000
1854000
1944000
2409000
2488000
1956000
2354000
2310000
2154000
1688000
1056000
2456000
2414000
2227000
1273000
2184000
2965000
3445000
2710000
2786000
1641000
1908000
1558000
1494000
1880000
1596000
2462000
1597000
2468000
2495000
2899000
1967000
2088000
1855000
1552000
893000

1976000

Station 20
Station_id AF3800_GAINS_ON_COLO_RIV_ABOVE_LEES_FERRY_AZ
Duration 1906 2003

18210000
21230000
11770000
21840000
14740000
15130000
19080000
14470000
21070000
14140000
19190000
23850000
15750000
12950000
21930000
22700000
18670000
18340000
14640000
13410000
16110000
18550000
17580000
21410000
15280000
8632000
17550000
12130000
6628000
12280000
14490000
14160000
17920000
11720000
9380000
18320000
19430000
13620000
15510000
13910000
11060000
15920000
15880000
16660000
13320000
12490000
20900000
11200000
8368000
9795000
11510000
20160000

16900000
9233000
11970000
9248000
17770000
9259000
10800000
18870000
11620000
11810000
13510000
14850000
15340000
15100000
12380000
19200000
13290000
16770000
11290000
5525000
14950000
17870000
17510000
8793000
16720000
24600000
25300000
21450000
22450000
16930000
11800000
10150000
9327000
12200000
10980000
18100000
10680000
20040000
14570000
21030000
17200000
16590000
11140000
10950000
6191000
10260000

APPENDIX D: EXAMPLE OF TRANSFORMATIONS

The logarithmic transformation coefficients for both annual and monthly flows for each site of the example data file Colorado_River.DAT are given below. Refer to Eq. (4.1) for detail.

Transformation coefficients for annual flows

Site	Type of Trans	Coefficients		Skewness Test		Filliben Test	
		a	b	0.3928	Result	0.9891	Result
1	Log	2324.1916	1	-0.0777	accept	0.9942	accept
2	Gamma	0	1	0.0656	accept	0.9983	accept
3	Gamma	0	1	0.0801	accept	0.9943	accept
4	Log	4334.4335	1	-0.0259	accept	0.9964	accept
5	Log	23.4228	1	-0.1336	accept	0.9927	accept
6	Log	884.0838	1	0.0920	accept	0.9946	accept
7	Log	636.9696	1	0.0329	accept	0.9943	accept
8	None	1	1	-0.0456	accept	0.9944	accept
9	Gamma	0	1	0.0338	accept	0.9958	accept
10	Gamma	0	1	0.0067	accept	0.9958	accept
11	Log	252.0259	1	-0.0475	accept	0.9977	accept
12	Log	1197.9786	1	0.0283	accept	0.9973	accept
13	Log	677.2791	1	0.0554	accept	0.9958	accept
14	Gamma	0	1	0.0356	accept	0.9964	accept
15	Log	0	1	-0.0376	accept	0.9944	accept
16	Gamma	0	1	0.0072	accept	0.9932	accept
17	Log	66.6643	1	0.0375	accept	0.9951	accept
18	Log	2540.7005	1	0.0114	accept	0.9949	accept
19	Log	194.098	1	-0.0514	accept	0.9967	accept
20	None	1	1	0.1947	accept	0.9774	REJECT
21	Log	-3.2543	1	-0.0148	accept	0.9967	accept
22	Log	46.0528	1	0.0554	accept	0.9948	accept
23	Power	457.3136	1.9	-0.0117	accept	0.9957	accept
24	Log	-55.4413	1	0.0024	accept	0.9958	accept
25	Log	1062.5804	1	-0.0409	accept	0.9974	accept
26	Gamma	0	1	-0.1730	accept	0.9905	accept
27	Log	0	1	-0.2582	accept	0.9921	accept
28	Gamma	0	1	0.0282	accept	0.9974	accept
29	Power	683.0857	1.3	0.0253	accept	0.9966	accept

Transformation coefficients for monthly flows (for month 1 only)

Site	Type of Trans	Coefficients		Skewness Test		Filliben Test	
		a	b	0.3928	Result	0.9891	Result
1	Log	33.7402	1	0.1596	accept	0.9922	accept
2	Log	21.8888	1	-0.0010	accept	0.9976	accept
3	Power	-0.3107	1.25	0.0906	accept	0.9945	accept
4	None	1	1	0.0109	accept	0.9951	accept
5	Log	2.3605	1	0.4676	REJECT	0.9733	REJECT
6	None	1	1	0.1894	accept	0.9813	REJECT
7	Log	4.1527	1	0.0881	accept	0.9941	accept
8	None	1	1	-0.0313	accept	0.9676	REJECT
9	Log	43.1103	1	0.2868	accept	0.9830	REJECT
10	None	1	1	0.4384	REJECT	0.9153	REJECT
11	Log	48.501	1	-0.0512	accept	0.9929	accept
12	Log	0	1	0.0543	accept	0.9964	accept
13	Gamma	0	1	0.1387	accept	0.9960	accept
14	Log	20.456	1	0.0524	accept	0.9922	accept
15	None	1	1	0.3179	accept	0.9836	REJECT
16	Power	111.0954	1.9	-0.0245	accept	0.9720	REJECT
17	Log	-0.7337	1	-0.0911	accept	0.9892	accept
18	Log	0	1	-0.2179	accept	0.9932	accept
19	Log	237.2225	1	0.2166	accept	0.9292	REJECT
20	None	1	1	0.1405	accept	0.9779	REJECT
21	Log	-0.3601	1	-0.0672	accept	0.9874	REJECT
22	Log	0.0009	1	-0.2150	accept	0.9900	accept
23	Power	42.5844	1.35	0.1123	accept	0.9752	REJECT
24	Log	-5.1589	1	0.2141	accept	0.9947	accept
25	Log	151.3734	1	0.1917	accept	0.9840	REJECT
26	Power	122.6741	1.9	0.1505	accept	0.9897	accept
27	Log	-0.0784	1	0.2529	accept	0.9782	REJECT
28	Log	185.4363	1	-0.0463	accept	0.9971	accept
29	Power	216.6031	1.9	-0.2606	accept	0.9878	REJECT